

High-Performance Process Manager Configuration Forms

HP88-600

Implementation Configuration Forms

High-Performance Process Manager Configuration Forms

HP88-600

Release 681

09/08

NOTE

This document was last updated for R681.
For information about those parameters introduced post R681,
refer to the PM Family Parameter Reference Dictionary.

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About This Publication

This publication supports TPS System network Release 681. TPS is the evolution of TDC 3000.

This manual consists of configuration forms. These forms are designed to facilitate the collection and organization of data for subsequent entry into the TPS system through the Universal Station.

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<u>Form Name</u>	<u>Form ID</u>
UCN Node Configuration	HP88-601A
HPM Node Specific Configuration	HP88-601B

This publication also contains the following point forms in numerical sequence. They are listed here in alphabetical order for convenience.

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HPM Configuration Form Instructions

FILLING OUT PAPER FORMS

Where to Find High-Performance Process Manager Parameter and Data Point Help Information

Refer to the *HPM Control Functions and Algorithms* and to the *HPM Parameter Reference Dictionary* Publications in the *Implementation/High-Performance Process Manager* binders 1 and 2 respectively.

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Universal Station's configuration display where values that have been written on the form and parameter options that have been circled on the form will be keyed in and selected, respectively.

HPM Analog Input Point

Unit:	Tagname:	Engineer:	Date:
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Point Assignment Display

NAME Tag Name	_____	range = Not applicable default = Blank
PNTFORM Point Form	FuLL Component	Point is fully displayed and alarmed Pt. Is partially displayed but not alarmed

If the point is of the form **Component**, parameters within boxes that have **thick, bold borders** do not apply.

Lines indicate **what parameter/s or Parameter-option/s** apply when an option is selected for a previous parameter.

PVSRCOPT PV Source Option	Only Auto All	N/A if PNTFORM = Component. PV Source selection is not available and Field wiring or memory fetch supplies PV. Selection from PVSOURCE is available And Operator/CL-program may supply PV.
PVSOURCE PV Source	Auto Man Sub	Field wiring or memory fetch supplies PV PV is supplied by Operator or program A value is substituted by a CL program

If parameters within a box do not Apply when some other parameter Option is selected, that box is Indicated by

NMODATTR Normal Mode Attribute	None Operator Program	N/A if NMODE = None Parameter NMODE sets "normal" mode Operator sets "normal" operating mode Program sets "normal" operating mode
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Lower-level alarm trip points must Have a value specified before Higher-level alarm trips points apply. The lines show the scheme.

Alarming Display N/A if PNTFORM = Component							
PVHITP PV High Alarm Trip Point	_____	range = PVLOTP - PVHHTP, & NaN default = NaN					
PVHIPP PV High Alarm Priority	NoAction Journal Low	High	Emergency	Printer	JnlPrint		
PVLOTP PV Low Alarm Trip Point	_____	range = PVLLTP - PVHITP, & NaN default = NaN					
PVLOPR PV Low Alarm Priority	NoAction Journal Low	High	Emergency	Printer	JnlPrint		
PVHHTP PV Hi Hi Alarm Trip Point	_____	range = PVHITP to PVEUHI, & NaN default = NaN					
PVHHPR PV High High Alarm Priority	NoAction Journal Low	High	Emergency	Printer	JnlPrint		
PVLLTP PV Low Low Alarm Trip Point	_____	range = PVEULO - PVLOTP, & NaN default = NaN					

ENTERING DATA FROM THE FORMS INTO THE SYSTEM

Select **NETWORK INTERFACE MODULE** from Engr's Main Menu to get NIM BUILD TYPE SELECT MENU

UCN NODE CONFIGURATION Form HP88-601A	HPM NODE SPECIFIC CONFIGURATION Form HP88-601B	PROCESS POINT BUILDING Forms HP88-602-635	NIM LIBRARY CONFIGURATION Form HP88-640
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Select a point type from this POINT BUILD SELECT MENU

Reg Ctl	AI/AO	DI/DO	Array	Pulse	DevCtl	Logic	Numeric	Proc Mod	Flag	Timer	Reg PV
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Select an algorithm type depending upon what is circled on the form for parameter CTLALGID

PidF	PidErfb	RatioCtl	AutoMan	ORSel	Summer
PidFF	Pid PosPr	PosProp	Switch	RampSoak	MulDiv

Select an algorithm type depending upon what is circled on the form for parameter PVALGID

DataAcq	FlowComp	MidOf3	HiLoAvg	Summer	VdtLdLag	Totalizr	GenLin	CalcuLtr
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HPM UCN Node Configuration

Status Message Text Configuration Display

N/A for PM; N/A if NMSGTXT =0

MSGTXT(1) Status Message Text #1		<i>range = Eight Character String default = Blank</i>
MSGTXT(2) Status Message Text #2		<i>range = Eight Character String default = Blank</i>
MSGTXT(3) Status Message Text #3		<i>range = Eight Character String default = Blank</i>
MSGTXT(4) Status Message Text #4		<i>range = Eight Character String default = Blank</i>
MSGTXT(5) Status Message Text #5		<i>range = Eight Character String default = Blank</i>
MSGTXT(6) Status Message Text #6		<i>range = Eight Character String default = Blank</i>
MSGTXT(7) Status Message Text #7		<i>range = Eight Character String default = Blank</i>
MSGTXT(8) Status Message Text #8		<i>range = Eight Character String default = Blank</i>
MSGTXT(9) Status Message Text #9		<i>range = Eight Character String default = Blank</i>
MSGTXT(10) Status Message Text #10		<i>range = Eight Character String default = Blank</i>
MSGTXT(11) Status Message Text #11		<i>range = Eight Character String default = Blank</i>
MSGTXT(12) Status Message Text #12		<i>range = Eight Character String default = Blank</i>
MSGTXT(13) Status Message Text #13		<i>range = Eight Character String default = Blank</i>
MSGTXT(14) Status Message Text #14		<i>range = Eight Character String default = Blank</i>
MSGTXT(15) Status Message Text #15		<i>range = Eight Character String default = Blank</i>

HPM NODE SPECIFIC Configuration

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	Engineer:	Date:
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Node Specific Configuration Display

NTWKNUM Network Number currently being configured	[.].	range = 1 to 20 default = Not applicable
NODENUM Node Number	[.].	range = 1 to 64 default = Not applicable
NODETYP Node Type	HPM PM, APM, LM or SM	An HPM Point Is Being Configured If a PM, APM, LM or SM is being configured, go to the appropriate forms for those devices.

Parameter	Description	Range	Default	Quantity N	PU Calculations N X PUS =	MU Calculations NX MUS =
NCTLSLOT	# Of Regulatory Control Slots	0 to 250	100		X 1 =	X 13 =
NFASTCTL	# Of Fast Reg. Control Slots	0 to 100 but ≤ NCTLSLOT	0		X 4 =	X 13 =
NPVSLOT	# Of Regulatory PV Slots	0 to 125	20		X 1 =	X 12 =
NFASTPV	# Of Fast Reg. PV Slots	0 to 100 but ≤ NPVSLOT	0		X 4 =	X 12 =
NLOGSLOT	# Of Logic Slots	0 to 400	25		X 1 =	X 15 =
NFASTLOG	# Of Fast Logic Slots	0 to 100 but ≤ NLOGSLOT	0		X 4 =	X 15 =
NDCSLOT	# Of Digital Composite Slots	0 to 999	150		X 0.1 =	X 15 =
NFASTDC	# Of Fast Digital Composite Slots	0 to 999 but ≤ NDCSLOT	0		X 0.4 =	X 15 =
NDEVSLOT		0 to 400	0		X 1 =	X 30 =
NFASTDEV	# Of Fast Device Control Slots	0 to 100 but ≤ NDEVSLOT	0		X 4 =	X 30 =
SEQPROC	Processing Units Per Proc. Mod. Data Point/Scan	1_PU - 2_PU	1_PU		X 0	N/A
NPMSLOT	# Of Process Module Slots	0 to 250	0		X 1 or 2 = (# of SeqProc)	X 15 =
NNUMERIC	# Of Numerics	0 to 16,384 (mult of 16)	1024		X 0	X 1/16 =
\$NNUMALM (R660 and later)	#Of Alarmable Numerics	0 to 2047 or ≤ NNUMERIC	0		X 0.025	X 1/4 =
NSTRING	# Of Strings	0 to 16,384 (mult of 16)	0		X 0	X 1/8 =
NTIME	# Of Times	0 to 4096 (mult of 32)	0		X 0	X 3/32 =
NARRSLOT	Number of Array Slots	0 to 500	0		X 0	X 8 =

HPM NODE SPECIFIC Configuration

	Engineer:	Date:
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Node Specific Configuration Display (Contd.)

Totals	PUs and MUs are explained in subsection 3.1.3 in the HPM Control Functions and Algorithms manual.		PUs	MUs
			Maximum = 800	Max. = 20,000
SCANPER	1.0	0.5	0.25	
SI Data HPMM CTL Scan Period				
SCANRATE	<u>Selection</u>	<u>RegCtl&RegPV</u>	<u>Logic & DigComp</u>	
HPMM'S Scan Rate in Times Per Second	Reg1Log1	1 per sec.	1 per sec.	
	Reg1Log2	1 per sec.	2 per sec.	
	Reg1Log4	1 per sec.	4 per sec.	
	Reg2Log2	2 per sec.	2 per sec.	
	Reg2Log4	2 per sec.	4 per sec.	
	Reg4Log4	4 per sec.	4 per sec.	
PKGOPT	Redun	Redundant HPMMs in same file; No redund.IOMs		
HPMM Hardware Packaging Option	Redun_2F	Redundant HPMMs in separate files; No redund. IOMS		
	Redun_IO	Redundant HPMMs (sep.files) & redund.I/O Modules		
	Nodefalt	No defaults for I/O Module/Card Positions		
NOTE The HPM can be operated in nonredundant mode, by selecting Redun, Redun_2F, Or No default and only inserting one set of HPMM cards. The card slots vacated by The second set of HPM cards can be used to house IOP cards, if necessary.				
DISP_SIM	On	If On, Simulation Mode indicators appear on		
Simulation Indicator Option	Off	Group and Detail displays (but only when the		
		Optional I/O simulator personality is loaded.		

HPM NODE SPECIFIC Configuration

	Engineer:	Date:
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IO Module Configuration Display - N/A if NTWKNUM = 0

The parameters listed below make up one entry or line in the IO Module Configuration Tables. The IO Module Numbers (Modnum) are predefined on the Data Entry Displays. Fill in a table (on the following pages) appropriate for the selected packaging option. Refer to the parameters below for descriptions/defaults.

If you elect to operate in nonredundant mode, you may configure IOP cards in the unused HPMM card slots. In that case, use the Nodefault configuration form to specify the IO Module assignments.

MODNUM I/O Module Number		<i>range = 1 to 40</i> <i>default = Not applicable</i>
IOMFILEA(#) IOM -A File Number		<i>range = 0 - 8</i> <i>default = 1(0 = not connected)</i>
IOMCARDA(#) IOM-A Card Number		<i>range = 0 - 15 (0=not connected)</i> <i>default = Varies with PKGOPT and MODNUM</i>
IOMTYPE(#) Module Type	FBUS Fieldbus PI Pulse Input HLAI High Level Analog Input SI IO Module Components AO_00 Analog Output AO_16 Digital Output DO_32 16 Pt. Analog Output 32 Pt. Digital Output	None Address not configured LLAI Low Level Analog Input DI Digital Input LLMUX Low Level Multiplexer (also includes RHMUX) STIM Smart Xmitter Intf. Modl DISOE Digital Input for SOE

FAILOPT Analog/Digital Output Option	Hold Unpower	The hardware output is maintained at its current setting The hardware output is unpowered
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FREQ6050 AC Line Period	FiftyHZ SixtyHZ	Line Frequency is 50 Hz Line Frequency is 60 Hz
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When Pkgopt = **Redun_IO** or **Nodefault**, each redundant IO Module also has the following options

IOREDPT(#) I/O Module Redundancy Option	Nonredun Redun	Not redundant Redundant
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IOMFILEB(#) Corresponding IOM-B File Number	<i>range = 0 - 8 (0 = not connected)</i> <i>default = Blank for Nodefault, 2-6 for Redun_IO</i>
IOMCARDB(#) Corresponding IOM-B Card Number	<i>range = 0 -15 (0 = not connected)</i> <i>default = Blank for Nodefault, See Pg 6-7 for Redun_IO</i>

HPM NODE SPECIFIC Configuration

IO Module Configuration Table

Engineer:	Date:
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IO Module Configuration Display – N/A if NTWKNUM = 0

For each IO Module, select the IO Module type (IOMTYPE) that applies. If the type selected is AO or DO, Select a failure option (FAILOPT). If the type is STIM or LLMUX, select an ac Line Frequency (FREQ6050).

****CAUTION – Use this part of the form only if PKGOPT on page 2 = Redun****

NOTE: IOMTYPE LLMUX also includes RHMUX.

MOD NUM	IOM-A FILE	IOM-A CARD	OIMTYPE										FAILOPT				IOMTYPE		FREQ6050	
1.	I11	I05	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
2.	I11	I06	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
3.	I11	I07	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
4.	I11	I11	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
5.	I11	I12	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
6.	I11	I13	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
7.	I11	I14	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
8.	I11	I15	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
9.	I21	I01	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
10.	I21	I02	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
11.	I21	I03	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
12.	I21	I04	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
13.	I21	I05	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
14.	I21	I06	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
15.	I21	I07	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
16.	I21	I08	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
17.	I21	I09	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
18.	I21	I10	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
19.	I21	I11	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
20.	I21	I12	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
21.	I21	I13	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
22.	I21	I14	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
23.	I21	I15	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
24.	I31	I01	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
25.	I31	I02	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
26.	I31	I03	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
27.	I31	I04	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
28.	I31	I05	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
29.	I31	I06	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
30.	I31	I07	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
31.	I31	I08	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
32.	I31	I09	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
33.	I31	I10	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
34.	I31	I11	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
35.	I31	I12	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
36.	I31	I13	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
37.	I31	I14	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
38.	I31	I15	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
39.	I31	I16	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
40.	I31	I17	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ

HPM NODE SPECIFIC Configuration

IO Module Configuration Table

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	Engineer:	Date:
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IO Module Configuration Display – N/A if NTWKNUM = 0

For each IO Module, select the IO Module type (IOMTYPE) that applies. If the type selected is AO or DO, select a failure option (FAILOPT). If the type is STIM or LLMUX, select an ac Line Frequency (FREQ6050).

****CAUTION – Use this part of the form only if PKGOPT on page 2 = Redun_2F ****

NOTE: IOMTYPE LLMUX also includes RHMUX.

MOD NUM	IOM-A FILE	IOM-A CARD	IOMTYPE										FAILOPT				IOMTYPE		FREQ6050	
1	I1I	I03I	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
2.	I1I	I04I	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
3.	I1I	I05	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
4.	I1I	I06	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
5.	I1I	I07	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
6.	I1I	I08	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
7.	I1I	I09	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
8.	I1I	I10I	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
9.	I1I	I11I	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
10.	I1I	I12I	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
11.	I1I	I13I	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
12.	I1I	I14I	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
13.	I1I	I15I	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
14.	I2I	I03I	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
15.	I2I	I04I	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
16.	I2I	I05	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
17.	I2I	I06	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
18.	I2I	I07	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
19.	I2I	I08	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
20.	I2I	I09	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
21.	I2I	I10I	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
22.	I2I	I11I	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
23.	I2I	I12I	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
24.	I2I	I13I	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
25.	I2I	I14I	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
26.	I2I	I15I	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
27.	I3I	I01I	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
28.	I3I	I02I	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
29.	I3I	I03I	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
30.	I3I	I04I	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
31.	I3I	I05I	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
32.	I3I	I06I	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
33.	I3I	I07I	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
34.	I3I	I08I	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
35.	I3I	I09I	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
36.	I3I	I10I	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
37.	I3I	I11I	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
38.	I3I	I12I	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
39.	I3I	I13I	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ
40.	I3I	I14I	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpwr	Hold	STIM	LLMUX	60HZ	50HZ

HPM NODE SPECIFIC Configuration

IO Module Configuration Table

Engineer:

Date:

IO Module Configuration Display – N/A if NTWKNUM = 0

For each IO Module, select the IO Module type (IOMTYPE) that applies. If there is a redundant IOM, check the Redun blank. If the type selected is AO or DO, select a failure option (FAILOPT). If the type is STIM OR LLMUX, select an ac Line Frequency (FREQ6050).

****CAUTION – Use this part of the form only if PKGOPT on page 2 = Redun_IO****

NOTE: IOMTYPE LLMUX also includes RHMUX.

MOD NUM	IOM-A FILE	IOM-A CARD	Redun ✓	IOM-B FILE	IOM-B CARD	IOMTYPE																FAILOPT		IOMTYPE		FREQ6050	
						REDUN																					
						↓																↓		↓		↓	
1.	<u>11</u>	<u>103</u>	--	<u>12</u>	<u>103</u>	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ				
2.	<u>11</u>	<u>104</u>	--	<u>12</u>	<u>104</u>	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ				
3.	<u>11</u>	<u>105</u>	--	<u>12</u>	<u>105</u>	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ				
4.	<u>11</u>	<u>106</u>	--	<u>12</u>	<u>106</u>	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ				
5.	<u>11</u>	<u>107</u>	--	<u>12</u>	<u>107</u>	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ				
6.	<u>11</u>	<u>108</u>	--	<u>12</u>	<u>108</u>	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ				
7.	<u>11</u>	<u>109</u>	--	<u>12</u>	<u>109</u>	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ				
8.	<u>11</u>	<u>110</u>	--	<u>12</u>	<u>110</u>	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ				
9.	<u>11</u>	<u>111</u>	--	<u>12</u>	<u>111</u>	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ				
10.	<u>11</u>	<u>112</u>	--	<u>12</u>	<u>112</u>	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ				
11.	<u>11</u>	<u>113</u>	--	<u>12</u>	<u>113</u>	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ				
12.	<u>11</u>	<u>114</u>	--	<u>12</u>	<u>114</u>	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ				
13.	<u>11</u>	<u>115</u>	--	<u>12</u>	<u>115</u>	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ				
14.	<u>13</u>	<u>101</u>	--	<u>14</u>	<u>101</u>	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ				
15.	<u>13</u>	<u>102</u>	--	<u>14</u>	<u>102</u>	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ				
16.	<u>13</u>	<u>103</u>	--	<u>14</u>	<u>103</u>	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ				
17.	<u>13</u>	<u>104</u>	--	<u>14</u>	<u>104</u>	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ				
18.	<u>13</u>	<u>105</u>	--	<u>14</u>	<u>105</u>	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ				
19.	<u>13</u>	<u>106</u>	--	<u>14</u>	<u>106</u>	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ				
20.	<u>13</u>	<u>107</u>	--	<u>14</u>	<u>107</u>	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ				

Continued on next sheet

HPM NODE SPECIFIC Configuration

IO Module Configuration Table

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Engineer:

Date:

IO Module Configuration Display – N/A if NTKWNUM = 0

For each IO Module, select the IO Module type (IOMTYPE) that applies. If there is a redundant IOM, check the Redun blank. If the type selected is AO or DO, select a failure option (FAILOPT). If the type is STIM OR LLMUX, select an ac Line Frequency (FREQ6050).

****CAUTION – Use this part of the form only if PKGOPT on page 2 = Redun_IO (Continued)****

NOTE: IOMTYPE LLMUX also includes RHMUX.

MOD NUM	IOM-A FILE	IOM-A CARD	Redun ✓	IOM-B FILE	IOM-B CARD	REDUN										IOMTYPE				FAILOPT		IOMTYPE		FREQ6050	
						↓														↓				↓	
						None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ		
21.	131	1081	--	141	1081	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ		
22.	131	1091	--	141	1091	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ		
23.	131	1101	--	141	1101	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ		
24.	131	1111	--	141	1111	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ		
25.	131	1121	--	141	1121	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ		
26.	131	1131	--	141	1131	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ		
27.	131	1141	--	141	1141	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ		
28.	131	1151	--	141	1151	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ		
29.	151	1011	--	161	1011	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ		
30.	151	1021	--	161	1021	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ		
31.	151	1031	--	161	1031	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ		
32.	151	1041	--	161	1041	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ		
33.	151	1051	--	161	1051	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ		
34.	151	1061	--	161	1061	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ		
35.	151	1071	--	161	1071	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ		
36.	151	1081	--	161	1081	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ		
37.	151	1091	--	161	1091	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ		
38.	151	1101	--	161	1101	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ		
39.	151	1111	--	161	1111	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ		
40.	151	1121	--	161	1121	None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ		

IO Module Configuration Table

	Engineer:	Date:
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IO Module Configuration Display – N/A if NTWKNUM = 0

If the type selected is AO or DO, select a failure option (FAILOPT). If the type is STIM or LLMUX, select an ac Line Frequency (FREQ6050).

****CAUTION – Use this part of the form only if PKGOPT on page 2 = Nodefault****

NOTE: IOMTYPE LLMUX also includes RHMUX.

MOD NUM	IOM-A FILE	IOM-A CARD	Redun ↓	IOM-B FILE	IOM-B CARD											IOMTYPE				FAILOPT						IOMTYPE		FREQ6050				
						REDUN																	↓		↓				↓		↓	
1.			—					None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ							
2.								None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ							
3.								None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ							
4.								None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ							
5.								None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ							
6.								None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ							
7.								None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ							
8.								None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ							
9.								None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ							
10.								None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ							
11.								None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ							
12.								None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ							
13.								None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ							
14.								None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ							
15.								None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ							
16.								None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ							
17.								None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ							
18.								None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ							
19.								None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ							
20.			—					None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ							
Continued on next sheet																																

HPM NODE SPECIFIC Configuration

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IO Module Configuration Table

	Engineer:	Date:
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IO Module Configuration Display – N/A if NTKWNUM = 0

For each IO Module, select the IO Module type (IOMTYPE) that applies. If there is a redundant IOM, check the Redun blank. Fill in the file and card location(s). If the type selected is AO or DO, select a failure option (FAILOPT). If the type is STIM OR LLMUX, select an ac Line Frequency (FREQ6050).

****CAUTION – Use this part of the form only if PKGOPT on page 2 = Nodefalt (Continued)****

NOTE: IOMTYPE LLMUX also includes RHMUX.

MOD NUM	IOM-A FILE	IOM-A CARD	Redun ✓	IOM-B FILE	IOM-B CARD	IOMTYPE										FAILOPT				IOMTYPE		FREQ6050	
						None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ
21.			—			None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ
22.			—			None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ
23.			—			None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ
24.			—			None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ
25.			—			None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ
26.			—			None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ
27.			—			None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ
28.			—			None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ
29.			—			None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ
30.			—			None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ
31.			—			None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ
32.			—			None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ
33.			—			None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ
34.			—			None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ
35.			—			None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ
36.			—			None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ
37.			—			None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ
38.			—			None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ
39.			—			None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ
40.			—			None	HLAI	LLAI	DI	PI	SI	FB	DISOE	AO	AO16	DO	DO32	Unpower	Hold	STIM	LLMUX	60HZ	50HZ

PNTMODTY	HLAI	-> pg 2-A	High Level Analog Input
Point's IO Module Type	STIM	-> pg 3-B	Smart Transmitter Intf. Module
	PI	-> pg 5-C	Pulse Input Module
	LLMUX	-> pg 2-D	Low Level Multiplexer; also applies to
		->	RHMUX, except that RHMUX does not
	LLAI	pg 4-E	apply to SENSRTYP of RTD
			Low Level Analog Input

HPM Analog Input Point

Unit:	Tagname:	Engineer:	Date:
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A

HLAI Only

PV Configuration Display

SENSRTYP Sensor Type	1_5_V P4_2_V 0_5_V Slidwire	+1 to +5 volts .4 to 2 volts 0 to 5 volts Slidewire Resistance Device
PVCHAR PV Characterization	Linear SqrRoot	Linear Squareroot
Btherm Etherm Jtherm Ktherm Rtherm Rptherm Stherm Ttherm DinRtd JisRtd NickIRtd CopprRtd	Thermocouple Type: B, E, J, K, R, RP, S, or T Resistance Temperature Device Type: Din,Jis,Nickel, or Copper C = Celcius; F = Farenheit R = Rankin; K = Kelvin	
PVTEMP Thermal EUS For PV	DegreesC DegreesF DegreesR DegreesK	
SLWSRCID Slidwire Voltage Slot ID (number)	1_._1	range = 1 – 16 default = 1

D

LLMUX Only

PV Configuration Display (continued)

NOTE: PV configuration display for LLMUX also applies to RHMUX IOP, except that RTD is not supported by RHMUX IOP.

SENSRTYP Sensor Type	Thermcpl RTD 0_100_MV	Thermocouple Resistance Temp.Device 0 to 100 millivolts
PVCHAR PV Characterization	Linear	Linear
Btherm Etherm Jtherm Ktherm Rtherm Rptherm Stherm Ttherm DinRtd JisRtd NickIRtd CopprRtd	Thermocouple Type: B, E, J, K, R, RP, S, or T Resistance Temperature Device Type: Din, Jis, Nickel, or Copper	
PIUOTDCF Open T/C Detect Enable Option	Off On	Open Thermocouple Detection (LLMUX-Thermcpl only)
PVTEMP Thermal EUS For PV	DegreesC DegreesR DegreesF DegreesK	C = Celcius; F = Farenheit R = Rankin; K = Kelvin
TCRNGOPT Thermocouple Range Option	Normal Extended	Use PVEXEULO's normal range table Use PVEXEULO's extended range table

HPM Analog Input Point

Unit:	Tagname:	Engineer:	Date:
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PV Configuration Display

STITAG Transmitter Tag ID	_ . _ . _ . _ . _ .	Device name stored in the Smart Xmtr. <i>default</i> = Blank
SENSRTYP Smart Transmitter Sensor Type	<pre> graph LR SPT_DP --- SPT_GP SPT_GP --- SPT_AP SPT_AP --- SFM SFM --- STT STT --- Linear STT --- SqrRoot </pre>	Differential Pressure Xmtr. Gauge Pressure Xmtr. Absolute Pressure Xmtr. Magnetic Flow Xmtr. Temperature Xmtr. Linear Squareroot
PVCHAR PV Characterization Option		

Btherm	B type thermocouple	NickIRtd	Nickel type Res. Temp. Device
Etherm	E type thermocouple	PT200Rtd	Point 200 Resistance Temp. Dev
Jtherm	J type thermocouple	PT500Rtd	Point 500 Resistance Temp. Dev.
Ktherm	D type thermocouple	CU10Rtd	Copper 10 Resistance Temp. Dev.
Rtherm	R type thermocouple	CU25Rtd	Copper 25 Resistance Temp. Dev.
Stherm	S type thermocouple	RTD_OHMS	Ohms Resistance Temp. Dev.
Ttherm	T type thermocouple	RHRAD	RH Radiomatic
Ntherm	N type thermocouple	W5W26TC	C type thermocouple
DinRtd	Din type Res. Temp. Device	W3W25TC	D type thermocouple
JisRtd	Jis type Res. Temp. Device	NINIMOTC	Nickle/nickle/Mol. thermocouple

STI_EU STI Engineering Units		Applies if – <u>SENSRTYP = SFM</u>			<u>SENSRTYP = STT AND</u>
<u>SENSRTYP = SPT DP/GP/AP</u>		CM_HR	BRL_DAY	CLUB_M	<u>PV CHAR =</u>
INH20	MBAR BAR	GAL_HR	CM_SEC	LITRES	Linear ——— { MV V
MMHG	G_SQCM	LIT_HR	PRCNT	MLITRES	
PSI	KG_SQCM	CC_HR	BLANK	FL_OZ	RTD_OHMS- OHMS
KPA	MMH20	CM_MIN	LBS	FEET	
MPA	INHG	GAL_MIN	KG	METERS	All Other ——— { DEG_C DEG_F DEG_K DEG_R
		LIT_MIN	TONS	MM	
		CC_MIN	GRAMS	INCHES	
		CM_DAY	OZ	KG_CUM	
		GAL_DAY	GAL	G_CUM	
		KGAL_DAY	BRL	LBS_CUFT	
				LBS_CUIN	

DECONF	PV PV_SV	Broadcast PV 3.6 time/sec.
DE Configuration Mode	PV_DB	Broadcast PV 3.2 times/sec., SV .46 X/Sec.
	PV_SV_DB	Broadcast PV 2.7 times/sec. Plus data base
		B/cast PV 2.4 X/Sec., SV .34 X/Sec+ d/base
URL		<i>range</i> = See Parameter Ref.Guide
Upper Range Limit	_._._._._._.__	<i>default</i> = NaN
URV		<i>range</i> = See Parameter Ref.Guide
Upper Range Value	_._._._._._.__	<i>default</i> = NaN
LRV		<i>range</i> = See Parameter Ref.Guide
Lower Range Value	_._._._._._.__	<i>default</i> = NaN
DAMPING		SPT=0-32, STT=0-102.3,SFM=0-100
Xmtr Filter Lag Time	_._._._._._.__	(discrete increments); <i>default</i> = 0.0

HPM Analog Input Point

Unit:	Tagname:	Engineer:	Date:
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Applies if SENSRTYP=STT

PIUOTDCF	Off	Thermocouple fault detection
Open TC Detection Enable Option	On	<i>range</i> = Off or On.
CJTA CT	On	On = Internal cold Jct.compensation active
Int. Cold Jct. Compensation	Off	Off = External cold Jct.compensation active

E

PV Configuration Display (continued) – Applies To All Point Types Except As Noted

LLAI Only

PV Configuration Display

SENSRTYP Sensor Type	1_5_V 0_5_V RTD Thermcpl 0_100_MV	+1 to +5 volts .0 to 5 volts Resistance Temp. Device Thermocouple 0 to 100 millivolts
PVCHAR PV Characterization	Linear SqrRoot	Linear Squareroot
	Btherm Etherm Jtherm Ktherm Rtherm Rptherm Stherm Ttherm	Thermocouple Type: B, E, J, K, R, RP, S, or T
	{ DinRtd JisRtd NicklRtd CopprRtd	Resistance Temperature Device Type: Din, Jis, Nickel, or Copper
PVTEMP Thermal EUS For PV	DegreesC DegreesR DegreesF DegreesK	C = Celcius; F = Farenheit R = Rankin; K = Kelvin
TCRNGOPT Thermocouple Range Option	Normal Extended	Use PVEXEULO's normal range table Use PVEXEULO's extended range table

Applies if SENSRTYP = Slidewire or 0-100mV

PVRAWHI PV Raw High Limit		<i>range</i> = <u>0 to 100 mv</u> <i>default</i> = 0 – 100 <i>units</i> = millivolts	<u>Slidewire</u> 0.0 - 1 1.0 Ratio
PVRAWLO PV Raw Low Limit		<i>range</i> = <u>0 to 100 mv</u> <i>default</i> = 0 – 100 <i>units</i> = millivolts	<u>Slidewire</u> 0.0 - 1 0 Ratio

Applies if PVCHAR = LINEAR or SQRROOT

INPTDIR Analog Input Direction	Direct Reverse	PVCALC increases as PVRAW increases PVCALC decreases as PVRAW increases
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HPM Analog Input Point

Unit:	Tagname:	Engineer:	Date:
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PVEUHI PV High Range In EUs	_ _ _ _ _ _ _	<i>range</i> = PVEULO to PVEXEUHI <i>default</i> = 100.0		
PVEULO PV Low Range In EUs	_ _ _ _ _ _ _	<i>range</i> = PVEXEULO to PVEUHI <i>default</i> = 0.0		
PVFORMAT PV Decimal Point Format	DO D1	D2 D3	D0 = 9999. D1 = 999.9	D2 = 99.99 D3 = 9.999

N/A if SENSRTYP = Thermcpl or RTD.

PVEXEUHI PV Extended Hi Range in EUs	_ _ _ _ _ _ _	<i>range</i> = >=PVEUHI <i>default</i> = 102.9
PVEXEULO PV Extended Lo Range in EUs	_ _ _ _ _ _ _	<i>range</i> = <=PVEULO <i>default</i> = -2.9

PV Configuration Display (continued) – Applies To All Point Types Except As Noted

PVCLAMP PV Clamping Option	NoClamp Clamp	No clamping of the PV value Clamp PV value at range extension limit
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N/A if SENSRTYP = THERMCPL or RTD

LOCUTOFF Low Signal CO For Flow Inputs	_ _ _ _ _ _ _	<i>range</i> = PVEULO – PVEUHI and NaN <i>default</i> = NaN
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C PI Only

C1 Scaling Constant 1	_ _ _ _ _ _ _	<i>range</i> = Real Number ≥ 0.0 <i>default</i> = 1.0
C2 Scaling Constant 2	_ _ _ _ _ _ _	<i>range</i> = Real Number ≥ 0.0 <i>default</i> = 1.0
TIMEBASE Input Time Base	Seconds Minutes Hours	PV & Setpoint Eng. Units per Second, Minute, or Hour

N/A if PNTFORM = Componnt.

PVSRCOPT PV Source Option	Only Auto	PV Source selection is not available and Field wiring or memory fetch supplies PV. Selection from PVSOURCE is available & Operator/CL-program may supply PV.
PVSOURCE PV Source	All Auto Man Sub	Field wiring/memory fetch supplies PV PV is supplied by Operator or program A value is substituted by a CL program

TF PV Filter Lag Time in Minutes	_ _ _ _	<i>range</i> = 0.0 to 60.0 (minutes) <i>default</i> = 0.0 (PV will not be delayed)
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PVTV PV Target Value in EUs	_ _ _ _ _ _ _	N/A if PNTFORM = Compnnt. <i>range</i> = PVEXEULO-PVEXEUHI, & NaN <i>default</i> = NaN
OVERVAL Overview Display Value	_ _ _ _	<i>range</i> = 0 to 100 <i>default</i> = 25

HPM Analog Input Point

Unit:	Tagname:	Engineer:	Date:
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Alarming Display – N/A if PNTFORM = Componnt

As you enter alarm trip points and alarm priorities in the next section please refer to the following alarm Priority definitions and the note.

NoAction	Alarm is not reported to the system and not annunciated (R670 & later)
Journal	Alarm is historized, not reported to US, not annunciated
Low	Alarm is historized, reported to US, and annunciated
High	Alarm is historized, reported to US, and annunciated
Emergency	Alarm is historized, reported to US, and annunciated
Printer	Not historized, reported to the printer, not annunciated
JnlPrint	Alarm historized, reported to the printer, not annunciated

Note: If you do not enter a trip point value for a given pair of trip-points and alarm-priorities, do not Circle an alarm priority option for that pair (i.e., no trip point, no alarm priority).

Alarming Display – N/A if PNTFORM = Componnt.

\$AUXUNIT Auxiliary Unit	_ _	<i>range</i> = Any configured LCN Unit ID <i>default</i> = Null (--)
\$ALDLOPT (R620 and later) Alarm Delay Option	None OffDlyLo OffDlyHi OffDlyEm OnDlyLo OnDlyHi OnDlyEm	Off Delay Lo Off Delay Hi Off Delay Emergency On Delay Lo On Delay Hi On Delay Emergency
\$DLBADAL (R681 and later) Delay for BADPV	None Offdelay	<i>range</i> = Offdelay, None <i>default</i> = None
\$DLTIME (R620 and later) Delay Time	_ _ _ _	<i>range</i> = 0 to 9999 seconds <i>default</i> = 0 seconds
\$REALARM (R660 and later) Re-alarming Interval in minutes	_ _ _ _	<i>range</i> = 0 to 9999 minutes <i>default</i> = 0 minutes
PVALDB PV Alarm Deadband In %	Half One Two Three	Four Five EU
PVALDBEU PV Alarm Deadband in EU	_ _ _ _ _ _ _	<i>range</i> >= 0.0 <i>default</i> = 1.0
PVHITP PV High Alarm Trip Point	_ _ _ _ _ _ _	<i>range</i> = PVLOTP - PVHHTP, & NaN <i>default</i> = NaN
PVHIPR PV High Alarm Priority	NoAction (R670 and later) Journal Low	High Emergency Printer JnlPrint
\$HISELCT (R681 and later) PV High Selective Cutout	Cutout Allow	<i>range</i> = Cutout, Allow <i>default</i> = Cutout

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Unit:	Tagname:	Engineer:	Date:
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PVLOTP PV Low Alarm Trip Point	_____								<i>range</i> = PVLLTP - PVHITP, & NaN <i>default</i> = NaN
PVLOPR PV Low Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint		
\$LOSELCT (R681 and later) PV Low Selective Cutout	Cutout Allow								<i>range</i> = Cutout, Allow <i>default</i> = Cutout
PVHHTP PV Hi Hi Alarm Trip Point	_____								<i>range</i> = PVHITP to PVEXEUHI, & NaN <i>default</i> = NaN
PVHHPR PV High High Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint		
\$HHSELCT (R681 and later) PV High High Selective Cutout	Cutout Allow								<i>range</i> = Cutout, Allow <i>default</i> = Cutout
PVLLTP PV Low Low Alarm Trip Point	_____								<i>range</i> = PVEXEULO – PVLOTP, & NaN <i>default</i> = NaN
PVLLPR PV Low Low Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint		
\$LLSELCT (R681 and later) PV Low Low Selective Cutout	Cutout Allow								<i>range</i> = Cutout, Allow <i>default</i> = Cutout
PVROCPTP PV Pos Rate Of Chg Trip Point	_____								<i>range</i> = >=0.0, andNaN <i>default</i> = NaN
PVROCPPR PV Pos Rate Of Chg Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint		

\$OLDEVDS Open-Line Detect Event Disable	NOOTDETC	Value must be selected. Point operation is not affected by this function.
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HPM Analog Output Point

Unit:	Tagname:	Engineer:	Date:
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Point Assignment Display

NAME Tag Name		<i>range</i> = Not applicable <i>default</i> = Blank
NODETYP Node Type	HPM PM, APM LM or SM	An HPM Point Is Being Configured If a PM, APM, LM or SM is being configured, go to the appropriate forms for those devices.
PNTFORM Point Form	FuLL Componnt	Point is fully displayed and alarmed Pt. is partially displayed but not alarmed

PTDESC Point Descriptor		<i>range</i> = Not applicable <i>default</i> = Blank
EUDESC Engineering Units Descriptor		<i>range</i> = Not applicable <i>default</i> = Blank
KEYWORD Keyword Descriptor		<i>range</i> = Not applicable <i>default</i> = Blank
ASSOCDSP Associated Display		<i>range</i> = Not applicable <i>default</i> = Blank
\$CDETAIL (R620 and later) Point Custom Display		<i>range</i> = Not applicable <i>default</i> = Blank
UNIT Process Unit Point Is Assigned To		<i>range</i> =Two Alphanumeric Characters <i>default</i> = Not applicable
NTWKNUM NIM's UCN That This Point Is In		<i>range</i> = 1 to 20 <i>default</i> = Not applicable
NODENUM HPMM's Address On The UCN		<i>range</i> = 1 to 64 <i>default</i> = Not applicable
MODNUM I/O Module Number		<i>range</i> = 1 to 40 <i>default</i> =Not applicable
SLOTNUM Slot Number		<i>range</i> = 1 – 8 or 1 – 16 <i>default</i> =Not applicable

N/A if PNTFORM = Componnt.

PRIMMOD Primary Module ID		<i>range</i> = Not Applicable <i>default</i> = Null
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HPM Analog Output Point

Unit:	Tagname:	Engineer:	Date:
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Point Assignment Display (continued)

PNTMODTY	AO	Standard AO IOP
Point Module Type	AO_16	High Density AO IOP

Operating Configuration Display

OPTDIR	Direct	For final OP: 0%=>4ma; 100% =>20ma
AO Direct/Reverse Action	Reverse	For final OP: 0% =>20ma; 100% =>4ma
OPCHAR	On Off	I/O coordinates define OP char. curve
Output Characterization Option	Enable	I/O coordinates do not apply
\$OLDEVDS		Allows OUTPUTFL soft failure event when open line detected in output.
Open-Line Detect Event Disable	Disable	
\$OPTOL		Disables OUTPUTFL soft failure event when open line detected in output.
Output Tolerance in %	_ _ _ _ _ _ _	range = 0.0 to 10.6.9 default = 0.0
OUTIND (R620 and later)	Direct	4 - 20mA = 0 – 100%
Reverse Output Indicator	Reverse	20 – 4mA = 0 – 100%

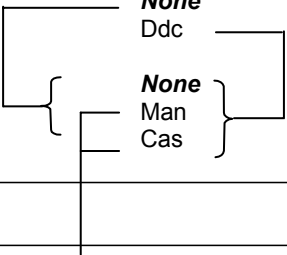
OPIN0	-6.9	range = Fixed at -6.9 percent default = Not applicable
Input Coordinate # 0		
OPIN1	_ _ _ _ _ _ _	range = >prev. coord, < next coord default = Not applicable
Input Coordinate # 1		
OPIN2	_ _ _ _ _ _ _	range = >prev. coord, < next coord default = Not applicable
Input Coordinate # 2		
OPIN3	_ _ _ _ _ _ _	range = >prev. coord, < next coord default = Not applicable
Input Coordinate # 3		
OPIN4	_ _ _ _ _ _ _	range = >prev. coord, < next coord default = Not applicable
Input Coordinate # 4		
OPIN5	106.9	range = Fixed at 106.9 percent default = Not applicable
Input Coordinate # 5		
OPOUT0	-6.9	range = Fixed at -6.9 percent default = Not applicable
Output Coordinate # 0		
OPOUT1	_ _ _ _ _ _ _	range = >prev. coord, < next coord default = Not applicable
Output Coordinate # 1		
OPOUT2	_ _ _ _ _ _ _	range = >prev.coord, < next coord default = Not applicable
Output Coordinate # 2		
OPOUT3	_ _ _ _ _ _ _	range = >prev.coord, < next coord default = Not applicable
Output Coordinate # 3		
OPOUT4	_ _ _ _ _ _ _	range = >prev.coord, < next coord default = Not applicable
Output Coordinate # 4		
OPOUT5	106.9	range = Fixed at 106.9 percent default = Not applicable
Output Coordinate # 5		

HPM Analog Output Point

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Unit:	Tagname:	Engineer:	Date:
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Mode Configuration Display – N/A if PNTFORM = Componnt.

RCASOPT Remote Cascade Option		None Ddc	No cascade mode of any type is allowed In Cas, AM point controls this point's OP
NMODE AnalOut Point's Normal Mode		None Man Cas	No configured "normal" operating mode Manual is the configured "normal" mode If RCASOPT equals Ddc, Cascade may be configured as the "normal" mode.
N/A if NMODE = None			
NMODATTR Normal Mode Attribute		None Operator Program	Parameter NMODE sets "normal" mode Operator sets "normal" operating mode Program sets "normal" operating mode
MODEPERM Mode Permissive		Permit NotPerm	Operator can change this point's mode Optr. cannot change this point's mode
\$MODESEL (R630 and later) Mode Selection List		Man Cas Prog	Man and Cas mode, and Prog mode attribute are allowed for this data point.

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[illegible]

HPM Auto Manual CTL Algorithm

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CTLALGID ControlAlgorithm Identifier	Null AutoMan	No affect Auto Manual
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PV Configuration Display – N/A if PNTFORM = Component

OVERVAL Overview Display Value	_ _ _ _	<i>range = 0 to 100</i> <i>default = 25</i>
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Mode Configuration Display

BADCTLOP Bad PV Mode Shed Option	No_Shed Shed Hold ShedLow Shed High Shed Safe	
RCASOPT Remote Cascade Option	None Ddc	Local cascade is the only valid cascade AM strategy stores to OP. No OP limits.
SHEDTIME Remote Cascade Shed Time	_ _ _ _	<i>range = 0 to 1000 (seconds)</i> <i>default = 0(mode shed disabled)</i>
SHEDMODE Shed Mode	Man } BCas }	This point will shed to Manual, This point will shed to BackupCas
NMODE RegCtl Point's Normal Mode	None } Man } Cas } Bcas }	No configured "normal" operating mode Manual is the "normal" operating mode Cascade is "normal" operating mode Backup Cas is "normal" operating mode
NMODATTR (N/A if NMODE = None) Normal Mode Attribute	None Operator Program	Parameter NMODE sets "normal" mode Operator sets "normal" operating mode Program sets "normal" operating mode
MODEPERM Mode Permissive	Permit NotPerm	Operator can change this point's mode Optr. cannot change this point's mode
\$MODESEL (R630 and later) Mode Selection List	Man Prog Cas Bcas	Man, Cas and Bcas mode, and Prog mode attribute are allowed for this data point.
\$BADOCMD (R680 and later) Bad Output Connection Mode	SHEDMAN 15_SEC 60_SEC NO_SHED	Point will shed to Manual immediately Point will shed to Manual after 15 seconds Point will shed to Manual after 60 seconds Do not shed mode
EXTSWOPT External Mode Switching	None Ems	No external mode switching is allowed Ext. source can change point's mode

Reg Control Auto/Man Algo Display

CTLEQN Control Equation Type	EqA EqB	$CV = X1 + B + BI$ $CV = X1 + (K \cdot X2) + BI$
B Output Bias	_ _ _ _	<i>range = Any Real number</i> <i>default = 0.0</i>
K Gain Constant For X2 Input	_ _ _ _	<i>range = Not applicable</i> <i>default = 1.0</i>
RATE1 Rate At Which Bias Ramps Down	_ _ _ _	<i>range = >=0.0, NaN</i> <i>default = 0.0</i>
XEUHI X Input High Range in EUs	_ _ _ _	<i>range = >=XEULO</i> <i>default = 100.0</i>

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<i>Output Configuration Display</i>		
NOCOPTS		
# Of Control Output Connections	_	range = 0 to 4 default = 1
CODSTN (Control-Output-Connection Destination)		
<u>Output #</u>		<u>CODSTN (point.parameter)</u>
1	_____	
2	_____	
3	_____	
4	_____	
OPHILM	_____	range = OPLOLM - 106.9 & NaN
Output High Limit In Percent		default = 105.0
OPLOLM	_____	range = -6.9 - OPHILM, & NaN
Output Low Limit In Percent		default = -5.0
SAFEOP	_____	range = -6.9 to to 106.9 & NaN
Safe Output Limit In Percent		default = NaN
OPMCHLM	_____	range = >=0.0, and NaN
Output Minimum Change In %		default = 0.0
OPROCLM	_____	range = >=0.0, and NaN
OP Rate Of Change Limit In %		default = NaN
\$OPTOL	_____	range = 0.0 to 106.9
Output Tolerance in %		default = 0.0
OUTIND (R620 and later)	Direct	4 - 20mA = 0 – 100%
Reverse Output Indicator	Reverse	20 - 4mA = 0 – 100%
		Note: With R670 and later, when this parameter is set to Reverse, the values of the displayed output and the related output parameters are changed. Refer to the definition of the OUTIND parameter in the PM Family Parameter Dictionary, XPM09.

HPM Auto Manual CTL Algorithm

Unit:	Tagname:	Engineer:	Date:
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Alarming Display - N/A if PNTFORM = Componnt

As you enter alarm priorities in the next section please refer to the following

NoAction	Alarm is not reported to the system and not annunciated (R670 & later)
Journal	Alarm is historized, not reported to US, not annunciated
Low	Alarm is historized, reported to US, and annunciated
High	Alarm is historized, reported to US, and annunciated
Emergency	Alarm is historized, reported to US, and annunciated Not
Printer	historized, reported to the printer, not annunciated Alarm
JnlPrint	historized, reported to the printer, not annunciated

Alarming Display - N/A if PNTFORM = Componnt

\$AUXUNIT	_____	range = Any configured LCN Unit ID default = Null (- -)
Auxiliary Unit		
\$DLBADAL (R681 and later)	None	range = Offdelay, None
Delay for BADCTL	Offdelay	default = None
DLTIME (R620 and later)	_____	range = 0 to 9999 seconds default = 0 seconds
Delay Time		
\$REALARM (R660 and later)	_____	range = 0 to 9999 minutes default = 0 minutes
Re-alarming Interval		
BADOCOPT	Off	Bad Output Connection Alarm is not initially permitted
BADOC Alarm Option	On	Bad Output Connection Alarm is Initially permitted
BADOCPR	NoAction	Journal Low High Emergency Printer JnlPrint
BADOC Alarm Priority	(R670 and later)	

OPALDB	_____	range = 0 - 25.0 but ≤ (OPHITP- OPLOTP)/2 default = 5.0 Output Units
Output Alarm Deadband		
OPHITP	_____	range = OPLOTP TO OPHILM and NaN default = NaN
Output High Alarm Trip Point		
OPHIPR	NoAction	Journal Low High Emergency JnlPrint Printer
Output High Alarm Priority	(R670 and later)	
OPLOTP	_____	range = OPLOLM TO OPHITP and NaN default = NaN
Output Low Alarm Trip Point		
OPLOPR	NoAction	Journal Low High Emergency JnlPrint Printer
Output Low Alarm Priority	(R670 and later)	
BADCTLPR	NoAction	Journal Low High Emergency JnlPrint Printer
Bad-Control Alarm Priority	(R670 and later)	
ALENBST		<u>Displayed</u> <u>Logged</u> <u>Reported To EIP</u>
Point Alarm Enable Status	Enable	Yes Yes Yes
	Disable	No Yes Yes
	Inhibit	No No No

Unit:	Tagname:	Engineer:	Date:
<i>Point Assignment Display</i>			
NAME Tag Name		<i>range</i> = Not applicable <i>default</i> = Blank	
NODETYP Node Type		PM APM HPM PNTFORM Point Form	
		A PM Point Is Being Configured An APM Point Is Being Configured An HPM Point Is Being Configured Point is fully displayed and alarmed Pt. is partially displayed but not alarmed	
PTDESC Point Descriptor		<i>range</i> = Not applicable <i>default</i> = Blank	
EUDESC Engineering Units Descriptor		<i>range</i> = Not applicable <i>default</i> = Blank	
KEYWORD Keyword Descriptor		<i>range</i> = Not applicable <i>default</i> = Blank	
ASSOCDSP Associated Display		<i>range</i> = Not applicable <i>default</i> = Blank	
\$CDETAIL (R620 and later) Point Custom Display		<i>range</i> = Not applicable <i>default</i> = Blank	
UNIT Process Unit Point Is Assigned To		<i>range</i> =Two Alphanumeric Characters <i>default</i> = Not applicable	
NTWKNUM NIM's UCN That This Point Is In		<i>range</i> = 1 to 20 <i>default</i> = Not applicable	
NODENUM HPMM's Address On The UCN		<i>range</i> = 1 to 64 <i>default</i> = Not applicable	
MODNUM I/O Module Number		<i>range</i> = 0 <i>default</i> = 0 (Control Processor Point)	
SLOTNUM Slot Number		<i>range</i> = 1–125 (But <=NPVSLOT) <i>default</i> =Not applicable	
N/A if PNTFORM = Componnt.			
PRIMMOD Primary Module ID		<i>range</i> =Not Applicable <i>default</i> = Null	
USERID User ID Reservation		Sixteen Character String <i>default</i> = dashes	
PVALGID PV Algorithm Identifier		Null No affect Calculator	

HPM Calculator PV Algorithm

Unit:	Tagname:	Engineer:	Date:
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PV Configuration Display

PVEUHI PV High Range In Engr Units	_ _ _ _ _ _ _	<i>range</i> =PVEULO to PVEXEUHI <i>default</i> = 100.0
PVEULO PV Low Range In Engr Units	_ _ _ _ _ _ _	<i>range</i> = PVEXEULO to PVEUHI <i>default</i> = 0.0
PVFORMAT PV Decimal Point Format	DO D2 D1 D3	D0=9999. D2=99.99 D1=999.9 D3 = 9.999
PVEXEUHI PV Extended Hi Range in EUs	_ _ _ _ _ _ _	<i>range</i> = >=PVEUHI <i>default</i> = 102.9
PVEXEULO PV Extended Lo Range in EUs	_ _ _ _ _ _ _	<i>range</i> = <=PVEULO <i>default</i> = -2.9
PVCLAMP PV Clamping Option	NoClamp Clamp	No clamping of the PV value Clamp PV value at range extension limit

N/A if PNTFORM = Componnt.

PVSRCOPT PV Source Option	Only Auto All	PV Source selection is not available Selection from PVSOURCE is available
PVSOURCE PV Source	Auto Man Sub	Field wiring/memory fetch supplies PV PV is supplied by Operator or program A value is substituted by a CL program

PV Configuration Display

TF PV Filter Lag Time in Minutes	_ _ _ _ _ _ _	<i>range</i> = 0.0 to 60.0 (minutes) <i>default</i> = 0.0 (PV will not be delayed)
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PVTV PV Target Value in EUs	_ _ _ _ _ _ _	<i>range</i> = PVEXEULO-PVEXEUHI, & NaN <i>default</i> = NaN
OVERVAL Overview Display Value	_ _ _ _ _ _ _	<i>range</i> = 0 to 100 <i>default</i> = 25

Reg PV Calculator Algo Display

N # Inputs To Calcultr Algo	_	<i>range</i> = 1 to 6 <i>default</i> = 2
C1 Calculator Constant # 1	_ _ _ _ _ _ _	<i>range</i> = N/A <i>default</i> = 0.0
C2 Calculator Constant # 2	_ _ _ _ _ _ _	<i>range</i> = N/A <i>default</i> = 0.0
C3 Calculator Constant # 3	_ _ _ _ _ _ _	<i>range</i> = N/A <i>default</i> = 0.0
C4 Calculator Constant # 4	_ _ _ _ _ _ _	<i>range</i> = N/A <i>default</i> = 0.0

HPM Calculator PV Algorithm

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Unit:	Tagname:	Engineer:	Date:
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Reg PV Calculator Algo Display (Continued)

CALCEXP		<i>range = Up to 40 Characters</i>
Calculator Expression		<i>default = "blank"</i>
.....		
PISRC (PV-Input-Connection Source)	PISRC (point.parameter)	Destination Para
Input#		-meter (PIDSTN)
1		P1
2		P2
3		P3
4		P4
5		P5
6		P6

Alarming Display - N/A if PNTFORM = Componnt

As you enter alarm trip points and alarm priorities in the next section please refer to the following alarm priority definitions and the note.

NoAction Alarm is not reported to the system and not annunciated (R670 & later)
Journal Alarm is historized, not reported to US, not annunciated
Low Alarm is historized, reported to US, and annunciated
High Alarm is historized, reported to US, and annunciated
Emergency Alarm is historized, reported to US, and annunciated Not
Printer historized, reported to the printer, not annunciated Alarm
JnlPrint historized, reported to the printer, not annunciated

Note: If you do not enter a trip point value for a given pair of trip-points and alarm-priorities, do not circle an alarm priority option for that pair (i.e., no trip point, no alarm priority).

Alarming Display - N/A if PNTFORM = Componnt

\$AUXUNIT	..	<i>range = Any configured LCN Unit ID</i>
Auxiliary Unit		<i>default = Null (- -)</i>
\$ALDLOPT (R620 and later)	None	
Alarm Delay Option	OffDlyLo	Off Delay Lo
	OffDlyHi	Off Delay Hi
	OffDlyEm	Off Delay Emergency
	OnDlyLo	On Delay Lo
	OnDlyHi	On Delay Hi
	OnDlyEm	On Delay Emergency
\$DLBADAL (R681 and later)	None	<i>range = Offdelay, None</i>
Delay for BADPV	Offdelay	<i>default = None</i>
DLTIME (R620 and later)		<i>range = 0 to 9999 seconds</i>
Delay Time		<i>default = 0 seconds</i>
\$REALARM (R660 and later)		<i>range = 0 to 9999 minutes</i>
Re-alarming Interval		<i>default = 0 minutes</i>

HPM Calculator PV Algorithm

Unit:	Tagname:	Engineer:	Date:
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Alarming Display (Continued)

	Half	One	Two	Three	Four	Five	EU	
PVALDB PV Alarm Deadband In %								
PVALDBEU PV Alarm Deadband In EU								range > = 0.0 default = 1.0
PVHITP PV High Alarm Trip Point								range = PVLOTP - PVHHTP, & NaN default = NaN
PVHIPR PV High Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
\$HISELCT (R681 and later) PV High Selective Cutout	Cutout Allow							range = Cutout, Allow default = Cutout
PVLOTP PV Low Alarm Trip Point								range = PVLLTP - PVHITP, & NaN default = NaN
PVLOPR PV Low Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
\$LOSELCT (R681 and later) PV Low Selective Cutout	Cutout Allow							range = Cutout, Allow default = Cutout
PVHHTP PV Hi Hi Alarm Trip Point								range = PVHITP to PVEUHI, & NaN default = NaN
PVHHPR PV High High Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
\$HHSELCT (R681 and later) PV High High Selective Cutout	Cutout Allow							range = Cutout, Allow default = Cutout
PVLLTP PV Low Low Alarm Trip Point								range = PVEULO - PVLOTP, & NaN default = NaN
PVLLPR PV Low Low Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
\$LLSELCT (R681 and later) PV Low Low Selective Cutout	Cutout Allow							range = Cutout, Allow default = Cutout
PVSGCHTP PV Significant Change Alarm Trip Point								range = >=0.0, and NaN default = NaN

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Unit:	Tagname:	Engineer:	Date:
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PVROCTP PV Pos Rate Of Change Trip Point	_ _ _ _ _ _ _	<i>range = >=0.0, and NaN default = NaN</i>						
PVROCPPR PV Pos Rate Of Chg Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
\$RPSLCT (R681 and later) PV Positive Rate of Change Selective Cutout	Cutout Allow	<i>range = Cutout, Allow default = Cutout</i>						
PVROCNTP PV Neg Rate Of Change Trip Point	_ _ _ _ _ _ _	<i>range = >=0.0, and NaN default = NaN</i>						
PVROCNPR PV Neg Rate Of Chg Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
\$RNSLCT (R681 and later) PV Negative Rate of Change Selective Cutout	Cutout Allow	<i>range = Cutout, Allow default = Cutout</i>						
BADPVPR Bad-PV Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
ALENBST Point Alarm Enable Status		Enable	<u>Displayed</u>	<u>Logged</u>	<u>Reported To EIP</u>			
		Disable	Yes	Yes	Yes			
		Inhibit	No	Yes No	Yes			
			No		No			

HPM Data Acquisition PV Algorithm

Unit:	Tagname:	Engineer:	Date:
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Point Assignment Display

NAME Tag Name		_____	range = Not applicable default = Blank
NODETYPE Node Type	PM APM		A PM Point Is Being Configured An APM Point Is Being Configured An HPM Point Is Being Configured
	HPM		
PNTFORM Point Form	FULL Componnt		Point is fully displayed and alarmed Pt. is partially displayed but not alarmed

PTDESC Point Descriptor		_____	range = Not applicable default = Blank
EUDESC Engineering Units Descriptor		_____	range = Not applicable default = Blank
KEYWORD Keyword Descriptor		_____	range = Not applicable default = Blank
ASSOCDSP Associated Display		_____	range = Not applicable default = Blank
\$CDETAIL (R620 and later) Point Custom Display		_____	range = Not applicable default = Blank
UNIT Process Unit Point Is Assigned To	_____	_____	range = Two Alphanumeric Characters default = Not applicable
NTWKNUM NIM's UCN That This Point Is In	_____	_____	range = 1 to 20 default = Not applicable
NODENUM HPMM's Address On The UCN	_____	_____	range = 1 to 64 default = Not applicable
MODNUM I/O Module Number	_____	_____	range = 0 default = 0 (Control Processor Point)
SLOTNUM Slot Number	_____	_____	range = 1-125 (but <= NPVSLOT) default = Not applicable

N/A if PNTFORM = Componnt.

PRIMMOD Primary Module ID	_____	range = Not Applicable default = Null
USERID User ID Reservation	_____	Sixteen Character String default = dashes

PVALGID PVAlgorithm Identifier	NULL DataAcq	No affect Data Acquisition
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HPM Data Acquisition PV Algorithm

Unit:	Tagname:	Engineer:	Date:
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PV Configuration Display

PVEUHI PV High Range In Engr Units	<u> </u>	<i>range</i> = PVEULO to PVEXEUHI <i>default</i> = 100.0
PVEULO PV Low Range In Engr Units	<u> </u>	<i>range</i> = PVEXEULO to PVEUHI <i>default</i> = 0.0
PVFORMAT PV Decimal Point Format	DO D2 D1 D3	D0 = 9999. D2 = 99.99 D1 = 999.9 D3 = 9.999
PVEXEUHI PV Extended Hi Range in EUs	<u> </u>	<i>range</i> = >=PVEUHI <i>default</i> = 102.9
PVEXEULO PV Extended Lo Range in EUs	<u> </u>	<i>range</i> = <=PVEULO <i>default</i> = -2.9
PV CLAMP PV Clamping Option	NoClamp Clamp	No clamping of the PV value Clamp PV value at range extension limit

PV Configuration Display (continued)

PV Configuration Display (Continued)		
PVSRCOPT PV Source Option	Only Auto All	PV Source selection is not available Selection from PVSOURCE is available
PVSOURCE PV Source	Auto Man Sub	Field wiring/memory fetch supplies PV PV is supplied by Operator or program A value is substituted by a CL program

TF PV Filter Lag Time in Minutes	0.0	<i>range</i> = 0.0 to 60.0 (minutes) <i>default</i> = 0.0 (PV will not be delayed)
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PVTV PV Target Value in EUs	_._._._._._.__	<i>range</i> = PVEXEULO-PVEXEUHI, & NaN <i>default</i> = NaN
OVERVAL Overview Display Value	_._.__	<i>range</i> = 0 to 100 <i>default</i> = 25

Reg PV Data Acq Algo Display

PISRC (PV-Input-Connection Source) <u>Input#</u>	PISRC (point.parameter)	<u>Destination Parameter (PIDSTN)</u>
1	_____	P1

HPM Data Acquisition PV Algorithm

Unit:	Tagname:	Engineer:	Date:
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Alarming Display – N/A if PNTFORM = Component

PVHHTP PV Hi Hi Alarm Trip Point	_____							<i>range = PVHITP to PVEUHI, & NaN default = NaN</i>
PVHHPR PV High High Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
\$HHSELCT (R681 and later) PV High High Selective Cutout	Cutout Allow							<i>range = Cutout, Allow default = Cutout</i>
PVLLTP PV Low Low Alarm Trip Point	_____							<i>range = PVEULO - PVLOTP, & NaN default = NaN</i>
PVLLPR PV Low Low Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
\$LLSELCT (R681 and later) PV Low Low Selective Cutout	Cutout Allow							<i>range = Cutout, Allow default = Cutout</i>
PVSGCHTP PV Significant Change Alarm Trip Point	_____							<i>range = >=0.0, and NaN default = NaN</i>
PVROCTP PV Pos Rate Of Change Trip Point	_____							<i>range = >=0.0, and NaN default = NaN</i>
PVROCPPR PV Pos Rate Of Chg Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
\$RPSELCT (R681 and later) PV Positive Rate of Change Selective Cutout	Cutout Allow							<i>range = Cutout, Allow default = Cutout</i>
PVROCNTP PV Neg Rate Of Change Trip Point	_____							<i>range = >=0.0, and NaN default = NaN</i>
PVROCNPR PV Neg Rate Of Chg Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
\$RNSELCT (R681 and later) PV Negative Rate of Change Selective Cutout	Cutout Allow							<i>range = Cutout, Allow default = Cutout</i>
BADPVPR Bad-PV Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
ALENBST Point Alarm Enable Status	Enable Disable Inhibit			<u>Displayed</u> Yes No No	<u>Logged</u> Yes Yes No	<u>Reported To EIP</u> Yes Yes No		

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USER ID	_ _ _ _ _ _ _ _ _ _ _ _ _ _ _	<i>range</i> = Sixteen Character String <i>default</i> = Dashes
User ID Reservation		

HPM Digital Composite Point

Unit:	Tagname:	Engineer:	Date:
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Point Assignment Display

NOSTATES	2	<i>default</i> = Not applicable
Number Of Digital States	3	2 digital switch states can be configured
		3 digital switch states can be configured
NODINPTS	0	0 digin connections can be configured
# Of Digital Input Connections	1	1 digin connections can be configured
	2	2 digin connections can be configured
NODOPTS	0	0 digout connections can be configured
# Of Digital Output Connections	1	1 digout connections can be configured
	2	2 digout connections can be configured
	3	3 digout connections can be configured

State Configuration Display

N/A if PNTFORM = Comppont.

PVTXTOPT PV Text Option	Off On	<i>range</i> = Not applicable <i>default</i> = "Off"			
STATETXT () State - Descriptor Text	(1) _ . _ . _ . _ . _ .	<i>range</i> = Not applicable			
	(0) _ . _ . _ . _ . _ .	<i>default</i> = "On"			
	(2) _ . _ . _ . _ . _ .	<i>default</i> = "Off"			
BOXCLR (1) State 1 Box Color	Red	Green (Top box default)	White	Black	
	Cyan	Yellow	Blue	Magenta	
BOXCLR (0) State 0 Box Color	Red	Green	White	Black	
	Cyan	Yellow (Middle box default)	Blue	Magenta	
BOXCLR (2) State 2 Box Color	Red (Lower box default)	Green	White	Black	
	Cyan	Yellow	Blue	Magenta	
MOVPTXT Moving PV Text	_ . _ . _ . _ . _ .	<i>range</i> = Eight character string <i>default</i> = "MOVING"			
BADPVTXT Bad PV Text	_ . _ . _ . _ . _ .	<i>range</i> = Eight character string <i>default</i> = "BAD"			

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PV / Input Configuration Display (Continued) - N/A if NODINPTS = 0		
PVSRCOPT PV Source Option	Only Auto	PV Source selection is not available and Field wiring or memory fetch supplies PV.
	All	Selection from PVSOURCE is available & Operator/CL-program may supply PV.
PVSOURCE PV Source	Auto	Field wiring /memory -fetch supplies PV
	Man	PV is supplied by Operator or program
	Sub	A value is substituted by a CL program
	Track	PV tracks the output

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HPM Digital Composite Point

Unit:	Tagname:	Engineer:	Date:
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Output Configuration Display – N/A if NODOPTS = 0

		<u>NOSTATES</u>	<u>NODOPTS</u>
ST1_OP1	Off	2 - 3	1 - 3
State 1 - Output 1	On		
ST1_OP2	Off	2 - 3	2 - 3
State 1 - Output 2	On		
ST1_OP3	Off	2 - 3	3
State 1 - Output 3	On		
ST0_OP1	Off	1 - 3	1 - 3
State 0 - Output 1	On		
ST0_OP2	Off	1 - 3	2 - 3
State 0 - Output 2	On		
ST0_OP3	Off	1 - 3	3
State 0 - Output 3	On		
ST2_OP1	Off	3	1 - 3
State 2 - Output 1	On		
ST2_OP2	Off	3	2 - 3
State 2 - Output 2	On		
ST2_OP3	Off	3	3
State 2 - Output 3	On		
DODSTN (Digital Output Connection Destination)			
Output#	<u>DODSTN (point.parameter)</u>	<u>NODOPTS</u>	
1		1	
2		2	
3		3	
PULSEWTH <i>range = 0.0 to 60.0</i>			
Pulse Width For DigComp Point Outputs <i>default = 1.0</i>			

Applies if NOSTATES = 3, and MOMSTATE = None

STCHGOPT	None	<i>range = Not Applicable</i>
State Change Option	State 0	<i>default = None</i>
PAUSETIM		<i>range = 0 to 1000 seconds</i>
State 0 Pause Time		<i>default = 0</i>

SEALOPT	None	<i>range = Not Applicable</i>
Seal-in Circuit Option	Sealin	<i>default = None</i>

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Unit:	Tagname:	Engineer:	Date:
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Maintenance Display

MAINTOPT Maintenance Support	Off On	range = Not Applicable default = Off
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Maintenance Statistics

<u>Maximum Hours Allowed in Each State</u>		<u>Maximum Number of Transitions into State</u>	
STATE 1 MAXTIM1H	_ _ _ _ _ _ _	MAXTRAN1	_ _ _ _ _ _ _
STATE 0 MAXTIM0H	_ _ _ _ _ _ _	MAXTRAN0	_ _ _ _ _ _ _
STATE 2 MAXTIM2H	_ _ _ _ _ _ _	MAXTRAN2	_ _ _ _ _ _ _
range = N/A default = 0		range = 0 (No Limit) default = 0.0	

Mode Configuration Display - N/A if NODOPTS = 0

NMODATTR Normal Mode Attribute	None Operator Program	Parameter NMODE sets "normal" mode Operator sets "normal" operating mode Program sets "normal" operating mode
MODEPERM Mode Permissive	Permit NotPerm	Operator can change this point's mode Optr. cannot change this point's mode
\$MODESEL (R630 and later) Mode Selection List	Man Prog	Man mode and Prog mode attribute are allowed for this data point.
OROPT Interlock Bypass Option	Off On	Optr. cannot bypass this point's interlocks Operator can bypass this point's interlocks If BYPASS = ON
BYPASS Interlock Bypass State	Off On	Outputs not allowed to ignore interlocks Stores to OP & SO are allowed to ignore the interlocks

Alarming Display - N/A if PNTFORM = Component or if NODINPTS = 0

\$AUXUNIT Auxiliary Unit	_ _	range = Any configured LCN Unit ID default = Null (- -)
\$REALARM Re-alarming Interval (R660 and later)	_ _ _ _	range = 0 to 9999 minutes default = 0 minutes
PVNORMAL PV Value's Normal State Text	_ _ _ _ _ _ _	range = STATETXT(0) – STATETXT(2) default = None
FBTIME Feedback Time	_ _ _ _	range = 1 to 1000 default = 0

N/A if FBTIME = 0.

CMDFALTM Command Failure Timeout	_ _ _ _	range = 0 to 1000 seconds default = 0												
ALENBST Alarm Enable Status	Enable Disable Inhibit	<table> <tr> <th><u>Displayed</u></th><th><u>Logged</u></th><th><u>Reported to EIP</u></th></tr> <tr> <td>Yes</td><td>Yes</td><td>Yes</td></tr> <tr> <td>No</td><td>Yes No</td><td>Yes</td></tr> <tr> <td>No</td><td>No</td><td>No</td></tr> </table>	<u>Displayed</u>	<u>Logged</u>	<u>Reported to EIP</u>	Yes	Yes	Yes	No	Yes No	Yes	No	No	No
<u>Displayed</u>	<u>Logged</u>	<u>Reported to EIP</u>												
Yes	Yes	Yes												
No	Yes No	Yes												
No	No	No												

HPM Digital Composite Point

Unit:	Tagname:	Engineer:	Date:
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Alarming Display- N/A if PNTFORM = Componnt. or if NODINPTS = 0

SI0ALOPT Safety Override Alarm	None Auto_Rtn Cnfm_Rqd	No alarms are generated Alarms generated, return to normal when override clears Alarms generated, confirmed before returning to normal If override clears. SIOALPPR must $\neq$ NoAction or Journal.
SI0DESC SI0 Alarm Descriptor	_ _ _ _ _ _ _ _ _	<i>range</i> = 8 Alphanumeric Characters <i>default</i> = blank
OVRDALOP Override Alarm Option	None Auto_Rtn Cnfm_Rqd	No alarms are generated Alarms generated, return to normal when override clears Alarms generated, confirmed before returning to normal If override clears. OVRDALPR must $\neq$ NoAction or Journal.
I1DESC I1 Alarm Descriptor	_ _ _ _ _ _ _ _ _	<i>range</i> = 8 Alphanumeric Characters <i>default</i> = blank
I0DESC I0 Alarm Descriptor	_ _ _ _ _ _ _ _ _	<i>range</i> = 8 Alphanumeric Characters <i>default</i> = blank
I2DESC I2 Alarm Descriptor	_ _ _ _ _ _ _ _ _	<i>range</i> = 8 Alphanumeric Characters <i>default</i> = blank

BADPVPR Bad PV Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
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N/A if FBTIME = 0.

CMDDISPR Command Disagree Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
OFFNRMPR Offnormal Alarm Priority	NoAction (R670 and later) Journal Low High Emergency Printer JnlPrint	Alarm not reported to the system and not annunciated Alarm is historized, not reported to US, not annunciated Alarm is historized, reported to US, and annunciated Alarm is historized, reported to US, and annunciated Alarm is historized, reported to US, and annunciated Not historized, reported to the printer, not annunciated Alarm historized, reported to the printer, not annunciated					
\$OFSELCT (R681 and later) Off Normal Selective Cutout	Cutout Allow	<i>range</i> = Cutout, Allow <i>default</i> = Cutout					

HPM Digital Composite Point

Unit:	Tagname:	Engineer:	Date:
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N/A if SI0ALOPT = None.

SI0ALPR Override SI0 Alarm Priority	NoAction	Journal	Low	High	Emergency	Printer	JnlPrint
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N/A if OVRDALOP = None.

OVRDALPR Override Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
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EVTOPT Event Reporting Option	None Eip	No Reporting Digital PV changes invoke Event-Initiated Processing
EIPPCODE Event Init Proc Destination	<i>range</i> = Not applicable <i>default</i> = Null	

HPM Digital Input, Status Type Point

Unit:	Tagname:	Engineer:	Date:
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Point Assignment Display – Begin DI Status Type configuration

NAME Tag Name	_____	<i>range</i> = Not applicable <i>default</i> = Blank
NODETYPE Node Type	HPM PM, APM, LM or SM	An HPM Point Is Being Configured If a PM, APM, LM or SM is being configured, go to the appropriate forms for those devices.
PNTFORM Point Form	Full Componnt	Point is fully displayed and alarmed Pt. is partially displayed but not alarmed

[illegible]

PRIMMOD Primary Module ID		range = Not Applicable default = Null
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HPM Digital Input, Status Type Point

Unit:	Tagname:	Engineer:	Date:
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PNTMODTY APM Module Type	DI DISOE	Digital Input Module High Resolution DI Module
DITYPE Digital Input Type	Status Latched Accum	Status input (open/closed contacts) Steady-state input Pulsed input

Operating Configuration Display - N/A if PNTFORM = Componnt.

STATETXT (1) State 1-Description Text	_ _ _ _ _ _ _	<i>range</i> = Not applicable <i>default</i> => "On"		
STATETXT (0) State 0-Description Text	_ _ _ _ _ _ _	<i>default</i> = Not applicable <i>default</i> = >"Off"		
BOXCLR (1) State 1 Box Color	Red Cyan	Green (Upper box default) Yellow	White Blue	Black Magenta
BOXCLR (0) State 0 Box Color	Red Cyan	Green Yellow (Lower box default)	White Blue	Black Magenta
INPTDIR Digital Input Direction	Direct Reverse		Open=>pvraw=off; Closed=>pvaw=on Closed=>pvraw=off; Open=>pvraw=on	
PVSRCOPT PV Source Option	Only Auto		PV Source selection is not available and field wiring or memory fetch supplies PV. Selection from PVSOURCE is available & Operator/CL-program may supply PV.	
PVSOURCE PV Source	All Auto Man Sub		Field wiring or memory fetch supplies PV PV is supplied by Operator or program A value is substituted by a CL program	

Applies if PNTMODTY = DISOE

DEBOUNCE Debounce Time in Seconds	_ _	<i>range</i> = 0-50 Milliseconds <i>default</i> = 10 Milliseconds
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HPM Digital Input, Status Type Point

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Unit:	Tagname:	Engineer:	Date:
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Alarming Display – N/A if PNTFORM = Componnt.

As you enter alarm priorities in the next section please refer to the following.

NoAction	Alarm is not reported to the system and not annunciated (R670 & later)
Journal	Alarm is historized, not reported to US, not annunciated
Low	Alarm is historized, reported to US, and annunciated
High	Alarm is historized, reported to US, and annunciated
Emergency	Alarm is historized, reported to US, and annunciated Not
Printer	historized, reported to the printer, not annunciated Alarm
JnlPrint	historized, reported to the printer, not annunciated

\$AUXUNIT *range = Any configured LCN Unit ID*
Auxiliary Unit *default = Null (--)*

\$REALARM (R660 and later) *range = 0 to 9999 minutes*
Re-alarming Interval *default = 0 minutes*

ALMOPT ☐ **None** No affect
Alarming Option ☐ Offnorml Alarm if PV and PVNORMAL differs
☐ ChngOfSt Change Of State Reporting (option)

PVNORMAL *range = STATETXT(0) – STATETX(1)*
PV Value's Normal State Text *default = Text equal to "Off"*

\$DLYOPT (R660 and later) ☐ **None** No alarms are to be delayed
Alarm Delay Option ☐ PmOffDly Suppress alarm events up to configured
☐ NmOffDly DLYTIME after first alarm
☐ NmOnDly Delay RTN for configured DLYTIME
Delay alarm event for configured DLYTIME
DLY TIME *range = 0 to 60*
Alarm Delay In Seconds *default = 5 (seconds)*

OFFNRMPR OFFNORMAL Alarm Priority or **NoAction** Journal Low High Emergency Printer JnlPrint
CHOFSTPR Alarm Priority (R670 and later)
\$OFSELCT (R681 and later) **Cutout** *range = Cutout, Allow*
Off Normal Selective Cutout Allow *default = Cutout*

ALENBST ☐ **Enable** Displayed Logged Reported to EIP
Point Alarm Enable Status ☐ Disable Yes Yes Yes
☐ Inhibit No Yes No Yes
No No No

EVTOPT ☐ **None** Neither Eip or Soe is allowed
Event Reporting Option ☐ Eip Process special is triggered in AM/CG
☐ Soe Point will notify Seq. Of Events process
☐ Eip_Soe EIP and Soe are both allowed

EIPPCODE *range = Not applicable*
Event Init Proc Destination *default = Null*

PVCHGDLY *range = 0 – 60 Seconds*
PV Change Delay in Seconds *default = 0*

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PNTMODTY APM Module Type	DI DISOE	Digital Input Module High ResolutionDI Module
DITYPE Digital Input Type	Status Latched Accum	Status input (open/closed contacts) Steady-state input Pulsed input

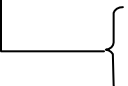
HPM Digital Input, Latched Type Point

Unit:	Tagname:	Engineer:	Date:
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Operating Configuration Display – N/A if PNTFORM = Component.

STATETXT (1) State 1-Description Text		<i>range</i> = Not applicable <i>default</i> =>"On"		
STATETXT (0) State 0-Description Text		<i>range</i> = Not applicable <i>default</i> =>"Off"		
BOXCLR (1) State 1 Box Color	Red Cyan	Green (Upper box default) Yellow	White Blue	Black Magenta
BOXCLR (0) State 0 Box Color	Red Cyan	Green Yellow (Lower box default)	White Blue	Black Magenta

Operating Configuration Display - N/A if PNTFORM = Component

Operating Configuration Display		
INPTDIR Digital Input Direction	Direct Reverse	Open=>pvraw=off; Closed=>pvraw=on Closed=>pvraw=off; Open=>pvraw=on
PVSRCOPT PV Source Option	Only Auto All	PV Source selection is not available and Field wiring or memory fetch supplies PV. Selection from PVSOURCE is available & Operator/CL-program may supply PV.
PVSOURCE PV Source	 Auto Man Sub	Field wiring or memory fetch supplies PV. PV is supplied by Operator or program A value is substituted by a CL program

Applies if PNTMODTY = DISOE

DEBOUNCE Debounce Time in Secondss	<code>[_._]</code>	<i>range</i> = 0 - 50 Milliseconds <i>default</i> = 10 Milliseconds
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Alarming Display - N/A if PNTFORM = Component

	<i>Alarming Display - N/A if PNTFORM = Component</i>	
EVTOPT Event Reporting Option	None Eip	Neither Eip or Soe is allowed Process special is triggered in AM/CG
EIPPCODE Event Init Proc Destination	<i>range = Not applicable</i> <i>default = Null</i>	

Applies if PNTMODTY = DISOE

PVCHGDLY PV Change Delay in Seconds	<u> . </u>	range = 0 – 60 Seconds default = 0
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HPM Digital Input, Accum Type Point

Unit:	Tagname:	Engineer:	Date:
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N/A if PNTFORM = Componnt.

PRIMMOD		<i>range = Not Applicable</i>
Primary Module ID		<i>default = Null</i>

PNTMODTY	DI	Digital Input Module
PM Module Type	DISOE	High Res. DI Module (N/A for Accum Point)
DITYPE	Status	Status input (open/closed contacts)
Digital Input Type	Latched	Steady-state input
	Accum	Pulsed input

Operating Configuration Display - N/A if PNTFORM = Componnt

INPTDIR	Direct	Open=>pvraw=off; Closed=>pvraw=on
Digital Input Direction	Reverse	Closed=>pvraw=off; Open=>pvraw=on
COUNTDWN	Off	<i>range = Off-On</i>
Accumulator Count Down Flag	On	<i>default = Off</i>
AVTV		<i>range = > = 0.0 and NaN</i>
Accumulated Value Target Value		<i>default = 0</i>
RESETVAL		<i>range = > = 0</i>
Accumulator Reset Value		<i>default = 0</i>
OVERVAL	. . .	<i>range = 0 to 100</i>
Overview Display Value		<i>default = 25</i>

1 of 2 DigOut Points - *Point Assignment Display*

Operating Configuration Display

OPTDIR DO Direct/Reverse Action	Direct Reverse	For OP pulse = "On", OP = "On" signal For OP pulse = "Off", OP = "Off" signal
PERIOD Pulse Width Modulation Cycle Time	_. _ . _ . _	<i>range = 1.0 to 120.0</i> <i>default =10.0 (Seconds)</i>

HPM Flow Compensation PV Algorithm

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Unit:	Tagname:	Engineer:	Date:
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Point Assignment Display

NAME Tag Name 		range = Not applicable default = Blank
NODETYPE Node Type	PM APM HPM	A PM Point Is Being Configured An APM Point Is Being Configured An HPM Point Is Being Configured
PNTFORM Point Form	FuLL Componnt	Point is fully displayed and alarmed Pt. is partially displayed but not alarmed

PTDESC Point Descriptor 		range = Not applicable default = Blank
.....		
EUDESC Engineering Units Descriptor 		range = Not applicable default = Blank
KEYWORD Keyword Descriptor 		range = Not applicable default = Blank
ASSOCDSP Associated Display 		range = Not applicable default = Blank
\$CDETAIL (R620 and later) Point Custom Display 		range = Not applicable default = Blank
UNIT Process Unit Point Is Assigned To ..		range = Two Alphanumeric Characters default = Not applicable
NTWKNUM NIM's UCN That This Point Is In ..		range = 1 to 20 default = Not applicable
NODENUM HPMM's Address On The UCN ..		range = 1 to 64 default = Not applicable
MODNUM I/O Module Number ..		range = 0 default = 0 (Control Processor Point)
SLOTNUM Slot Number ..		range = 1-125 (but <=NPVSLOT) default = Not applicable

PRIMMOD Primary Module ID 	range = Not Applicable default = Null
USERID User ID Reservation 	Sixteen Character String default = dashes

PV Algorithm Display

PVALGID PV Algorithm Identifier	Null FlowComp	No affect Flow Compensation
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HPM Flow Compensation PV Algorithm

Unit:	Tagname:	Engineer:	Date:
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PV Configuration Display

PVEUHI PV High Range In Engr Units	_ _ _ _ _ _ _	<i>range = PVEULO to PVEXEUHI</i> <i>default = 100.0</i>
PVEULO PV Low Range In Engr Units	_ _ _ _ _ _ _	<i>range = PVEXEULO to PVEUHI</i> <i>default = 0.0</i>
PVFORMAT PV Decimal Point Format	DO D2 D1 D3	D0=9999. D2=99.99 D1=999.9 D3 = 9.999
PVEXEUHI PV Extended Hi Range in EUs	_ _ _ _ _ _ _ _ _ _ _ _ _	<i>range = >=PVEUHI</i> <i>default = 102.9</i>
PVEXEULO PV Extended Lo Range in EUs	_ _ _ _ _ _ _ _ _ _ _ _ _	<i>range = <=PVEULO</i> <i>default = -2.9</i>
PVCLAMP PV Clamping Option	NoClamp Clamp	No clamping of the PV value Clamp PV value at range extension limit

N/A if PNTFORM = Componnt.

PVSRCOPT PV Source Option	Only Auto All	PV Source selection is not available Selection from PVSOURCE is available
PVSOURCE PV Source	Auto Man Sub	Field wiring /memory -fetch supplies PV PV is supplied by Operator or program A value is substituted by a CL program

PV Configuration Display (Continued)

TF PV Filter Lag Time in Minutes	_ _ _ _ _ _ _	<i>range = 0.0 to 60.0 (minutes)</i> <i>default = 0.0 (PV will not be delayed)</i>
PVTV PV Target Value in EUs	_ _ _ _ _ _ _ _ _ _ _ _ _	<i>range = PVEXEULO-PVEXEUHI, & NaN</i> <i>default = NaN</i>
OVERVAL Overview Display Value	_ _ _ _ _ _ _	<i>range = 0 to 100</i> <i>default = 25</i>

HPM Flow Compensation PV Algorithm

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Unit:	Tagname:	Engineer:	Date:
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Reg PV Flow Comp Algo Display

PVEQN	EqA	Inputs	Type of compensation
PV Equation Type	EqB	F and G	Mass/vol. Flow of liquid
	EqC	F, P, and T	Mass flow of gas
	EqD	F, G, P, and T	Mass flow of gas
	EqE	F, G, P, and T	Volumetric flow of gas
		F, P, T, X, and Q	Mass flow of steam
PVCHAR	Linear		Linear PV Characterization
PV Characterization	SqrRoot		Square Root PV Characterization
C	_ _ _ _ _ _ _		range = N/A default = 1.0
Scaling Constant			
C1	_ _ _ _ _ _ _		range = N/A default = 1.0
Correction Factor			
C2	_ _ _ _ _ _ _		range >= 0.1 default = 1.0
Correction Factor			
COMPILM			range = COMPLLM to 10.0, & NaN default = 1.25
Compensation Term High Limit	_ _ _ _		
COMPLLM			range = 0.0 to COMPILM, and NaN default = 0.8
Compensation Term Low Limit	_ _ _ _		

Reference Constants			PV Equation =				
			A	B	C	D	E
RG	_ _ _ _ _ _ _	range = Real # default = 1.0	9		9	9	
Reference Specific Gravity							
RP	_ _ _ _ _ _ _	range = Real # default = 1.0		9	9	9	9
Reference Pressure							
RT	_ _ _ _ _ _ _	range = Real # default = 1.0		9	9	9	9
Reference Temperature							
RX	_ _ _ _ _ _ _	range = Real # default = 1.0					9
Reference Steam Compressability							
RQ	_ _ _ _ _ _ _	range = Real # default = 1.0					9
Reference Steam Quality Factor							
P0 (Flow Comp)	_ _ _ _ _ _ _	range = Real # default = 0.0		9	9	9	9
Zero Reference For Pressure							
T0	_ _ _ _ _ _ _	range = Real # default = 0.0		9	9	9	9
Zero Reference for Temperature							

HPM Flow Compensation PV Algorithm

Unit:	Tagname:	Engineer:	Date:
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Reg PV Flow Comp Algo Display (Continued)

PISRC(n) Input#	(PV-Input-Connection Source)					PISRC (point.parameter)	Destination (PIDSTN)
Equation =	A	B	C	D	E		
n =	1	1	1	1	1		F
n =	2		2	2			G
n =		2	3	3	2		P
n =		3	4	4	3		T
n =					4		X
n =					5		Q

Alarming Display – N/A if PNTFORM = Componnt

As you enter alarm trip points and alarm priorities in the next section please refer to the following alarm Priority definitions and the note.

NoAction	Alarm is not reported to the system and not annunciated (R670 & later)
Journal	Alarm is historized, not reported to US, not annunciated
Low	Alarm is historized, reported to US, and annunciated
High	Alarm is historized, reported to US, and annunciated
Emergency	Alarm is historized, reported to US, and annunciated Not
Printer	historized, reported to the printer, not annunciated Alarm
JnlPrint	historized, reported to the printer, not annunciated

Note: If you do not enter a trip point value for a given pair of trip-points and alarm-priorities, do not Circle an alarm priority option for that pair (i.e., no trip point, no alarm priority).

Alarming Display - N/A if PNTFORM = Componnt

\$AUXUNIT Auxiliary Unit	...	range = Any configured LCN Unit ID default = Null (- -)
\$ALDLOPT (R620 and later) Alarm Delay Option	None OffDlyLo OffDlyHi OffDlyEm OnDlyLo OnDlyHi OnDlyEm	Off Delay Lo Off Delay Hi Off Delay Emergency On Delay Lo On Delay Hi On Delay Emergency
\$DLBADAL (R681 and later) Delay for BADPV	None Offdelay	range = Offdelay, None default = None
DLYTIME (R620 and later) Delay Time	...	range = 0 to 9999 seconds default = 0 seconds
\$REALARM Re-alarming Interval (R660 and later)	...	range = 0 to 9999 minutes default = 0 minutes
PVALDB PV Alarm Deadband In %	Half One Two Three Four Five EU	
PVALDBEU PV Alarm Deadband In EU	...	range > = 0.0 default = 1.0
PVHITP PV High Alarm Trip Point	...	range = PVLOTP - PVHHTP, & NaN default = NaN

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[illegible]

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[illegible]

HPM General Linearization PV Algorithm

Unit:	Tagname:	Engineer:	Date:
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PVAlgorithm Display

PVALGID PV Algorithm Identifier	Null GenLin	No affect General Linearity
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PV Configuration Display

PVEUHI PV High Range In Engr Units	_ _ _ _ _ _ _	<i>range</i> = PVEULO to PVEXEUHI <i>default</i> = 100.0
PVEULO PV Low Range In Engr Units	_ _ _ _ _ _ _	<i>range</i> = PVEXEULO to PVEUHI <i>default</i> = 0.0
PVFORMAT PV Decimal Point Format	DO D1	D2 D3 D0 = 9999. D1 = 999.9 D2 = 99.99 D3 = 9.999
PVEXEUHI PV Extended Hi Range in EUs	_ _ _ _ _ _ _ _ _	<i>range</i> = >=PVEUHI <i>default</i> = 102.9
PVEXEULO PV Extended Lo Range in EUs	_ _ _ _ _ _ _ _ _	<i>range</i> = <=PVEULO <i>default</i> = -2.9
PV CLAMP PV Clamping Option	NoClamp Clamp	No clamping of the PV value Clamp PV value at range extension limit

. N/A if PNTFORM = Componnt

PVSRCOPT PV Source Option	Only Auto All	PV Source selection is not available Selection from PVSOURCE is available
PVSOURCE PV Source	Auto Man Sub	Field wiring /memory -fetch supplies PV PV is supplied by Operator or program A value is substituted by a CL program

PV Configuration Display (continued)

TF PV Filter Lag Time in Minutes	_ _ _ _ _	<i>range</i> = 0.0 to 60.0 (minutes) <i>default</i> = 0.0 (PV will not be delayed)
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N/A if PNTFORM = Componnt.

PVTV PV Target Value in EUs	_ _ _ _ _ _ _ _ _	<i>range</i> = PVEXEULO-PVEXEUHI, & NaN <i>default</i> = NaN
OVERVAL Overview Display Value	_ _ _ _	<i>range</i> = 0 to 100 <i>default</i> = 25

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Unit:	Tagname:	Engineer:	Date:
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Req PV General/Linear Alqo Display

[illegible]

Alarming Display – N/A if PNTFORM = Component

As you enter alarm trip points and alarm priorities in the next section please refer to the following alarm priority definitions and the note.

NoAction	Alarm is not reported to the system and not annunciated (R670 & later)
Journal	Alarm is historized, not reported to US, not annunciated
Low	Alarm is historized, reported to US, and annunciated
High	Alarm is historized, reported to US, and annunciated
Emergency	Alarm is historized, reported to US, and annunciated
Printer	Not historized, reported to the printer, not annunciated
JnlPrint	historized, reported to the printer, not annunciated

Note: If you do not enter a trip point value for a given pair of trip-points and alarm-priorities, do not circle an alarm priority option for that pair (i.e., no trip point, no alarm priority).

HPM General Linearization PV Algorithm

Unit:	Tagname:	Engineer:	Date:
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Alarming Display - N/A if PNTFORM = Component

\$AUXUNIT Auxiliary Unit	_____	<i>range</i> = Any configured LCN Unit ID <i>default</i> = Null (- -)
\$ALDLOPT (R620 and later) Alarm Delay Option	None OffDlyLo OffDlyHi OffDlyEm OnDlyLo OnDlyHi OnDlyEm	Off Delay Lo Off Delay Hi Off Delay Emergency On Delay Lo On Delay Hi On Delay Emergency
\$DLBADAL (R681 and later) Delay for BADPV	None Offdelay	<i>range</i> = Offdelay, None <i>default</i> = None
\$DLTIME (R620 and later) Delay Time	_____	<i>range</i> = 0 to 9999 seconds <i>default</i> = 0 seconds
\$REALARM (R660 and later) Re-alarming Interval	_____	<i>range</i> = 0 to 9999 minutes <i>default</i> = 0 minutes
PVALDB PV Alarm Deadband In %	Half One Two Three Four Five EU	
PVALDBEU PV Alarm Deadband In EU	_____	<i>range</i> > = 0.0 <i>default</i> = 1.0
PVHITP PV High Alarm Trip Point	_____	<i>range</i> = PVLOTP - PVHHTP, & NaN <i>default</i> = NaN
PVHIPR PV High Alarm Priority	NoAction Journal Low	High Emergency Printer JnlPrint
\$HISELCT (R681 and later) PV High Selective Cutout	Cutout Allow	<i>range</i> = Cutout, Allow <i>default</i> = Cutout
PVLOTP PV Low Alarm Trip Point	_____	<i>range</i> = PVLLTP - PVHITP, & NaN <i>default</i> = NaN
PVLOPR PV Low Alarm Priority	NoAction Journal Low	High Emergency Printer JnlPrint
\$LOSELCT (R681 and later) PV Low Selective Cutout	Cutout Allow	<i>range</i> = Cutout, Allow <i>default</i> = Cutout
PVHHTP PV Hi Hi Alarm Trip Point	_____	<i>range</i> = PVHITP to PVEUHI, & NaN <i>default</i> = NaN
PVHHPR PV High High Alarm Priority	NoAction Journal Low	High Emergency Printer JnlPrint
\$HHSELCT (R681 and later) PV High High Selective Cutout	Cutout Allow	<i>range</i> = Cutout, Allow <i>default</i> = Cutout
PVLLTP PV Low Low Alarm Trip Point	_____	<i>range</i> = PVEULO - PVLOTP, & NaN <i>default</i> = NaN

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Unit:	Tagname:	Engineer:	Date:
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[illegible]

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PVALGID	Null	No affect
PV Algorithm Identifier	HiLoAvg	High Low Average Selector

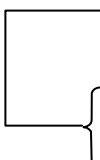
HPM High Low Average PV Algorithm

Unit:	Tagname:	Engineer:	Date:
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PV Configuration Display

PVEUHI PV High Range In Engr Units	_ _ _ _ _ _ _	<i>range</i> = PVEULO to PVEXEUHI <i>default</i> = 100.0
PVEULO PV Low Range In Engr Units	_ _ _ _ _ _ _	<i>range</i> = PVEXEULO to PVEUHI <i>default</i> = 0.0
PVFORMAT PV Decimal Point Format	DO D2 D1 D3	D0 = 9999. D2 = 99.99 D1 = 999.9 D3 = 9.999
PVEXEUHI PV Extended Hi Range in EUs	_ _ _ _ _ _ _ _ _ _ _ _ _	<i>range</i> = >=PVEUHI <i>default</i> = 102.9
PVEXEULO PV Extended Lo Range in EUs	_ _ _ _ _ _ _ _ _ _ _ _ _	<i>range</i> = <=PVEULO <i>default</i> = -2.9
PVCLAMP PV Clamping Option	NoClamp Clamp	No clamping of the PV value Clamp PV value at range extension limit

N/A if PNTFORM = Componnt.

PVSRCOPT PV Source Option		Only Auto All	PV Source selection is not available Selection from PVSOURCE is available
PVSOURCE PV Source			

PV Configuration Display (Continued)

TF PV Filter Lag Time in Minutes	_ _ _ _ _	<i>range</i> = 0.0 to 60.0 (minutes) <i>default</i> = 0.0 (PV will not be delayed)
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N/A if PNTFORM = Componnt.

PVTV PV Target Value in EUs	_ _ _ _ _ _ _ _ _ _ _ _ _	<i>range</i> = PVEXEULO-PVEXEUHI, & NaN <i>default</i> = NaN
OVERVAL Overview Display Value	_ _ _ _ _	<i>range</i> = 0 to 100 <i>default</i> = 25

HPM High Low Average PV Algorithm

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Unit:	Tagname:	Engineer:	Date:
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Reg PV High/Low Avg Algo Display

PVEQN PV Equation Type	EqA EqB EqC	Select & identify highest of up to 6 inputs Select & identify lowest of up to 6 inputs Calculate the average of up to 6 inputs
N # Inputs To HiLoAvg Algo		range = 2 to 6 default = 2
NMIN Minimum No. Of Good Inputs		range = 1 to N (# Inputs parameter) default = 1
FRCPERM Forced Input Permissive	Off On	Optr. Cannot select input to HiLoAvg Operator can select input to HiLoAvg
FSELIN Input To Be Force Selected	SelectP1 SelectP2 SelectP3 SelectP4 SelectP5 SelectP6	Input P1 is the forced input Input P2 is the forced input Input P3 is the forced input Input P4 is the forced input Input P5 is the forced input Input P6 is the forced input

<u>Input#</u>	<u>PISRC (point.parameter)</u>	<u>Destination Para-meter (PIDSTN)</u>
1		P1
2		P2
3		P3
4		P4
5		P5
6		P6

Alarming Display – N/A if PNTFORM = Comppnt

As you enter alarm trip points and alarm priorities in the next section please refer to the following alarm Priority definitions and the note.

NoAction	Alarm is not reported to the system and not annunciated (R670 & later)
Journal	Alarm is historized, not reported to US, not annunciated
Low	Alarm is historized, reported to US, and annunciated
High	Alarm is historized, reported to US, and annunciated
Emergency	Alarm is historized, reported to US, and annunciated
Printer	Not historized, reported to the printer, not annunciated
JnlPrint	Alarm historized, reported to the printer, not annunciated

Note: If you do not enter a trip point value for a given pair of trip-points and alarm-priorities, do not Circle an alarm priority option for that pair (i.e., no trip point, no alarm priority).

Alarming Display – N/A if PNTFORM = Comppnt

\$AUXUNIT Auxiliary Unit	..	range = Any configured LCN Unit ID default = Null (- -)
\$ALDLOPT (R620 and later) Alarm Delay Option	None OffDlyLo OffDlyHi OffDlyEm OnDlyLo OnDlyHi OnDlyEm	Off Delay Lo Off Delay Hi Off Delay Emergency On Delay Lo On Delay Hi On Delay Emergency

HPM High Low Average PV Algorithm

Unit:	Tagname:	Engineer:	Date:
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Alarming Display - N/A if PNTFORM = Componnt

\$DLBADAL (R681 and later) Delay for BADPV	None Offdelay	<i>range</i> = Offdelay, None <i>default</i> = None
DLTIME (R620 and later) Delay Time	_ _ _ _	<i>range</i> = 0 to 9999 seconds <i>default</i> = 0 seconds
\$REALARM Re-alarming Interval (R660 and later)	_ _ _ _	<i>range</i> = 0 to 9999 minutes <i>default</i> = 0 minutes
PVALDB PV Alarm Deadband In %	Half One Two Three Four Five EU	
PVALDBEU PV Alarm Deadband In EU	_ _ _ _	<i>range</i> > = 0.0 <i>default</i> = 1.0
PVHITP PV High Alarm Trip Point	_ _ _ _ _ _ _	<i>range</i> = PVLOTP - PVHHTP, & NaN <i>default</i> = NaN
PVHIPR PV High Alarm Priority	NoAction Journal Low High Emergency Printer JnlPrint (R670 and later)	
\$HISELCT (681 and later) PV High Selective Cutout	Cutout Allow	<i>range</i> = Cutout, Allow <i>default</i> = Cutout
PVLOTP PV Low Alarm Trip Point	_ _ _ _ _ _ _	<i>range</i> = PVLLTP - PVHITP, & NaN <i>default</i> = NaN
PVLOPR PV Low Alarm Priority	NoAction Journal Low High Emergency Printer JnlPrint (R670 and later)	
\$LOSELCT (681 and later) PV Low Selective Cutout	Cutout Allow	<i>range</i> = Cutout, Allow <i>default</i> = Cutout
PVHHTP PV Hi Hi Alarm Trip Point	_ _ _ _ _ _ _	<i>range</i> = PVHITP to PVEUHI, & NaN <i>default</i> = NaN
PVHHPR PV High High Alarm Priority	NoAction Journal Low High Emergency Printer JnlPrint (R670 and later)	
\$HHSELCT (681 and later) PV High High Selective Cutout	Cutout Allow	<i>range</i> = Cutout, Allow <i>default</i> = Cutout
PVLLTP PV Low Low Alarm Trip Point	_ _ _ _ _ _ _	<i>range</i> = PVEULO - PVLOTP, & NaN <i>default</i> = NaN
PVLLPR PV Low Low Alarm Priority	NoAction Journal Low High Emergency Printer JnlPrint (R670 and later)	
\$LLSELCT (681 and later) PV Low Low Selective Cutout	Cutout Allow	<i>range</i> = Cutout, Allow <i>default</i> = Cutout

HPM High Low Average PV Algorithm

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Unit:	Tagname:	Engineer:	Date:
PVSGCHTP PV Significant Change Alarm Trip Point			
			range = ≥ 0.0 , and NaN default = NaN
PVROCPTP PV Pos Rate Of Change Trip Point			
			range = ≥ 0.0 , and NaN default = NaN
PVROCPPR PV Pos Rate Of Chg Alarm Priority			
		NoAction Journal Low	High Emergency Printer JnlPrint (R670 and later)
\$RPSELCT (R681 and later) PV Positive Rate of Change Selective Cutout			
		Cutout Allow	range = Cutout, Allow default = Cutout
PVROCNTP PV Neg Rate Of Change Trip Point			
			range = ≥ 0.0 , and NaN default = NaN
PVROCNPR PV Neg Rate Of Chg Alarm Priority			
		NoAction Journal Low	High Emergency Printer JnlPrint (R670 and later)
\$RNSELCT (R681 and later) PV Negative Rate of Change Selective Cutout			
		Cutout Allow	range = Cutout, Allow default = Cutout
BADPVPR Bad-PV Alarm Priority			
		NoAction Journal Low	High Emergency Printer JnlPrint (R670 and later)
ALENBST Point Alarm Enable Status			
		Enable	<u>Displayed</u> <u>Logged</u> <u>Reported To EIP</u>
		Disable	Yes Yes Yes
		Inhibit	No Yes No Yes
			No No No

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OVERVAL	_. .	<i>range = 0 to 100</i>
Overview Display Value		<i>default = 25</i>

HPM Incremental Summer CTL Algorithm

Unit:	Tagname:	Engineer:	Date:
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Mode Configuration Display

Parameter	Value	Value	Description
RCASOPT Remote Cascade Option	None Ddc		Local cascade is the only valid cascade AM strategy stores to OP. No OP limits.
SHEDTIME Remote Cascade Shed Time			<i>range</i> = 0 to 1000 (seconds) <i>default</i> = 0 (mode shed disabled)
SHEDMODE Shed Mode	Man Bcas		Supervisory point will shed to Manual Supervisory point will shed to BackupCas
NMODE RegCtl Point's Normal Mode	None Man Cas Bcas		No configured "normal" operating mode Manual is the "normal" operating mode Cascade is "normal" operating mode Backup Cas is "normal" operating mode
NMODATTR Normal Mode Attribute	None Operator Program		Parameter NMODE sets "normal" mode Operator sets "normal" operating mode Program sets "normal" operating mode
MODEPERM Mode Permissive	Permit NotPerm		Operator can change this point's mode Optr. cannot change this point's mode
\$MODESEL (R630 and later) Mode Selection List	Man Cas	Prog Bcas	Man, Cas and Bcas mode, and Prog mode attribute are allowed for this data point.
\$BADOCMD (R680 and later) Bad Output Connection Mode	\$HEDMAN 15_SEC 60_SEC NO_SHED		Point will shed to Manual immediately Point will shed to Manual after 15 seconds Point will shed to Manual after 60 seconds Do not shed mode
EXTSWOPT External Mode Switching	None Ems		No external mode switching is allowed Ext. source can change point's mode

Reg Control Incr/Sum Algo Display

M # Of Initializable Control Algorithm Inputs	_	range = 2 to 4 default = 2
XEUHI X Input High Range In EUs	_ _ _ _ _ _ _	range = >= XEULO default = 100.0
XEULO X Input Low Range in EUs	_ _ _ _ _ _ _	range = <= XEUHI default = 0.0
K1 (IncrSum) Gain Constant For X1 Input	_ _ _ _ _ _ _	range = >=0.0 default = 1.0
K2 (IncrSum) Gain Constant For X2 Input	_ _ _ _ _ _ _	range = >=0.0 default = 1.0
K3 (IncrSum) Gain Constant For X3 Input	_ _ _ _ _ _ _	range = >=0.0 default = 1.0
K4 (IncrSum) Gain Constant For X4 Input	_ _ _ _ _ _ _	range = >=0.0 default = 1.0

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Reg Control Incr/Sum Algo Display (Continued)

NOCINPTS	_	<i>range = 0 to 4</i>
# Of Control Input Connections		<i>default = 0</i>
CISRC (Control-Input-Connection Source)	<u>CISRC</u> (point.parameter)	<u>Destination Para-</u> <u>-meter (CIDSTN)</u>
<u>Input #</u>		
1	_____.	_____.
2	_____.	_____.
3	_____.	_____.
4	_____.	_____.

Output Configuration Display

NOCOPTS		<i>range = 0 to 4</i>
# Of Control Output Connections	_	<i>default = 1</i>
CODSTN (Control-Output-Connection Destination)		
<u>Output #</u>	<u>CODSTN</u> (point.parameter)	
1	_._._._._._._._._._._._._._._._._.	
2	_._._._._._._._._._._._._._._._._.	
3	_._._._._._._._._._._._._._._._._.	
4	_._._._._._._._._._._._._._._._._.	
OPHILM		<i>range = OPLOLM – 106.9 & NaN</i>
Output High Limit In Percent	_._._._._	<i>default = 105.0</i>
OPLOLM		<i>range = -6.9 – OPHILM, & NaN</i>
Output Low Limit In Percent	_._._._._	<i>default = -5.0</i>
SAFEOP		<i>range = -6.9 to to 106.9 & NaN</i>
Safe Output Limit In Percent	_._._._._	<i>default = NaN</i>
OPMCHLM		<i>range = >=0.0, and NaN</i>
Output Minimum Change In %	_._._._._	<i>default = 0.0</i>
OPROCLM		<i>range = >=0.0, and NaN</i>
OP Rate Of Change Limit In %	_._._._._	<i>default = NaN</i>
\$OPTOL		<i>range = 0.0 to 106.9</i>
Output Tolerance in %	_._._._._	<i>default = 0.0</i>

HPM Incremental Summer CTL Algorithm

Unit:	Tagname:	Engineer:	Date:
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Output Configuration Display (Continued)

OUTIND (R620 and later)	Direct	4 - 20mA = 0 – 100%
Reverse Output Indicator	Reverse	20 - 4mA = 0 – 100%
Note: With R670 and later, when this parameter is set to Reverse, the values of the displayed output and the related output parameters are changed. Refer to the definition of the OUTIND parameter in the PM Family Parameter Dictionary, XPM09.		

Alarming Display - N/A if PNTFORM = Componnt (Continued)

As you enter alarm priorities in the next section please refer to the following.

NoAction	Alarm is not reported to the system and not annunciated (R670 & later)
Journal	Alarm is historized, not reported to US, not annunciated
Low	Alarm is historized, reported to US, and annunciated
High	Alarm is historized, reported to US, and annunciated
Emergency	Alarm is historized, reported to US, and annunciated
Printer	Not historized, reported to the printer, not annunciated
JnlPrint	Alarm historized, reported to the printer, not annunciated

Alarming Display - N/A if PNTFORM = Componnt

\$AUXUNIT		<i>range</i> = Any configured LCN Unit ID <i>default</i> = Null (- -)
Auxiliary Unit		
\$ REALARM (R660 and Later)		<i>range</i> = 0 to 9999 minutes <i>default</i> = 0 minutes
Re-alarming Interval		
BADOCOPT	Off	Bad Output Connection Alarm is not initially permitted
BADOC Alarm Option	On	Bad Output Connection Alarm is Initially permitted
BADOCPR	NoAction	Journal Low High Emergency JnlPrint Printer
BADOC Alarm Priority	(R670 and later)	

OPALDB		<i>range</i> = 0 - 25.0 but $\leq$ (OPHITP-OPLOTP)/2 <i>default</i> = 5.0 Output Units
Output Alarm Deadband		
OPHITP		<i>range</i> = OPLOTP to OPHILM and NaN <i>default</i> = NaN
Output High Alarm Trip Point		
OPHIPR	NoAction	Journal Low High Emergency JnlPrint Printer
Output High Alarm Priority	(R670 and later)	
OPLOTP		<i>range</i> = OPLOLM to OPHITP and NaN <i>default</i> = NaN
Output Low Alarm Trip Point		
OPLOPR	NoAction	Journal Low High Emergency JnlPrint Printer
Output Low Alarm Priority	(R670 and later)	

ALENBST		<u>Displayed</u>	<u>Logged</u>	<u>Reported To EIP</u>
Point Alarm Enable Status	Enable	Yes	Yes	Yes
	Disable	No	Yes No	Yes
	Inhibit	No		No

LOGMIX	12_24_4	12 inputs	24 logic blocks	4 outputs
Logic Mix	12_16_8	12 inputs	16 logic blocks	8 outputs
	12_8_12	12 inputs	8 logic blocks	12 outputs

HPM Logic Point

Unit:	Tagname:	Engineer:	Date:
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Input Connections Display

LISRC (Logic Point Inputs)		
Input	Source Point.Parameter	
L1		
L2		
L3		
L4		
L5		
L6		
L7		
L8		
L9		
L10		
L11		
L12		

LIBADOPT	Off	"Off" state is substituted for bad input
Logic Bad Input Handling Option	On	"On" state is substituted for bad input
	Hold	Last good value is subed. for bad input

Input Connections Display (Continued)

Logic Flags And Numerics - Initial Values
Numerics 1-8 are used as local variables. Entry is a real number. default = NAN

NN1		NN2	
NN3		NN4	
NN5		NN6	
NN7		NN8	

Flags 7-12 are used as local variables

FL7	Off	On	FL8	Off	On
FL9	Off	On	FL10	Off	On
FL11	Off	On	FL12	Off	On

HPM Logic Point

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Logic Output Connections Display

The following 2 parameters are not to be configured but are to be used for making choice when filling out the output-connections table below:

LOSRC	L1...L12	L(external input) = a logical or a real value
Logic-Output-Conn Source	FL1...FL12	FL(flag) = either a 1(On) or a 0(Off)
	NN1...NN8	NN(numeric) = data type of Real
	SO1..SO24	SO(Status Output) = Boolean (0 or 1)

LOENBL	L1...L12	L(external input) = a logical or a real value
Logic Output Enable	FL1,..FL12	FL(flag) = either a 1(On) or a 0(Off)
	SO1..SO16	SO(Status Output) = Boolean (0 or 1)

Logic Output Destination Table LODSTN (Logic-Output-Connection Destination) Destination

<u>LOSRC</u> defaults = FL1	<u>Pint.Parameter</u> defaults = blank	<u>LOENBL</u> defaults = FL2
01		
02		
03		
04		
Enter data in lines 5 through 8 if LOGMIX = 12_16_8 or 12_8_12		
05		
06		
07		
08		
Enter data in lines 9 through 12 if LOGMIX = 12_8_12		
09		
10		
11		
12		

HPM Logic Point

Unit:	Tagname:	Engineer:	Date:
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Logic Descriptors Display – N/A if PNTFORM = Componnt.

NODESC

Number of Generic Descriptors

|_|

range = 0 to 12

default = 0

The following parameter is not to be configured but is to be used for making choices when filling out the logic descriptors table below:

PRMDESC(1-12)

Parameter Descriptor Assignment

L1...L12

FL1..FL12

NN1..NN8

SO1..SO24

Logic point inputs

Flags

Numerics

Status Outputs

Logic Descriptors Table

<u>Descr#</u>	<u>Generic Descriptors</u> (defaults = blanks)	<u>Descr. Assign</u> (defaults = SO1)
1	GENDESC (1) _.	PRMDESC (1) _.
2	GENDESC (2) _.	PRMDESC (2) _.
3	GENDESC (3) _.	PRMDESC (3) _.
4	GENDESC (4) _.	PRMDESC (4) _.
5	GENDESC (5) _.	PRMDESC (5) _.
6	GENDESC (6) _.	PRMDESC (6) _.
7	GENDESC (7) _.	PRMDESC (7) _.
8	GENDESC (8) _.	PRMDESC (8) _.
9	GENDESC (9) _.	PRMDESC (9) _.
10	GENDESC (10) _.	PRMDESC (10) _.
11	GENDESC (11) _.	PRMDESC (11) _.
12	GENDESC (12) _.	PRMDESC (12) _.

HPM Logic Point

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Use the appropriate section of this form for each Logic Block (n) required. Circle the Algorithm type (e.g., AND, OR, etc.) and enter/select the parameters that apply for that algorithm.

In the following tables, valid entries for S1, S2, S3, or S4, are: L1 - L12, SO1 - SO24, or FL1 - FL12. Valid entries for R1, or R2 are: L1 - L12, and NN1 - NN8. The Logic Block outputs are SO1 through SO16 or SO1 through SO24.

LOGALGID (n) Logic Block n = ____ Enter 1-24 for LOGMIX 12-24-4; Enter 1-16 for LOGMIX 12-16-8

NULL **AND** **OR** **NAND** **NOR**

S1(n)	___.___	Default = L1	S1Rev(n)	OFF	ON
S2(n)	___.___	Default = L1	S2Rev(n)	OFF	ON
S3(n)	___.___	Default = L1	S3Rev(n)	OFF	ON

LOGALGID (n) Logic Block n = ____ Enter 1-24 for LOGMIX 12-24-4; Enter 1-16 for LOGMIX 12-16-8

NULL **NOT**

S1(n)	___.___	Default = L1
-------	---------	--------------

LOGALGID (n) Logic Block n = ____ Enter 1-24 for LOGMIX 12-24-4; Enter 1-16 for LOGMIX 12-16-8

NULL **XOR**

S1(n)	___.___	Default = L1
S2(n)	___.___	Default = L1

HPM Logic Point

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LOGALGID (n) Logic Block n = ____ Enter 1-24 for LOGMIX 12-24-4; Enter 1-16 for LOGMIX 12-16-8

NULL **QOR2** **QOR3**

S1(n) |_.|_.|_.|_.| Default = L1

S2(n) |_.|_.|_.|_.| Default = L1

S3(n) |_.|_.|_.|_.| Default = L1

S4(n) |_.|_.|_.|_.| Default = L1

LOGALGID (n) Logic Block n = ____ Enter 1-24 for LOGMIX 12-24-4; Enter 1-16 for LOGMIX 12-16-8

NULL **SWITCH** **FLIPFLOP** **CHDETECT**

S1(n) |_.|_.|_.|_.| Default = L1

S2(n) |_.|_.|_.|_.| Default = L1

S3(n) |_.|_.|_.|_.| Default = L1

LOGALGID (n) Logic Block n = ____ Enter 1-24 for LOGMIX 12-24-4; Enter 1-16 for LOGMIX 12-16-8

NULL **EQ** **NE** **GT** **GE** **LT** **LE**

R1(n) |_.|_.|_.|_.| Default = L1

R2(n) |_.|_.|_.|_.| Default = L1

Deadband (n) |_.|_.|_.|_.| range = >=0
default = 1.0

LOGALGID (n) Logic Block n = ____ Enter 1-24 for LOGMIX 12-24-4; Enter 1-16 for LOGMIX 12-16-8

NULL **CHECKBAD**

R1(n) |_.|_.|_.|_.| Default = L1

HPM Logic Point

Unit:	Tagname:	Engineer:	Date:
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LOGALGID (n) Logic Block n = ____ Enter 1-24 for LOGMIX 12-24-4; Enter 1-16 for LOGMIX 12-16-8

NULL **PULSE** **MINPULSE** **MAXPULSE**

S1(n) |_. _ . _ . _| Default = L1

Dlytime(n) |_. _ . _ . _| range = 0-8000
default = 1.0

LOGALGID (n) Logic Block n = ____ Enter 1-24 for LOGMIX 12-24-4; Enter 1-16 for LOGMIX 12-16-8

NULL **DELAY**

S1(n) |_. _ . _ . _| Default = L1

LOGALGID (n) Logic Block n = ____ Enter 1-24 for LOGMIX 12-24-4; Enter 1-16 for LOGMIX 12-16-8

NULL **ONLY** **OFFDLY**

S1(n) |_. _ . _ . _| Default = L1

Dlytime(n) |_. _ . _ . _| range = 0-8000
default = 1.0

LOGALGID (n) Logic Block n = ____ Enter 1-24 for LOGMIX 12-24-4; Enter 1-16 for LOGMIX 12-16-8

NULL **WATCHDOG**

Dlytime(n) |_. _ . _ . _| range = 0-8000
default = 1.0

LOGALGID (n) Logic Block n = ____ Enter 1-24 for LOGMIX 12-24-4; Enter 1-16 for LOGMIX 12-16-8

NULL **DISCREP3**

S1(n) |_. _ . _ . _| Default = L1

S2(n) |_. _ . _ . _| Default = L1

S3(n) |_. _ . _ . _| Default = L1

Dlytime(n) |_. _ . _ . _| range = 0-8000
default = 1.0

PVALGID PV Algorithm Identifier	<i>Null</i> MidOf3	No affect Middle of Three
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HPM Middle of 3 PV Algorithm

Unit:	Tagname:	Engineer:	Date:
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PV Configuration Display

PVEUHI PV High Range In Engr Units	_ _ _ _ _ _ _	<i>range</i> = PVEULO to PVEXEUHI <i>default</i> = 100.0
PVEULO PV Low Range In Engr Units	_ _ _ _ _ _ _	<i>range</i> = PVEXEULO to PVEUHI <i>default</i> = 0.0
PVFORMAT PV Decimal Point Format	DO D2 D1 D3	D0=9999. D2=99.99 D1=999.9 D3 = 9.999
PVEXEUHI PV Extended Hi Range in EUs	_ _ _ _ _ _ _ _ _ _ _ _ _	<i>range</i> = >=PVEUHI <i>default</i> = 102.9
PVEXEULO PV Extended Lo Range in EUs	_ _ _ _ _ _ _ _ _ _ _ _ _	<i>range</i> = <=PVEULO <i>default</i> = -2.9
PV CLAMP PV Clamping Option	NoClamp Clamp	No clamping of the PV value Clamp PV value at range extension limit
PVSRCOPT PV Source Option	Only Auto All	PV Source selection is not available Selection from PVSOURCE is available
PVSOURCE PV Source	Auto Man Sub	Field wiring /memory-fetch supplies PV PV is supplied by Operator or program A value is substituted by a CL program
TF PV Filter Lag Time in Minutes	_ _ _ _ _ _ _	<i>range</i> = 0.0 to 60.0 (minutes) <i>default</i> = 0.0 (PV will not be delayed)
PVTV PV Target Value in EUs	_ _ _ _ _ _ _ _ _ _ _ _ _	<i>range</i> = PVEXEULO-PVEXEUHI, & NaN <i>default</i> = NaN
OVERVAL Overview Display Value	_ _ _ _ _ _ _	<i>range</i> = 0 to 100 <i>default</i> = 25

Reg PV Middle Of 3 Algo Display

PVEQN	EqA	Highest good input when 1 or 2 are bad
PV Equation Type	EqB	Lowest good input when 1 or 2 are bad
	EqC	Average of any and all good inputs

PISRC (PV-Input-Connection Source)	<u>PISRC</u> (point.parameter)	<u>Destination Parameter (PIDSTN)</u>
<u>Input#</u>		
1		P1
2		P2
3		P3

HPM Middle of 3 PV Algorithm

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Unit:	Tagname:	Engineer:	Date:
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Alarming Display – N/A if PNTFORM = Componnt
As you enter alarm trip points and alarm priorities in the next section please refer to the following alarm Priority definitions and the note.

NoAction	Alarm is not reported to the system and not annunciated (R670 & later)
Journal	Alarm is historized, not reported to US, not annunciated
Low	Alarm is historized, reported to US, and annunciated
High	Alarm is historized, reported to US, and annunciated
Emergency	Alarm is historized, reported to US, and annunciated
Printer	Not historized, reported to the printer, not annunciated
JnlPrint	Alarm historized, reported to the printer, not annunciated

Note: If you do not enter a trip point value for a given pair of trip-points and alarm-priorities, do not Circle an alarm priority option for that pair (that is, no trip point, no alarm priority).

Alarming Display - N/A if PNTFORM = Componnt

\$AUXUNIT Auxiliary Unit	_ _ _	<i>range = Any configured LCN Unit ID default = Null (- -)</i>
\$ALDLOPT (R620 and later) Alarm Delay Option	None OffDlyLo OffDlyHi OffDlyEm OnDlyLo OnDlyHi OnDlyEm	Off Delay Lo Off Delay Hi Off Delay Emergency On Delay Lo On Delay Hi On Delay Emergency
\$DLBADAL (R681 and later) Delay for BADPV	None Offdelay	<i>range = Offdelay, None default = None</i>
\$DLTIME (R620 and later) Delay Time	_ _ _	<i>range = 0 to 9999 seconds default = 0 seconds</i>
\$REALARM Re-alarming Interval (R660 and later)	_ _ _	<i>range = 0 to 9999 minutes default = 0 minutes</i>
PVALDB PV Alarm Deadband In %	Half One Two Three Four Five EU	} <i>range > = 0.0 default = 1.0</i>
PVALDBEU PV Alarm Deadband In EU	_ _ _	
PVHITP PV High Alarm Trip Point	_ _ _ _ _ _ _	<i>range = PVLOTP - PVHITP, & NaN default = NaN</i>
PVHIPR PV High Alarm Priority	NoAction Journal Low	High Emergency JnlPrint Printer
\$HISELCT (R681 and later) PV High Selective Cutout	Cutout Allow	<i>range = Cutout, Allow default = Cutout</i>
PVLOTP PV Low Alarm Trip Point	_ _ _ _ _ _ _	<i>range = PVLLTP - PVHITP, & NaN default = NaN</i>

Warning Display Point (R670 and later)									
Point Name	Point Description	Point Type	Journal	Low	High	Emergency	JnlPrint	Printer	
PVLOPR	PV Low Alarm Priority	NoAction (R670 and later)							
\$LOSELCT (R681 and later)	PV Low Selective Cutout	Cutout Allow				<i>range</i> = Cutout, Allow <i>default</i> = Cutout			
PVHHTP(RegCtl)	PV Hi Hi Alarm Trip Point					<i>range</i> = PVHITP to PVEUHI, & NaN <i>default</i> = NaN			
PVHHPR	PV High High Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	JnlPrint	Printer	
\$HHSELCT (R681 and later)	PV High High Selective Cutout	Cutout Allow				<i>range</i> = Cutout, Allow <i>default</i> = Cutout			
PVLLTP(RegCtl)	PV Low Low Alarm Trip Point					<i>range</i> = PVEULO - PVLOTP, & NaN <i>default</i> = NaN			
PVLLPR	PV Low Low Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	JnlPrint	Printer	
\$LLSELCT (R681 and later)	PV Low Low Selective Cutout	Cutout Allow				<i>range</i> = Cutout, Allow <i>default</i> = Cutout			
PVSGCHTP	PV Significant Change Alarm Trip Point					<i>range</i> = >=0.0, and NaN <i>default</i> = NaN			
PVROCTP	PV Pos Rate Of Change Trip Point					<i>range</i> = >=0.0, and NaN <i>default</i> = NaN			
PVROCPPR	PV Pos Rate Of Chg Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	JnlPrint	Printer	
\$RPSELCT (R681 and later)	PV Positive Rate of Change Selective Cutout	Cutout Allow				<i>range</i> = Cutout, Allow <i>default</i> = Cutout			
PVROCNTP	PV Neg Rate Of Change Trip Point					<i>range</i> = >=0.0, and NaN <i>default</i> = NaN			
PVROCNPR	PV Neg Rate Of Chg Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	JnlPrint	Printer	
\$RNSELCT (R681 and later)	PV Negative Rate of Change Selective Cutout	Cutout Allow				<i>range</i> = Cutout, Allow <i>default</i> = Cutout			
BADPVPR	Bad-PV Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	JnlPrint	Printer	
ALENBST	Point Alarm Enable Status				<u>Displayed</u>	<u>Logged</u>	<u>Reported To EIP</u>		
		Enable			Yes	Yes	Yes		
		Disable			No	Yes	Yes		
		Inhibit			No	No	No		

HPM Override Selector CTL Algorithm

Unit:	Tagname:	Engineer:	Date:
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Point Assignment Display

NAME Tag Name		<i>range</i> = Not applicable <i>default</i> = Blank
NODETYP Node Type	PM APM HPM	A PM Point Is Being Configured An APM Point Is Being Configured An HPM Point Is Being Configured
PNTFORM Point Form	FuLL Componnt	Point is fully displayed and alarmed Pt. is partially displayed but not alarmed

[illegible]

N/A if PNTFORM = Componnt.

PRIMMOD		<i>range = Not Applicable</i>
Primary Module ID	_._._._._._._._._._._._._._.	<i>default = Null</i>
USERID		Sixteen Character String
User ID Reservation	_._._._._._._._._._._._._._.	<i>default = dashes</i>

HPM Override Selector CTL Algorithm

Unit:	Tagname:	Engineer:	Date:
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Control Algorithm Display

CTLALGID Control Algorithm Identifier	NULL OrSel	No affect Override Selector
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PV Configuration Display

OVERVAL Overview Display Value	. . .	N/A if PNTFORM = Componnt. <i>range</i> = 0 to 100 <i>default</i> = 25
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Mode Configuration Display

BADCTLOP Bad PV Mode Shed Option	No_Shed Shed Hold Shed Low Shed High Shed Safe	See APM Parameter Reference Dictionary
RCASOPT Remote Cascade Option	None Ddc	Local cascade is the only valid cascade AM strategy stores to OP. No OP limits.
SHEDTIME Remote Cascade Shed Time		<i>range</i> = 0 to 1000 (seconds) <i>default</i> = 0 (mode shed disabled)
SHEDMODE Shed Mode	Man } BCas }	Supervisory point will shed to Manual Supervisory point will shed to BackupCas
NMODE RegCtl Point's Normal Mode	None } Man } Cas } Bcas }	No configured "normal" operating mode Manual is the "normal" operating mode Cascade is "normal" operating mode Backup Cas is "normal" operating mode
NMODATTR Normal Mode Attribute	None } Operator } Program }	Parameter NMODE sets "normal" mode Operator sets "normal" operating mode Program sets "normal" operating mode
MODEPERM Mode Permissive	Permit NotPerm	Operator can change this point's mode Optr. cannot change this point's mode
\$MODESEL (R630 and later) Mode Selection List	Man Prog Cas Bcas	Man, Cas and Bcas mode, and Prog mode attribute are allowed for this data point.
\$BADO CMD (R680 and later) Bad Output Connection Mode	SHEDMAN 15_SEC 60_SEC NO_SHED	Point will shed to Manual immediately Point will shed to Manual after 15 seconds Point will shed to Manual after 60 seconds Do not shed mode
EXTSWOPT External Mode Switching	None Ems	No external mode switching is allowed Ext. source can change point's mode

Reg Control OrSel Algo Display

CTLEQN Control Equation Type	EqA EqB	Selects the highest input Selects the lowest input
OROPT Override Feedback Option	Off On	Hi Lo selector without override feedback Hi Lo selector with override feedback
OROFFSET Override Offset	Off } On }	No override offset is applied Offset of Gain times Error is applied

HPM Override Selector CTL Algorithm

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Unit:	Tagname:	Engineer:	Date:
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Reg Control OrSel Algo Display (Continued)

M # Of Initializable Control Algorithm Inputs		range = 2 to 4 default = 2
XEUHI X Input High Range in EUs		range = >=XEULO default = 100.0
XEULO X Input Low Range In EUs		range = <= XEUHI default = 0.0
NOCINPTS #Of Control Input Connections		range = 0 to 4 default = 0
CISRC (Control-Input-Connection Source)		Destination Para
<u>Input#</u>	<u>CISRC (point.parameter)</u>	<u>meter (CIDSTN)</u>
1		
2		
3		
4		

Output Configuration Display

NOCOPTS # Of Control Output Connections		range = 0 to 4 default = 1
CODSTN (Control-Output-Connection Destination)		
<u>Output #</u>	<u>CODSTN (point.parameter)</u>	
1		
2		
3		
4		

OPHILM Output High Limit In Percent		range = OPLOLM – 106.9 & NaN default = 105.0
OPLOLM Output Low Limit In Percent		range = -6.9 – OPHILM, & NaN default = -5.0
SAFEOP Safe Output Limit In Percent		range = -6.9 to to 106.9 & NaN default = NaN
OPMCHLM Output Minimum Change In %		range = >=0.0, and NaN default = 0.0
OPROCLM OP Rate Of Change Limit In %		range = >=0.0, and NaN default = NaN
\$OPTOL Output Tolerance in %		range = 0.0 to 106.9 default = 0.0

HPM Override Selector CTL Algorithm

Unit:	Tagname:	Engineer:	Date:
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Output Configuration Display (Continued)

OUTIND (R620 and later)	Direct	4 - 20mA = 0 – 100%
Reverse Output Indicator	Reverse	20 - 4mA = 0 – 100%
Note: With R670 and later, when this parameter is set to Reverse, the values of the displayed output and the related output parameters are changed. Refer to the definition of the OUTIND parameter in the PM Family Parameter Dictionary, XPM09.		

Alarming Display - N/A if PNTFORM = Componnt

As you enter alarm priorities in the next section please refer to the following.

NoAction	Alarm is not reported to the system and not annunciated (R670 & later)
Journal	Alarm is historized, not reported to US, not annunciated
Low	Alarm is historized, reported to US, and annunciated
High	Alarm is historized, reported to US, and annunciated
Emergency	Alarm is historized, reported to US, and annunciated
Printer	Not historized, reported to the printer, not annunciated
JnlPrint	Alarm historized, reported to the printer, not annunciated

Alarming Display - N/A if PNTFORM = Componnt

\$AUXUNIT	_ _ _ _	range = Any configured LCN Unit ID default = Null (- -)
Auxiliary Unit		
\$DLBADAL (R681 and later)	None	range = Offdelay, None
Delay for BADCTL	Offdelay	default = None
DLTIME (R620 and later)	_ _ _ _	range = 0 to 9999 seconds
Delay Time		default = 0 seconds
\$REALARM (R660 and later)	_ _ _ _	range = 0 to 9999 minutes
Re-alarming Interval		default = 0 minutes
BADOCOPT	Off	Bad Output Connection Alarm is not initially permitted
BADOC Alarm Option	On	Bad Output Connection Alarm is Initially permitted
BADOCPR	NoAction	Journal Low High Emergency JnlPrint Printer
BADOC Alarm Priority	(R670 and later)	

OPALDB	_ _ _ _	range = 0 - 25.0 but $\leq$ (OPHITP-OPLOTP)/2 default = 5.0 Output Units
Output Alarm Deadband		
OPHITP	_ _ _ _	range = OPLOTP TO OPHILM and NaN default = NaN
Output High Alarm Trip Point		
OPHIPR	NoAction	Journal Low High Emergency JnlPrint Printer
Output High Alarm Priority	(R670 and later)	
OPLOTP	_ _ _ _	range = OPLOLM TO OPHITP and NaN default = NaN
Output Low Alarm Trip Point		
OPLOPR	NoAction	Journal Low High Emergency JnlPrint Printer
Output Low Alarm Priority	(R670 and later)	

HPM Override Selector CTL Algorithm

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Unit:	Tagname:	Engineer:	Date:
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BADCTLPR	NoAction	Journal	Low	High	Emergency	JnlPrint	Printer
Bad-control Alarm Priority	(R670 and later)						
ALENBST				<u>Displayed</u>	<u>Logged</u>	<u>Reported To EIP</u>	
Point Alarm Enable Status	Enable			Yes	Yes	Yes	
	Disable			No	Yes No	Yes	
	Inhibit			No		No	

CTLALGID Control Algorithm Identifier	<i>Null</i> Pid	No affect Proportional, Integral and Derivative
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HPM PID CTL Algorithm

Unit:	Tagname:	Engineer:	Date:
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PV Configuration Display

PVEUHI PV High Range In Engr Units	_ _ _ _ _ _ _	<i>range =</i> >=PVEULO <i>default =</i> 100.0
PVEULO PV Low Range In Engr Units	_ _ _ _ _ _ _	<i>range =</i> <= PVEUHI <i>default =</i> 0.0
PVFORMAT PV Decimal Point Format	DO D2 D1 D3	D0=9999. D2=99.99 D1=999.9 D3 = 9.999

PVSRCOPT PV Source Option	Only Auto All	PV Source selection is not available Selection from PVSOURCE is available
PVSOURCE PV Source	Auto Man Sub	Field-wiring/memory -fetch supplies PV PV is supplied by Operator or program A value is substituted by a CL program
OVERVAL Overview Display Value	_ _ _ _	<i>range =</i> 0 to 100 <i>default =</i> 25

Mode Configuration Display

BADCTLOP Bad PV Mode Shed Option	No_Shed Shed Hold Shed Low Shed High Shed Safe	See HPM Parameter Reference Dictionary
RCASOPT Remote Cascade Option	None Spc Ddc DdcRsp Rsp	Local cascade is the only valid cascade AM strategy stores to SP within SP limits AM Strategy stores to OP. No OP limits. AM strategy does Ddc and Rsp functions AM strategy stores to SP when this Point is in Auto and is being initialized By its secondary. SP limits are applied.
SHEDTIME Remote Cascade Shed Time	_ _ _ _	<i>range =</i> 0 to 1000 (seconds) <i>default =</i> 0 (mode shed disabled)
SHEDMODE (If Shedtime ≥ 1) Shed Mode	Man Auto BCas	Supervisory point will shed to Manual, Automatic, or Backup Cascade mode.
NMODE RegCtl Point's Normal Mode	None Man Auto Cas Bcas	No configured "normal" operating mode Manual is the "normal" operating mode Automatic is "normal" operating mode Cascade is "normal" operating mode Backup Cascade is "normal" operating mode
		N/A if NMODE = None
NMODATTR Normal Mode Attribute	None Operator Program	Parameter NMODE sets "normal" mode Operator sets "normal" operating mode Program sets "normal" operating mode

HPM PID CTL Algorithm

Unit:	Tagname:	Engineer:	Date:
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MODEPERM Mode Permissive	Permit NotPerm	Operator can change this point's mode Optr. cannot change this point's mode
\$MODESEL (R630 and later) Mode Selection List	Man Prog Auto Cas Bcas	Man, Auto, Cas and Bcas mode, and Prog mode attribute are allowed for this data point.
\$BADOCMD (R680 and later) Bad Output Connection Mode	\$HEDMAN 15_SEC 60_SEC NO_SHED	Point will shed to Manual immediately Point will shed to Manual after 15 seconds Point will shed to Manual after 60 seconds Do not shed mode
EXTSWOPT External Mode Switching	None Ems	No external mode switching is allowed Ext. source can change point's mode

Setpoint Display

SPHILM SP High Limit		<i>range</i> = SPLOLM – SPEUHI, and NaN <i>default</i> = NaN
SPLOLM SP Low Limit		<i>range</i> = SPEULO – SPHILM, and NaN <i>default</i> = NaN
\$SPTOL Set Point Tolerance		<i>range</i> = ≥ 0.0 <i>default</i> = 0.0
SP Set Point in Eng. Units of the PV		<i>range</i> = SPLOLM to SPHILM <i>default</i> = 0.0
SPOPT Set Point Option Asp is invalid if PNTFORM=Componnt.	None Tv Asp	No specialized options are available Target value processing Advisory deviation alarming
RBOPT Ratio Bias Option	NoRatBi FixRatBi AutoRat AutoBi	No ratio/bias is used to calculate the SP Fixed ratio (R) and fixed bias(B) are used R is back calculated during initialization B is back calculated during initialization
RTHILM Ratio High Limit		<i>range</i> = RTLOLM – 100.0, & NaN <i>default</i> = 100.0
RTLOLM Ratio Low Limit		<i>range</i> = 0.0001 - RTHILM, & NaN <i>default</i> = 0.0001
BSHILM Bias High Limit		<i>range</i> = $\geq$ BSLOLM, & NaN <i>default</i> = 50.0
BSLOLM Bias Low Limit		<i>range</i> = $\leq$ BSHILM, & NaN <i>default</i> = -50.0
RATIO Ratio		<i>range</i> = RTLOLM to RTHILM <i>default</i> = 1.0
BIAS Bias Parameter		<i>range</i> = BSLOLM to BSHILM <i>default</i> = 0.0

Unit:	Tagname:	Engineer:	Date:
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PIDFORM	Interact	(Proportional + Integral) x (Derivative)
PID Controller Form	Ideal	(Proportional) + (Integral) + (Derivative)
CTLEQN	EqA	P, I & D act on Error
Control Equation Type	EqB	P & I act on Error, D acts on PV
	EqC	I acts on Error, P & D act on PV
	EqD	Integral-only control

PVTRACK PV Tracking Option	No Track Track	SP will never be set equal to PV Man mode or init. causes SP to track PV
CTLACTN Control Action	Reverse Direct	As PV increases, Output decreases As PV decreases, Output increases As PV increases, Output increases As PV decreases, Output decreases

N/A if CTLEQN = EqD.

Reg Control Page Display (continued)

GAINOPT
Gain (K) Option

Ext
NonLin
Gap
Lin

K (Pid)
Overall Gain In % Per %

KLIN
Linear Gain Factor

KGAP
Gap Gain Factor

GAPHI
Gap High Limit

GAPLO
Gap Low Limit

NLGAIN
Non Linear Gain

NLFM
Non Linearity Form

KEXT
External Gain Modifier

T1
Integral Time In Minutes

T2
Derivative Time In Minutes

K = KLIN*KEXT (gain from external source)
K = KLIN*((NLFM)+(NLGAIN*[PVP-SPP]/[100])
K = KLIN*KGAP if PV>(SP-GAPLO) or if
PV<(SP+GAPHI), else K = KLIN.
K = the value entered by the user

range = 0.0 – 240.0
default = Not applicable

range = 0.0 – 240.0
default = 1.0

range = 0.0 to 1.0
default = 1.0

range = >=0.0
default = 0.0

range = >=0.0
default = 0.0

range = 0.0 – 240.0
default = 0.0

K can go to zero
K cannot go to < 1

range = 0.0 to 240.0
default = 1.0

range = 0.0 – 1440.0
default = 0.0 (minutes)

range = 0.0 – 1440.0
default = 0.0 (minutes)

HPM PID CTL Algorithm

Unit:	Tagname:	Engineer:	Date:
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NOCINPTS		<i>range = 0 to 2 default = 1</i>
# Of Control Input Connections		
CISRC (Control-Input-Connection Source)		<u>Destination Parameter</u>
<u>Input#</u>	<u>CISRC (point.parameter)</u>	<u>(CIDSTN)</u>
1		PVAUTO
2		SP

Output Configuration Display

NOCOPTS		<i>range = 0 to 4 default = 1</i>
# Of Control Output Connections		
CODSTN (Control-Output-Connection Destination)		
<u>Output #</u> (N/A if NOCOPTS = 0)	<u>CODSTN (point.parameter)</u>	
1		
2		
3		
4		

(N/A if NOCOPTS > 0)

CVEUHI		<i>range = > = CVEULO default = 100.0</i>
Output Engineering Unit High Range		
CVEULO		<i>range = > = CVEUHI default = 0.0</i>
Output Engineering Unit Low Range		

OPHILM		<i>range = OPLOLM - 106.9 & NaN default = 105.0</i>
Output High Limit In Percent		
OPLOLM		<i>range = -6.9 - OPHILM, & NaN default = -5.0</i>
Output Low Limit In Percent		
SAFEOP		<i>range = -6.9 to 106.9, & NaN default = NaN</i>
Safe Output Limit In Percent		
OPMCHLM		<i>range = > = 0.0, and NaN default = 0.0</i>
Output Minimum Change In %		
OPROCLM		<i>range = > = 0.0, and NaN default = NaN</i>
OP Rate Of Change Limit In %		
\$OPTOL		<i>range = 0.0 to 106.9 default = 0.0.</i>
Output Tolerance in %		
OUTIND (R620 and later)	Direct	4 - 20mA = 0 – 100%
Reverse Output Indicator	Reverse	20 – 4mA = 0 – 100%
Note: With R670 and later, when this parameter is set to Reverse, the values of the displayed output and the related output parameters are changed. Refer to the definition of the OUTIND parameter in the PM Family Parameter Dictionary, XPM09.		

HPM PID CTL Algorithm

Unit:	Tagname:	Engineer:	Date:
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Alarming Display - N/A if PNTFORM = Componnt

As you enter alarm trip points and alarm priorities in the next section please refer to the following alarm priority definitions and the note.

NoAction	Alarm is not reported to the system and not annunciated (R670 & later)
Journal	Alarm is historized, not reported to US, not annunciated
Low	Alarm is historized, reported to US, and annunciated
High	Alarm is historized, reported to US, and annunciated
Emergency	Alarm is historized, reported to US, and annunciated
Printer	Not historized, reported to the printer, not annunciated
JnlPrint	Alarm historized, reported to the printer, not annunciated

Note: If you do not enter a trip point value for a given pair of trip-points and alarm-priorities, do not circle an alarm priority option for that pair (that is, no trip point, no alarm priority).

Alarming Display - N/A if PNTFORM = Componnt

\$AUXUNIT Auxiliary Unit	_ _ _	<i>range</i> = Any configured LCN Unit ID <i>default</i> = Null (- -)
\$ALDLOPT (R620 and later) Alarm Delay Option	None OffDlyLo OffDlyHi OffDlyEm OnDlyLo OnDlyHi OnDlyEm	Off Delay Lo Off Delay Hi Off Delay Emergency On Delay Lo On Delay Hi On Delay Emergency
\$DLBADAL (R681 and later) Delay for BADPV	None Offdelay	<i>range</i> = Offdelay, None <i>default</i> = None
DLTIME (R620 and later) Delay Time	_ _ _ _	<i>range</i> = 0 to 9999 seconds <i>default</i> = 0 seconds
\$REALARM (R660 and later) Re-alarming Interval	_ _ _ _	<i>range</i> = 0 to 9999 minutes <i>default</i> = 0 minutes
BADOCOPT BADOC Alarm Option	Off On	Bad Output Connection Alarm is not initially permitted Bad Output Connection Alarm is Initially permitted
BADOCPR BADOC Alarm Priority	NoAction Journal Low High Emergency JnlPrint Printer (R670 and later)	
OPALDB Output Alarm Deadband	_ _ _ _ _ _ _	<i>range</i> = 0 - 25.0 but $\leq$ (OPHITP-OPLOTP)/2 <i>default</i> = 5.0 Output Units
OPHITP Output High Alarm Trip Point	_ _ _ _ _ _ _	<i>range</i> = OPLOTP TO OPHILM and NaN <i>default</i> = NaN
OPHIPR Output High Alarm Priority	NoAction Journal Low High Emergency JnlPrint Printer (R670 and later)	
OPLOTP Output Low Alarm Trip Point	_ _ _ _ _ _ _	<i>range</i> = OPLOLM TO OPHITP and NaN <i>default</i> = NaN
OPLOPR Output Low Alarm Priority	NoAction Journal Low High Emergency JnlPrint Printer (R670 and later)	

HPM PID CTL Algorithm

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Unit:	Tagname:	Engineer:	Date:
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PVALDB PV Alarm Deadband In %	Half	One	Two	Three	Four	Five	EU	
PVALDBEU PV Alarm Deadband In EU	_ _ _ _ _ _ _							range > = 0.0 default = 1.0
PVHITP PV High Alarm Trip Point	_ _ _ _ _ _ _							range = PVLOTP - PVHHTP, & NaN default = NaN
PVHIPR PV High Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	JnlPrint	Printer	
\$HISELCT (R681 and later) PV High Selective Cutout	Cutout Allow							range = Cutout, Allow default = Cutout
PVLOTP PV Low Alarm Trip Point	_ _ _ _ _ _ _							range = PVLLTP - PVHITP, & NaN default = NaN
PVLOPR PV Low Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	JnlPrint	Printer	
\$LOSELCT (R681 and later) PV Low Selective Cutout	Cutout Allow							range = Cutout, Allow default = Cutout

Alarming Display – N/A if PNTFORM = Component

PVHHTP PV Hi Hi Alarm Trip Point	_ _ _ _ _ _ _							range = PVHITP to PVEUHI, & NaN default = NaN
PVHHPR PV High High Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
\$HHSELCT (R681 and later) PV High High Selective Cutout	Cutout Allow							range = Cutout, Allow default = Cutout
PVLLTP PV Low Low Alarm Trip Point	_ _ _ _ _ _ _							range = PVEULO – PVLOTP, & NaN default = NaN
PVLLPR PV Low Low Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
\$LLSELCT (R681 and later) PV Low Low Selective Cutout	Cutout Allow							range = Cutout, Allow default = Cutout
PVSGCHTP PV Signif Chg Alarm Trip Point	_ _ _ _ _ _ _							range = >=0.0, and NaN default = NaN
PVROCTP PV Pos Rate Of Chg Trip Point	_ _ _ _ _ _ _							range = >=0.0, and NaN default = NaN
PVROCPPR PV Pos Rate Of Chg Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	

[illegible]

HPM PID with External Reset Feedback CTL Algorithm

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Unit:	Tagname:	Engineer:	Date:
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Point Assignment Display

NAME Tag Name		<i>range = Not applicable</i> <i>default = Blank</i>
NODETYPE Node Type	PM APM HPM	A PM Point Is Being Configured An APM Point Is Being Configured An HPM Point Is Being Configured
PNTFORM Point Form	FuLL Componnt	Point is fully displayed and alarmed Pt. is partially displayed but not alarmed

PTDESC Point Descriptor		<i>range = Not applicable</i> <i>default = Blank</i>
EUDESC Engineering Units Descriptor		<i>range = Not applicable</i> <i>default = Blank</i>
KEYWORD Keyword Descriptor		<i>range = Not applicable</i> <i>default = Blank</i>
ASSOCDSP Associated Display		<i>range = Not applicable</i> <i>default = Blank</i>
\$CDETAIL (R620 and later) Point Custom Display		<i>range = Not applicable</i> <i>default = Blank</i>
UNIT Process Unit Point Is Assigned To		<i>range = Two Alphanumeric Characters</i> <i>default = Not applicable</i>
NTWKNUM NIM's UCN That This Point Is In		<i>range = 1 to 20</i> <i>default = Not applicable</i>
NODENUM HPMM's Address On The UCN		<i>range = 1 to 64</i> <i>default = Not applicable</i>
MODNUM I/O Module Number		<i>range = 0</i> <i>default = 0 (Control Processor Point)</i>
SLOTNUM Slot Number		<i>range = 1-250 (But <=NCTLSLOT)</i> <i>default = Not applicable</i>

N/A if PNTFORM = Componnt.

PRIMMOD Primary Module ID		<i>range = Not Applicable</i> <i>default = Null</i>
USERID User ID Reservation		Sixteen Character String <i>default = dashes</i>

Control Algorithm Display

CTLALGID Control Algorithm Identifier	Null PidErfb	No affect PID with External Reset Feedback
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PV Configuration Display

PVEUHI PV High Range In Engr Units		<i>range = >=PVEULO</i> <i>default = 100.0</i>
PVEULO PV Low Range In Engr Units		<i>range = <= PVEUHI</i> <i>default = 0.0</i>
PVFORMAT PV Decimal Point Format	DO D1	D2 D3 D0=9999. D1=999.9 D2=99.99 D3 = 9.999

HPM PID with External Reset Feedback CTL Algorithm

Unit:	Tagname:	Engineer:	Date:
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PVSRCOPT PV Source Option	Only Auto All	PV Source selection is not available Selection from PVSOURCE is available
PVSOURCE PV Source	Auto Man Sub	Field-wiring /memory -fetch supplies PV PV is supplied by Operator or program A value is substituted by a CL program
OVERVAL Overview Display Value	. . .	range = 0 to 100 default = 25

Mode Configuration Display

BADCTLOP Bad PV Mode Shed Option	No_Shed Shed Hold Shed Low Shed High Shed Safe	range = See HPM Parameter Reference Dictionary default = No_Shed
RCASOPT Remote Cascade Option	None Spc Ddc DdcRsp Rsp	Local cascade is the only valid cascade AM strategy stores to SP within SP limits AM Strategy stores to OP. No OP limits. AM strategy does Ddc and Rsp functions AM strategy stores to SP when this point is in Auto and is being initialized by its secondary. SP limits are applied.
SHEDTIME Remote Cascade Shed Time	. . .	range = 0 to 1000 (seconds) default = 0 (mode shed disabled)
SHEDMODE Shed Mode	Man Auto Bcas	Supervisory point will shed to Manual, Automatic, or Backup Cascade mode.
NMODE RegCtl Point's Normal Mode	None Man Auto Cas Bcas	No configured "normal" operating mode Manual is the "normal" operating mode Automatic is "normal" operating mode Cascade is "normal" operating mode Backup Cascade is "normal" operating mode

N/A if NMODE =None

NMODATTR Normal Mode Attribute	None Operator Program	Parameter NMODE sets "normal" mode Operator sets "normal" operating mode Program sets "normal" operating mode
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MODEPERM Mode Permissive	Permit NotPerm	Operator can change this point's mode Optr. cannot change this point's mode
\$MODESEL (R630 and later) Mode Selection List	Man Prog Auto Cas Bcas	Man, Auto, Cas and Bcas mode, and Prog mode attribute are allowed for this data point.
\$BADOCMD (R680 and later) Bad Output Connection Mode	SHEDMAN 15_SEC 60_SEC NO_SHED	Point will shed to Manual immediately Point will shed to Manual after 15 seconds Point will shed to Manual after 60 seconds Do not shed mode

HPM PID with External Reset

Feedback CTL Algorithm

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Unit:	Tagname:	Engineer:	Date:
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EXTSWOPT External Mode Switching	None Ems	No external mode switching is allowed Ext. source can change point's mode
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Setpoint Display

SPHILM SP High Limit		range = SPLOLM – SPEUHI, and NaN default = NaN
SPLOLM SP Low Limit		range = SPEULO – SPHILM, and NaN default = NaN
\$SPTOL Set Point Tolerance		range = ≥ 0.0 default = 0.0
SP Set Point In Engineering Units Of The PV		range = SPLOLM to SPHILM default = 0.0
SPOPT Set Point Option Asp is invalid if PNTFORM=Componnt.	None Tv Asp	No specialized options are available Target value processing Advisory deviation alarming
RBOPT Ratio Bias Option	NoRatBi FixRatBi AutoRat AutoBi	No ratio/bias is used to calculate the SP Fixed ratio (R) and fixed bias(B) are used R is back calculated during initialization B is back calculated during initialization
RTHILM Ratio High Limit	{ }	range = RTLOLM – 100.0, & NaN default = 100.0
RTLOLM Ratio Low Limit	{ }	range = 0.0001- RTHILM, & NaN default = 0.0001
BSHILM Bias High Limit	{ }	range = $\geq$ BSLOLM, & NaN default = 50.0
BSLOLM Bias Low Limit	{ }	range = $\leq$ BSHILM, & NaN default = -50.0
RATIO Ratio	{ }	range = RTLOLM to RTHILM default = 1.0
BIAS Bias Parameter	{ }	range = BSLOLM to BSHILM default = 0.0

Reg Control PidErFb Algo Display

PIDFORM PID Controller Form	Interact Ideal	(Proportional + Integral) x (Derivative) (Proportional) +(Integral) + (Derivative)
CTLEQN(Pid) Control Equation Type	EqA EqB EqC EqD	P, I & D act on Error P & I act on Error, D acts on PV I acts on Error, P & D act on PV Integral-only control
PVTRACK PV Tracking Option	No Track Track	SP will never be set equal to PV Man mode or init. Causes SP to track PV

HPM PID with External Reset Feedback CTL Algorithm

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Reg Control PidErfb Algo Display (continued)

N/A if CTLEQN = EqD.

CTLACTN Control Action	Reverse Direct	As PV increases, Output decreases As PV increases, Output increases
GAINOPT Gain (K) Option	Ext NonLin Gap Lin	K = K1*KEXT (gain from external source) K = K1*((NLFM)+(NLGAIN*[PVP-SPP]/[100])) K = KLIN*KGAP if PV>(SP-GAPLO) or if PV<(SP+GAPHI), else K = KLIN. K = the value entered by the user
K (Pid) Overall Gain In % Per %	_ _ _ _ _ _ _	range = 0.0 – 240.0 default = Not applicable
KLIN Linear Gain Factor	_ _ _ _ _ _ _	range = 0.0 – 240.0 default = 1.0
KGAP Gap Gain Factor	_ _ _ _	range = 0.0 to 1.0 default = 1.0
GAPHI Gap High Limit	_ _ _ _ _ _ _	range = >=0.0 default = 0.0
GAPLO Gap Low Limit	_ _ _ _ _ _ _	range = >=0.0 default = 0.0
NLGAIN Non Linear Gain	_ _ _ _ _ _ _	range = 0.0 – 240.0 default = 0.0
NLFM Non Linearity Form	NLFM = 0 NLFM = 1	K can go to zero K cannot go to < 1
KEXT External Gain Modifier	_ _ _ _ _ _ _	range = 0.0 to 240.0 default = 1.0
T1 Integral Time In Minutes	T1 _ _ _ _ _ _ _	range = 0.0 to 1440.0 default = 0.0 (minutes)
T2 Derivative Time In Minutes	T2 _ _ _ _ _ _ _	range = 0.0 to 1440.0 default = 0.0 (minutes)
K1 (PidErfb) External Reset Feedback Gain Factor	_ _ _ _	range = 0.0 to 1.0 default = 1.0

HPM PID with External Reset

Feedback CTL Algorithm

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Unit:	Tagname:	Engineer:	Date:
NOCINPTS #Of Control Input Connections		range = 3 to 4 default = 3	
CISRC (Control-Input-Connection Source)			Destination Parameter (CIDSTN)
<u>Input#</u>	<u>CISRC (point.parameter)</u>		
1		PVAUTO	
2		RFB	
3		TRFB	
4		SP	

Output Configuration Display

NOCOPTS # Of Control Output Connections		range = 0 to 4 default = 1
CODSTN (Control-Output-Connection Destination)		
<u>Output #</u>	<u>CODSTN (point.parameter)</u>	
1		
2		
3		
4		
CVEUHI CV High Limit In EUs		range = ≥ 0.0 default = 100.00
CVEULO CV Low Limit In EUs		range = ≥ 0.0 default = 0.0
OPHILM Output High Limit In Percent		range = OPLOLM – 106.9 & NaN default = 105.0
OPLOLM Output Low Limit In Percent		range = -6.9 – OPHILM, & NaN default = -5.0
SAFEOP Safe Output Limit In Percent		range = -6.9 to to 106.9 & NaN default = NaN
OPMCHLM Output Minimum Change In %		range = ≥ 0.0 , and NaN default = 0.0
OPROCLM OP Rate Of Change Limit In %		range = ≥ 0.0 , and NaN default = NaN
\$OPTOL Output Tolerance in %		range = 0.0 to 106.9 default = 0.0

HPM PID with External Reset Feedback CTL Algorithm

Unit:	Tagname:	Engineer:	Date:
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OUTIND (R620 and later) Reverse Output Indicator	Direct Reverse	4 - 20mA = 0 – 100% 20 – 4mA = 0 – 100% Note: With R670 and later, when this parameter is set to Reverse, the values of the displayed output and the related output parameters are changed. Refer to the definition of the OUTIND parameter in the PM Family Parameter Dictionary, XPM09.
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Alarming Display - N/A if PNTFORM = Componnt

As you enter alarm trip points and alarm priorities in the next section please refer to the following alarm priority definitions and the note.

NoAction	Alarm is not reported to the system and not annunciated (R670 & later)
Journal	Alarm is historized, not reported to US, not annunciated
Low	Alarm is historized, reported to US, and annunciated
High	Alarm is historized, reported to US, and annunciated
Emergency	Alarm is historized, reported to US, and annunciated Not
Printer	historized, reported to the printer, not annunciated Alarm
JnlPrint	historized, reported to the printer, not annunciated

Note: If you do not enter a trip point value for a given pair of trip-points and alarm-priorities, do not circle an alarm priority option for that pair (that is, no trip point, no alarm priority).

Alarming Display - N/A if PNTFORM = Componnt

\$AUXUNIT Auxiliary Unit	_ _	<i>range</i> = Any configured LCN Unit ID <i>default</i> = Null (- -)
\$ALDLOPT (R620 and later) Alarm Delay Option	None OffDlyLo OffDlyHi OffDlyEm OnDlyLo OnDlyHi OnDlyEm	Off Delay Lo Off Delay Hi Off Delay Emergency On Delay Lo On Delay Hi On Delay Emergency
\$DLBADAL (R681 and later) Delay for BADPV	None Offdelay	<i>range</i> = Offdelay, None <i>default</i> = None
DLTIME (R620 and later) Delay Time	_ _ _	<i>range</i> = 0 to 9999 seconds <i>default</i> = 0 seconds
\$REALARM (R660 and later) Re-alarms Interval	_ _ _	<i>range</i> = 0 to 9999 minutes <i>default</i> = 0 minutes
BADOCOPT BADOC Alarm Option	Off On	Bad Output Connection Alarm is not initially permitted Bad Output Connection Alarm is Initially permitted
BADOCPR BADOC Alarm Priority	NoAction Journal Low High Emergency JnlPrint Printer (R670 and later)	
OPALDB Output Alarm Deadband	_ _ _ _	<i>range</i> = 0 - 25.0 but ≤ (OPHITP- OPLOTP)/2 <i>default</i> = 5.0 Output Units

HPM PID With External Reset

Feedback CTL Algorithm

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OPHITP Output High Alarm Trip Point		<i>range</i> = OPLOTP TO OPHILM and NaN <i>default</i> = NaN
OPHIPR Output High Alarm Priority	NoAction Journal Low High Emergency JnlPrint Printer (R670 and later)	
OPLOTP Output Low Alarm Trip Point		<i>range</i> = OPLOLM TO OPHITP and NaN <i>default</i> = NaN
OPLOPR Output Low Alarm Priority	NoAction Journal Low High Emergency JnlPrint Printer (R670 and later)	
PVALDB PV Alarm Deadband In %	Half One Two Three Four Five EU	
PVALDBEU PV Alarm Deadband In EU		<i>range</i> > = 0.0 <i>default</i> = 1.0
PVHITP PV High Alarm Trip Point		<i>range</i> = PVLOTP - PVHHTP, & NaN <i>default</i> = NaN
PVHIPR PV High Alarm Priority	NoAction Journal Low High Emergency JnlPrint Printer (R670 and later)	
\$HISELCT (R681 and later) PV High Selective Cutout	Cutout Allow	<i>range</i> = Cutout, Allow <i>default</i> = Cutout
PVLOTP PV Low Alarm Trip Point		<i>range</i> = PVLLTP - PVHITP, & NaN <i>default</i> = NaN
PVLOPR PV Low Alarm Priority	NoAction Journal Low High Emergency JnlPrint Printer (R670 and later)	
\$LOSELCT (R681 and later) PV Low Selective Cutout	Cutout Allow	<i>range</i> = Cutout, Allow <i>default</i> = Cutout
PVHHTP PV Hi Hi Alarm Trip Point		<i>range</i> = PVHITP to PVEUHI, & NaN <i>default</i> = NaN
PVHHPR PV High High Alarm Priority	NoAction Journal Low High Emergency Printer JnlPrint (R670 and later)	
\$HHSELCT (R681 and later) PV High High Selective Cutout	Cutout Allow	<i>range</i> = Cutout, Allow <i>default</i> = Cutout
PVLLTP PV Low Low Alarm Trip Point		<i>range</i> = PVEULO – PVLOTP, & NaN <i>default</i> = NaN
PVLLPR PV Low Low Alarm Priority	NoAction Journal Low High Emergency Printer JnlPrint (R670 and later)	
\$LLSELCT (R681 and later) PV Low Low Selective Cutout	Cutout Allow	<i>range</i> = Cutout, Allow <i>default</i> = Cutout

Alarming Display – N/A if PNTFORM = Component

Parameter	Default	Range	Unit	Alarm	Priority	Event	Printer	JnlPrint
PVSGCHTP PV Signif Chg Alarm Trip Point	_____	range = >=0.0, and NaN default = NaN						
PVROCPTP PV Pos Rate Of Chg Trip Point	_____	range = >=0.0, and NaN default = NaN						
PVROCPPR PV Pos Rate Of Chg Alarm Priority	NoAction (R670 and later)		Journal	Low	High	Emergency	Printer	JnlPrint
\$RPSLECT (R681 and later) PV Positive Rate of Change Selective Cutout	Cutout Allow	range = Cutout, Allow default = Cutout						
PVROCNTP PV Neg Rate Of Chg Trip Point	_____	range = >=0.0, and NaN default = NaN						
PVROCNP PV Neg Rate Of Chg Alarm Priority	NoAction (R670 and later)		Journal	Low	High	Emergency	Printer	JnlPrint
\$RNSLECT (R681 and later) PV Negative Rate of Change Selective Cutout	Cutout Allow	range = Cutout, Allow default = Cutout						
BADPVPR Bad-PV Alarm Priority	NoAction (R670 and later)		Journal	Low	High	Emergency	Printer	JnlPrint
DEVHITP Deviation High Alarm Trip Point	_____	range = >=0.0, and NaN default = NaN						
DEVHIPR Deviation High Alarm Priority	NoAction (R670 and later)		Journal	Low	High	Emergency	Printer	JnlPrint
\$DHSELECT (R681 and later) Deviation High Selective Cutout	Cutout Allow	range = Cutout, Allow default = Cutout						
DEVLOTP Deviation Low Alarm Trip Point	_____	range = >=0.0, and NaN default = NaN						
DEVLOPR Deviation Low Alarm Priority	NoAction (R670 and later)		Journal	Low	High	Emergency	Printer	JnlPrint
\$DLSELECT (R681 and later) Deviation Low Selective Cutout	Cutout Allow	range = Cutout, Allow default = Cutout						
ADVDEVTP Advisory Deviation Alarm Trip Point (When SPOPT = ASP)	_____	range = >=0.0, and NaN default = NaN						
ADVDEVPR Advisory Deviation Alarm Priority (When SPOPT = ASP)	NoAction (R670 and later)		Journal	Low	High	Emergency	Printer	JnlPrint
ALENBST Point Alarm Enable Status								
	Enable				Displayed	Logged	Reported To EIP	
	Disable				Yes	Yes	Yes	
	Inhibit				No	Yes	Yes	
					No	No	No	

HPM PID with Feedforward CTL Algorithm

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Point Assignment Display

NAME Tag Name	_____	<i>range</i> = Not applicable <i>default</i> = Blank
NODETYPE Node Type	PM APM HPM	A PM Point Is Being Configured An APM Point Is Being Configured An HPM Point Is Being Configured
PNTFORM Point Form	FuLL Componnt	Point is fully displayed and alarmed Pt. is partially displayed but not alarmed

PTDESC Point Descriptor	_____	<i>range</i> = Not applicable <i>default</i> = Blank
EUDESC Engineering Units Descriptor	_____	<i>range</i> = Not applicable <i>default</i> = Blank
KEYWORD Keyword Descriptor	_____	<i>range</i> = Not applicable <i>default</i> = Blank
ASSOCDSP Associated Display	_____	<i>range</i> = Not applicable <i>default</i> = Blank
\$CDETAIL (R620 and later) Point Custom Display	_____	<i>range</i> = Not applicable <i>default</i> = Blank
UNIT Process Unit Point Is Assigned To	____	<i>range</i> = Two Alphanumeric Characters <i>default</i> = Not applicable
NTWKNUM NIM's UCN That This Point Is In	____	<i>range</i> = 1 to 20 <i>default</i> = Not applicable
NODENUM HPMM's Address On The UCN	____	<i>range</i> = 1 to 64 <i>default</i> = Not applicable
MODNUM I/O Module Number	____	<i>range</i> = 0 <i>default</i> = 0 (Control Processor Point)
SLOTNUM Slot Number	____	<i>range</i> = 1-250 (but <= NCTLSLOT) <i>default</i> = Not applicable

N/A if PNTFORM = Componnt.

PRIMMOD Primary Module ID	_____	<i>range</i> = Not Applicable <i>default</i> = Null
USERID User ID Reservation	_____	Sixteen Character String <i>default</i> = dashes

Control Algorithm Display

CTLALGID Control Algorithm Identifier	NULL PidFF	No affect PID with Feedforward
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HPM PID with Feedforward CTL Algorithm

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PV Configuration Display

PVEUHI PV High Range In Engr Units	_ _ _ _	<i>range = >= PVEULO I</i> <i>default = 100.0</i>
PVEULO PV Low Range In Engr Units	_ _ _ _	<i>range = <= PVEUHI</i> <i>default = 0.0</i>
PVFORMAT PV Decimal Point Format	D0 D2 D1 D3	D0 = 9999. D2 = 99.99 D1 = 999.9 D3 = 9.999

PVSRCOPT PV Source Option	OnlyAuto All	PV source selection is not available Selection from PVSOURCE is available
PVSOURCE PV Source	Auto Man Sub	Field-wiring/memory-fetch supplies PV PV is supplied by Operator or program A value is substituted by a CL program
OVERVAL Overview Display Value	_ _ _	<i>range = 0 to 100</i> <i>default = 25</i>

Mode Configuration Display

BADCTLOP Bad PV Mode Shed Option	No_Shed Shed Hold Shed Low Shed High Shed Safe	<i>range = See HPM Parameter Reference Dictionary</i> <i>default = No_Shed</i>
RCASOPT Remote Cascade Option	None Spc Ddc DdcRsp Rsp	Local cascade is the only valid cascade AM strategy stores to SP within SP limits AM strategy stores to OP. No OP limits. AM strategy does Ddc and Rsp functions AM strategy stores to SP when this point is in Auto and is being initialized by its secondary. SP limits are applied.
SHEDTIME Remote Cascade Shed Time	_ _ _	<i>range = 0 to 1000 (seconds)</i> <i>default = 0 (mode shed disabled)</i>
SHEDMODE (If Shedtime >1) Shed Mode	Man Auto Bcas	Supervisory point will shed to Manual, Automatic, or Backup Cascade mode.
NMODE(RegCtl) RegCtl Point's Normal Mode	None Man Auto Cas Bcas	No configured "normal" operating mode Manual is the "normal" operating mode Automatic is "normal" operating mode Cascade is "normal" operating mode

HPM PID with Feedforward CTL Algorithm

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N/A if NMODE = None

NMODATTR Normal Mode Attribute	None Operator Program	Parameter NMODE sets "normal" mode Operator sets "normal" operating mode Program sets "normal" operating mode
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MODEPERM Mode Permissive	Permit NotPerm	Operator can change this point's mode Optr. cannot change this point's mode
\$MODESEL (R630 and later) Mode Selection List	Man Prog Auto Cas Bcas	Man, Auto, Cas and Bcas mode, and Prog mode attribute are allowed for this data point.
\$BADOCMD (R680 and later) Bad Output Connection Mode	\$HEDMAN 15_SEC 60_SEC NO_SHED	Point will shed to Manual immediately Point will shed to Manual after 15 seconds Point will shed to Manual after 60 seconds Do not shed mode
EXTSWOPT External Mode Switching	None Ems	No external mode switching is allowed Ext. source can change point's mode

Setpoint Display

SPHILM SP High Limit	_.	<i>range</i> = SPLOLM – SPEUHI, and NaN <i>default</i> = NaN
SPLOLM SP Low Limit	_.	<i>range</i> = SPEULO – SPHILM, and NaN <i>default</i> = NaN
\$SPTOL Set Point Tolerance	_.	<i>range</i> = >=0.0 <i>default</i> = 0.0
SP Set Point In Engineering Units Of The PV	_.	<i>range</i> = SPLOLM to SPHILM <i>default</i> = 0.0

HPM PID With Feedforward CTL Algorithm

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Setpoint Display (Continued)

SPOINT Set Point Option Asp is invalid if PNTFORM=Componnt.	None Tv Asp	No specialized option are available Target value processing Advisory deviation alarming
RBOPT Ratio Bias Option	NoRatBi FixRatBi AutoRat AutoBi	No ratio/bias is used to calculate the SP Fixed ratio (R) and fixed bias (B) are used R is back calculated during initialization B is back calculated during initialization
RTHILM Ratio High Limit	{ 	<i>range</i> = RTLOLM – 100.0, & NaN <i>default</i> = 100.0
RTLOLM Ratio Low Limit	{ 	<i>range</i> = 0.0001- RTHILM, & NaN <i>default</i> = 0.0001
BSHILM Bias High Limit	{ 	<i>range</i> = >=BSLOLM, & NaN <i>default</i> = 50.0
BSLOLM Bias Low Limit	{ 	<i>range</i> = <=BSHILM, & NaN <i>default</i> = 50.0
RATIO Ratio	{ 	<i>range</i> = RTLOLM to RTHILM <i>default</i> = 1.0
BIAS Bias Parameter	{ 	<i>range</i> = BSLOLM to BSHILM <i>default</i> = 0.0

Reg Control PidFf Algo Display

PIDFORM PID Controller Form	Interact Ideal	(Proportional + Integral) x (Derivative) (Proportional) + (Integral) + (Derivative)
CTLEQN(Pid) Control Equation Type	EqA EqB EqC EqD	P, I & D act on Error P & I act on Error, D acts on PV I acts on Error, P & D act on PV Integral-only control
PVTRACK PV Tracking Option	No Track Track	SP will never be set equal to PV Man mode or init. Causes SP to track PV
CTLACTN Control Action	Reverse Direct	As PV increases, Output decreases As PV increases, Output increases

HPM PID with Feedforward CTL Algorithm

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Reg Control Pid Algo Display (continued).

N/A if CTLEQN = EqD

GAINOPT Gain (K) Option	Ext NonLin Gap	K = KFF*KEXT (gain from external source) K = KFF*((NLFM)+(NLGAIN*[PVP-SPP]/[100]) K = KLIN*KGAP if PV>(SP-GAPLO) or if PV<(SP+GAPHI), else K = KLIN. K = the value entered by the user	
	Lin		
K(Pid) Overall Gain In % Per %	_ _ _ _	range = 0.0 – 240.0 default = Not applicable	
KLIN Linear Gain Factor	_ _ _ _	range = 0.0 – 240.0 default = 1.0	
KGAP Gap Gain Factor	_ _	range = 0.0 to 1.0 default = 1.0	
GAPHI Gap High Limit	_ _ _ _ _ _ _	range = >=0.0 default = 0.0	
GAPLO Gap Low Limit	_ _ _ _ _ _ _	range = >=0.0 default = 0.0	
NLGAIN Non Linear Gain	_ _ _ _	range = 0.0 – 240.0 default = 0.0	
NLFM Non Linearity Form	0 1	K can go to zero K cannot go to < 1	
KEXT External Gain Modifier	_ _ _ _	range = 0.0 to 240.0 default = 1.0	
T1 Integral Time In Minutes	_ _ _ _ _	range = 0.0 to 1440.0 default = 0.0 (minutes)	
T2 Derivative Time In Minutes	_ _ _ _ _	range = 0.0 to 1440.0 default = 0.0 (minutes)	
FFOPT Feed Forward Type	Add Multiply	(Scaled Feedforward) + (Feedback) (Feedback)x(scaled, Biased Feedfwd)	
BFF Feed Forward Input Bias	_ _ _ _ _ _ _	range = >=0.0 default = 0.0	
KFF Gain For Feed Forward Input	_ _ _	range = >=0.0 to <=1.0 default = 1.0	
NOCINPTS #Of Control Input Connections	_	range = 0 to 3 default = 2	

OPHILM		<i>range</i> = OPLOLM – 106.9 & NaN
Output High Limit In Percent	_ _ _ _ _ _ _	<i>default</i> = 105.0
OPLOLM		<i>range</i> = -6.9 – OPHILM, & NaN
Output Low Limit In Percent	_ _ _ _ _ _ _	<i>default</i> = -5.0
SAFEOP		<i>range</i> = -6.9 to 106.9, & NaN
Safe Output Limit In Percent	_ _ _ _ _ _ _	<i>default</i> = NaN
OPMCHLM		<i>range</i> = >=0.0, and NaN
Output Minimum Change In %	_ _ _ _ _ _ _	<i>default</i> = 0.0
OPROCLM		<i>range</i> = >=0.0, and NaN
OP Rate Of Change Limit In %	_ _ _ _ _ _ _	<i>default</i> = 0.0
\$OPTOL		<i>range</i> = 0.0 to 106.9
Output Tolerance in %	_ _ _ _ _ _ _	<i>default</i> = 0.0
OUTIND (R620 and later)	Direct	4 - 20mA = 0 - 100%
Reverse Output Indicator	Reverse	20 - 4mA = 0 - 100%
Note: With R670 and later, when this parameter is set to Reverse, the values of the displayed output and the related output parameters are changed. Refer to the definition of the OUTIND parameter in the PM Family Parameter Dictionary, XPM09.		

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Alarming Display – N/A if PNTFORM = Componnt
As you enter alarm trip points and alarm priorities in the next section please refer to the following alarm Priority definitions and the note.

NoAction	Alarm is not reported to the system and not annunciated (R670 & later)
Journal	Alarm is historized, not reported to US, not annunciated
Low	Alarm is historized, reported to US, and annunciated
High	Alarm is historized, reported to US, and annunciated
Emergency	Alarm is historized, reported to US, and annunciated
Printer	Not historized, reported to the printer, not annunciated
JnlPrint	Alarm historized, reported to the printer, not annunciated

Note: If you do not enter a trip point value for a given pair of trip-points and alarm-priorities, do not Circle an alarm priority option for that pair (that is, no trip point, no alarm priority).

Alarming Display – N/A if PNTFORM = Componnt

\$AUXUNIT Auxiliary Unit	_ _	<i>range = Any configured LCN Unit ID default = Null (--)</i>
\$ALDLOPT (R620 and later) Alarm Delay Option	None OffDlyLo OffDlyHi OffDlyEm OnDlyLo OnDlyHi OnDlyEm	Off Delay Lo Off Delay Hi Off Delay Emergency On Delay Lo On Delay Hi On Delay Emergency
\$DLBADAL (R681 and later) Delay for BADPV	None Offdelay	<i>range = Offdelay, None default = None</i>
DLTIME (R620 and later) Delay Time	_ _ _ _	<i>range = 0 to 9999 seconds default = 0 seconds</i>
\$REALARM (R660 and later) Re-arming Interval	_ _ _ _	<i>range = 0 to 9999 minutes default = 0 minutes</i>
BADOCPT BADOC Alarm Option	Off On	Bad Output Connection Alarm is not initially permitted Bad Output Connection Alarm is initially permitted
BADOCPR BADOC Alarm Priority	NoAction Journal Low High Emergency Printer JnlPrint (R670 and later)	
OPALDB Output Alarm Deadband	_ _ _ _ _ _ _	<i>range = 0 - 25.0 but ≤ (OPHITP-OPLOTP)/2 default = 5.0 Output Units</i>
OPHITP Output High Alarm Trip Point	_ _ _ _ _ _ _	<i>range = OPLOTP TO OPHILM and NaN default = NaN</i>
OPHIPR Output High Alarm Priority	NoAction Journal Low High Emergency Printer JnlPrint (R670 and later)	
OPLOTP Output Low Alarm Trip Point	_ _ _ _ _ _ _	<i>range = OPLOLM TO OPHITP and NaN default = NaN</i>
OPLOPR Output Low Alarm Priority	NoAction Journal Low High Emergency Printer JnlPrint (R670 and later)	
PVALDB PV Alarm Deadband In %	Half One Two Three Four Five EU	
PVALDBEU PV Alarm Deadband in EU	_ _ _ _	<i>range ≥ 0.0 default = 1.0</i>

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range = PVLOTP - PVHHTP, & NaN
default = NaN

Alarming Display – N/A if PNTFORM = Componnt. (Continued)

default = Cutout

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Unit:	Tagname:	Engineer:	Date:
BADPVPR Bad-PV Alarm Priority	<i>NoAction</i> (R670 and later)	Journal Low	High Emergency Printer JnlPrint
DEVHITP Deviation High Alarm Trip Point	_ _ _ _ _ _ _ _ _		<i>range = >=0.0, and NaN</i> <i>default = NaN</i>
DEHVHRP Deviation High Alarm Priority	<i>NoAction</i> (R670 and later)	Journal Low	High Emergency Printer JnlPrint
\$DHSELCT (R681 and later) Deviation High Selective Cutout	<i>Cutout</i> Allow		<i>range = Cutout, Allow</i> <i>default = Cutout</i>
DEVLOTP Deviation Low Alarm Trip Point	_ _ _ _ _ _ _ _ _		<i>range = >=0.0, and NaN</i> <i>default = NaN</i>
DEVLOPR Deviation Low Alarm Priority	<i>NoAction</i> (R670 and later)	Journal Low	High Emergency Printer JnlPrint
\$DLSELCT (R681 and later) Deviation Low Selective Cutout	<i>Cutout</i> Allow		<i>range = Cutout, Allow</i> <i>default = Cutout</i>
ADVDEVTP Advisory Deviation Alarm Trip Point (When SPOPT = ASP)	_ _ _ _ _ _ _ _ _		<i>range = >=0.0, and NaN</i> <i>default = NaN</i>
ADVDEVPR Advisory Deviation Alarm Priority (When SPOPT = ASP)	<i>NoAction</i> (R670 and later)	Journal Low	High Emergency Printer JnlPrint
ALENBST Point Alarm Enable Status	<i>Enable</i> Disable Inhibit		<u>Displayed</u> Yes No No <u>Logged</u> Yes No <u>Reported to EIP</u> Yes Yes No

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CTLALGID Control Algorithm Identifier	<i>Null</i> PosProp	No affect Position Proportional
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HPM Position Proportional CTL Algorithm

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PV Configuration Display

PVEUHI PV High Range In Engr Units	_ _ _ _	<i>range = > = PVEULO</i> <i>default = 100.0</i>
PVEULO PV Low Range In Engr Units	_ _ _ _	<i>range = < = to PVEUHI</i> <i>default = 0.0</i>
PVFORMAT PV Decimal Point Format	D0 D2 D1 D3	D0 = 9999. D2 = 99.99 D1 = 999.9 D3 = 9.999

PVSRCOPT PV Source Option	OnlyAuto All	PV source selection is not available Selection from PVSOURCE is available
PVSOURCE PV Source	Auto Man Sub	Field-wiring/memory-fetch supplies PV PV is supplied by Operator or program A value is substituted by a CL program
OVERVAL Overview Display Value	_ _ _	<i>range = 0 to 100</i> <i>default = 25</i>

Mode Configuration Display

BADCTLOP Bad PV Mode Shed Option	No_Shed Shed Hold Shed Low Shed High Shed Safe	<i>range = Not applicable</i> <i>default = No_Shed</i>
RCASOPT Remote Cascade Option	None Spc	Local cascade is the only valid cascade AM strategy stores to SP within SP limits
SHEDTIME Remote Cascade Shed Time	_ _ _	<i>range = 0 to 1000 (seconds)</i> <i>default = 0 (mode shed disabled)</i>
SHEDMODE Shed Mode	Man Auto Bcas	Supervisory point will shed to Manual Supervisory point will shed to Automatic Supervisory point will shed to Backup Cascade mode
NMODE (RegCtl) RegCtl Point's Normal Mode	None Man Auto Cas Bcas	No configured "normal" operating mode Manual is the "normal" operating mode Automatic is "normal" operating mode Cascade is "normal" operating mode Backup Cascade is "normal" operating mode

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N/A if NMODE = None

NMODATTR	None	Parameter NMODE sets “normal” mode
Normal Mode Attribute	Operator	Operator sets “normal” operating mode
	Program	Program sets “normal” operating mode

MODEPERM Mode Permissive	<i>Permit</i> NotPerm	Operator can change this point's mode Optr. cannot change this point's mode
\$MODESEL (R630 and later) Mode Selection List	<i>Man</i> <i>Auto</i> <i>Cas</i>	<i>Prog</i> <i>Bcas</i> Man, Auto, Cas and Bcas mode, and Prog mode attribute are allowed for this data point.
\$BADOCMD (R680 and later) Bad Output Connection Mode	<i>\$HEDMAN</i> 15_SEC 60_SEC NO_SHED	Point will shed to Manual immediately Point will shed to Manual after 15 seconds Point will shed to Manual after 60 seconds Do not shed mode
EXTSWOPT External Mode Switching	<i>None</i> Ems	No external mode switching is allowed Ext. source can change point's mode

Setpoint Display

SPHILM SP High Limit		<i>range</i> = SPLOLM - SPEUHI, and NaN <i>default</i> = NaN
SPLOLM SP Low Limit		<i>range</i> = SPEULO -SPHILM, and NaN <i>default</i> = NaN
\$SPTOL Set Point Tolerance		<i>range</i> = ≥ 0.0 <i>default</i> = 0.0
SP Set Point In Engineering Units Of The PV		<i>range</i> = SPLOLM to SPHILM <i>default</i> = 0.0
SPOPT Set Point Option	None Tv Asp	No specialized options are available Target value processing Advisory deviation alarming

Reg Control Posn/Prop Algo Display

DEADBAND		<i>range</i> = 0.0 to 100.0
Dead Band In % Of Full Scale		<i>default</i> = 5.0%
CYCLETIM		<i>range</i> = 0.25 to 1000.0
PosProp Output Cycle Time In Seconds		<i>default</i> = 10.0 (seconds)
K		<i>range</i> = 0.0 to 10.0
Gain Value		<i>default</i> = 1.0
RAISRATE		<i>range</i> = >=0.0
Raise OP Stroke Rate In %/Sec		<i>default</i> = 100.0

HPM Position Proportional CTL Algorithm

Unit:	Tagname:	Engineer:	Date:
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Reg Control Posn/Prop Algo Display (Continued)

LOWRRATE				<i>range = >=0.0 percent/second</i>
Lower OP Stroke Rate In %/Sec				<i>default = 100.0 percent/second</i>
MAXPULSE				<i>range = Minpulse to 60, NaN</i>
Maximum Pulse Time Limit				<i>default = 60.0</i>
MINPULSE				<i>range = N/A</i>
Minimum Pulse Time Limit				<i>default = 0.0</i>
RP				<i>range = .01-100.0</i>
Pulse Ratio				<i>default = 1.0</i>
DEADTIME				<i>range = 0.0-60.0</i>
Raise Dead Time Value				<i>default = 0.0 Seconds</i>
RT				<i>range = .01-100.0</i>
Dead Time Ratio				<i>default = 1.0</i>
RAISDSTN				
Raise Output Pulse Destination		Point.Parameter		<i>range = ONPULSE-OFFPULSE</i>
				<i>default = Null.null</i>
LOWRDSTN				
Lower OP Pulse Destination		Point.Parameter		<i>range = ONPULSE-OFFPULSE</i>
				<i>default = Null.null</i>
MANOPTIM				<i>range = 0.0 to 60.0 seconds</i>
Manual Output Pulse Time				<i>default = 1.0 second</i>
OPHISRC				
OPHIFL Input Connection		Point.Parameter		<i>range = See Parameter Ref.Dictionary</i>
				<i>default = Null.null</i>
OPHOSRC				
OPHOFL Input Connection		Point.Parameter		<i>range = See Parameter Ref.Dictionary</i>
				<i>default = Null.null</i>
SAFOPCMD				<i>range = Not Applicable</i>
Safe OP Command	Idle Raise Lower			<i>default = Idle</i>
LMSRC				
Local Manual Source		Point.Parameter		<i>range = See Parameter Ref. Dictionary</i>
				<i>default = Null.null</i>
NOCINPTS				<i>range = 0 to 2</i>
# Of Control Input Connections				<i>default = 0</i>
CISRC (Control-Input-Connection Source)				
<u>Input#</u>		<u>CISRC (point.parameter)</u>		<u>Destination Parameter</u>
				<u>(CIDSTN)</u>
1				PVAUTO
2				SP

HPM Position Proportional CTL Algorithm

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Unit:	Tagname:	Engineer:	Date:
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Alarming Display - N/A if PNTFORM = Componnt

As you enter alarm trip points and alarm priorities in the next section please refer to the following alarm priority definitions and the note.

NoAction	Alarm is not reported to the system and not annunciated (R670 & later)
Journal	Alarm is historized, not reported to US, not annunciated
Low	Alarm is historized, reported to US, and annunciated
High	Alarm is historized, reported to US, and annunciated
Emergency	Alarm is historized, reported to US, and annunciated
Printer	Not historized, reported to the printer, not annunciated
JnlPrint	Alarm historized, reported to the printer, not annunciated

Note: If you do not enter a trip point value for a given pair of trip-points and alarm-priorities, do not circle an alarm priority option for that pair (that is, no trip point, no alarm priority).

Alarming Display - N/A if PNTFORM = Componnt

\$AUXUNIT Auxiliary Unit	_ _	<i>range</i> = Any configured LCN Unit ID <i>default</i> = Null (- -)
\$ALDLOPT (R620 and later) Alarm Delay Option	None OffDlyLo OffDlyHi OffDlyEm OnDlyLo OnDlyHi OnDlyEm	Off Delay Lo Off Delay Hi Off Delay Emergency On Delay Lo On Delay Hi On Delay Emergency
\$DLBADAL (R681 and later) Delay for BADPV	None Offdelay	<i>range</i> = Offdelay, None <i>default</i> = None
DLTIME (R620 and later) Delay Time	_ _ _ _	<i>range</i> = 0 to 9999 seconds <i>default</i> = 0 seconds
\$REALARM (R660 and later) Re-alarming Interval	_ _ _ _	<i>range</i> = 0 to 9999 minutes <i>default</i> = 0 minutes
BADOCOPT BADOC Alarm Option	Off On	Bad Output Connection Alarm is not initially permitted Bad Output Connection Alarm is initially permitted
BADOCPR BADOC Alarm Priority	NoAction Journal Low High Emergency Printer JnlPrint (R670 and later)	
PVALDB PV Alarm Deadband In %	Half One Two Three Four Five EU	
PVALDBEU PV Alarm Deadband In EU	_ _ _ _	<i>range</i> > = 0.0 <i>default</i> = 1.0
PVHITP PV High Alarm Trip Point	_ _ _ _ _ _ _	<i>range</i> = PVLOTP - PVHHTP, & NaN <i>default</i> = NaN
PVHIPR PV High Alarm Priority	NoAction Journal Low High Emergency JnlPrint Printer (R670 and later)	
\$HISELCT (R681 and later) PV High Selective Cutout	Cutout Allow	<i>range</i> = Cutout, Allow <i>default</i> = Cutout
PVLOTP PV Low Alarm Trip Point	_ _ _ _ _ _ _	<i>range</i> = PVLLTP - PVHITP, & NaN <i>default</i> = NaN

Unit:	Tagname:	Engineer:	Date:
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Alarming Display – N/A if PNTFORM = Component. (Continued)

Variable	Default	Units	Alarm	Priority	High	Emergency	JnlPrint	Printer
PVLOPR PV Low Alarm Priority	NoAction (R670 and later)	Journal	Low		High	Emergency	JnlPrint	Printer
\$LOSELCT (R681 and later) PV Low Selective Cutout	Cutout Allow					<i>range</i> = Cutout, Allow <i>default</i> = Cutout		
PVHHTP PV Hi Hi Alarm Trip Point	_____					<i>range</i> = PVHITP to PVEUHI, & NaN <i>default</i> = NaN		
PVHHPR PV High High Alarm Priority	NoAction (R670 and later)	Journal	Low		High	Emergency	Printer	JnlPrint
\$HHSELCT (R681 and later) PV High High Selective Cutout	Cutout Allow					<i>range</i> = Cutout, Allow <i>default</i> = Cutout		
PVLLTP PV Low Low Alarm Trip Point	_____					<i>range</i> = PVEULO - PVLOTP, & NaN <i>default</i> = NaN		
PVLLPR PV Low Low Alarm Priority	NoAction (R670 and later)	Journal	Low		High	Emergency	Printer	JnlPrint
\$LLSELCT (R681 and later) PV Low Low Selective Cutout	Cutout Allow					<i>range</i> = Cutout, Allow <i>default</i> = Cutout		
PVSGCHTP PV Significant Change Alarm Trip Point	_____					<i>range</i> = >=0.0, and NaN <i>default</i> = NaN		
PVROCTP PV Pos Rate Of Chg Trip Point	_____					<i>range</i> = >=0.0, and NaN <i>default</i> = NaN		
PVROCPPR PV Pos Rate Of Chg Alarm Priority	NoAction (R670 and later)	Journal	Low		High	Emergency	Printer	JnlPrint
\$RPSELCT (R681 and later) PV Positive Rate of Change Selective Cutout	Cutout Allow					<i>range</i> = Cutout, Allow <i>default</i> = Cutout		
PVROCNTP PV Neg Rate Of Chg Trip Point	_____					<i>range</i> = >=0.0, and NaN <i>default</i> = NaN		
PVROCNP PV Neg Rate Of Chg Alarm Priority	NoAction (R670 and later)	Journal	Low		High	Emergency	Printer	JnlPrint
\$RNSELCT (R681 and later) PV Negative Rate of Change Selective Cutout	Cutout Allow					<i>range</i> = Cutout, Allow <i>default</i> = Cutout		

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Unit:	Tagname:	Engineer:	Date:
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[illegible]

PVEUHI	_____		<i>range = >= PVEULO</i>	
PV High Range In Engr Units	_____		<i>default = 100.0</i>	
PVEULO	_____		<i>range = <= PVEUHI</i>	
PV Low Range In Engr Units	_____		<i>default = 0.0</i>	
PVFORMAT	DO	D2	D0=9999.	D2=99.99
PV Decimal Point Format	D1	D3	D1=999.9	D3 = 9.999

NOCINPTS	_	<i>range = 0 to 1</i>
# Of Control Input Connections		<i>default = 1</i>
CISRC (Control-Input-Connection Source)		<u>Destination Parameter</u>
<u>Input #</u>	<u>CISRC (point.parameter)</u>	<u>(CIDSTN)</u>
1	_	PV

Note: If you do not enter a trip point value for a given pair of trip-points and alarm-priorities, do not circle an alarm priority option for that pair (that is, no trip point, no alarm priority).

HPM Ramp Soak CTL Algorithm

Unit:	Tagname:	Engineer:	Date:
\$AUXUNIT Auxiliary Unit _ _ _ _ <i>range</i> = Any configured LCN Unit ID <i>default</i> = Null(--)			
\$ALDLOPT (R620 or later) Alarm Delay Option None OffDlyLo OffDlyHi OffDlyEm OnDlyLo OnDlyHi OnDlyEm Off Delay Lo Off Delay Hi Off Delay Emergency On Delay Lo On Delay Hi On Delay Emergency			
DLTIME (R620 or later) Delay Time _ _ _ _ <i>range</i> = 0 to 9999 seconds <i>default</i> = 0 seconds			
\$REALARM (R660 and later) Re-alarms Interval _ _ _ _ <i>range</i> = 0 to 9999 minutes <i>default</i> = 0 minutes			
BADOCPT BADOC Alarm Option Off On Bad Output Connection Alarm is not initially permitted Bad Output Connection Alarm is initially permitted			
BADOCPR BADOC Alarm Priority NoAction (R670 and later) Journal Low High Emergency Printer JnlPrint			
DEVHITP Deviation High Alarm Trip Point _ _ _ _ _ _ _ _ _ _ <i>range</i> = >=0.0, and NaN <i>default</i> = NaN			
DEVHIPR Deviation High Alarm Priority NoAction (R670 and later) Journal Low High Emergency Printer JnlPrint			
\$DHSELCT (R681 and later) High Selective Cutout Cutout Allow <i>range</i> = Cutout, Allow <i>default</i> = Cutout			
DEVLOTP Deviation Low Alarm Trip Point _ _ _ _ _ _ _ _ _ _ <i>range</i> = >=0.0, and NaN <i>default</i> = NaN			
DEVLOPR Deviation Low Alarm Priority NoAction (R670 and later) Journal Low High Emergency Printer JnlPrint			
\$DLSELCT (R681 and later) Deviation Low Selective Cutout Cutout Allow <i>range</i> = Cutout, Allow <i>default</i> = Cutout			
ALENBST Point Alarm Enable Status Enable Disable Inhibit <u>Displayed</u> Yes No No <u>Logged</u> Yes Yes No <u>Reported to EIP</u> Yes Yes No			

HPM Ramp Soak CTL Algorithm

Unit:	Tagname:	Engineer:	Date:
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Reg Control Ramp/Soak Algo Display (continued from Page 2)

Ramp Rates and Soak Times/Values

Pair Number	Ramp Rate (RATEX) <i>range = >0.0(EUs); default = NaN</i>	Soak Time (SOAKTX) <i>range = 0.0 – 120.0; default = 0.0</i>	Soak Value (SOAKVX) <i>range = >0.0 (EUs); default = NaN</i>
1	_ _ _ _ _ _ _ _ _	_ _ _ _	_ _ _ _ _ _ _ _ _
2	_ _ _ _ _ _ _ _ _	_ _ _ _	_ _ _ _ _ _ _ _ _
3	_ _ _ _ _ _ _ _ _	_ _ _ _	_ _ _ _ _ _ _ _ _
4	_ _ _ _ _ _ _ _ _	_ _ _ _	_ _ _ _ _ _ _ _ _
5	_ _ _ _ _ _ _ _ _	_ _ _ _	_ _ _ _ _ _ _ _ _
6	_ _ _ _ _ _ _ _ _	_ _ _ _	_ _ _ _ _ _ _ _ _
7	_ _ _ _ _ _ _ _ _	_ _ _ _	_ _ _ _ _ _ _ _ _
8	_ _ _ _ _ _ _ _ _	_ _ _ _	_ _ _ _ _ _ _ _ _
9	_ _ _ _ _ _ _ _ _	_ _ _ _	_ _ _ _ _ _ _ _ _
10	_ _ _ _ _ _ _ _ _	_ _ _ _	_ _ _ _ _ _ _ _ _
11	_ _ _ _ _ _ _ _ _	_ _ _ _	_ _ _ _ _ _ _ _ _
12	_ _ _ _ _ _ _ _ _	_ _ _ _	_ _ _ _ _ _ _ _ _

HPM Ramp Soak CTL Algorithm

Unit:	Tagname:	Engineer:	Date:
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Mark Functions			
<u>Segment Number 1</u>		<u>Segment Number 2</u>	
Begin Time (S1BGNTM)	End Time (S1ENDTM)	Begin Time (S2BGNTM)	End Time (S2ENDTM)
<i>range = 0.0 – 120 (Mins)</i>	<i>range = 0.0 – 120 (Mins)</i>	<i>range = 0.0 – 120 (Mins)</i>	<i>range = 0.0 – 120 (Mins)</i>
<i>default = 0.0</i>	<i>default = 0.0</i>	<i>default = 0.0</i>	<i>default = 0.0</i>
_ _ _ _ _ _ _	_ _ _ _ _ _ _	_ _ _ _ _ _ _	_ _ _ _ _ _ _
<u>Segment ID (S1SEGID)</u>		<u>Segment ID (S2SEGID)</u>	
RAMP1	SOAK1		SOAK1
RAMP2	SOAK2		SOAK2
RAMP3	SOAK3		SOAK3
RAMP4	SOAK4		SOAK4
RAMP5	SOAK5		SOAK5
RAMP6	SOAK6		SOAK6
RAMP7	SOAK7		SOAK7
RAMP8	SOAK8		SOAK8
RAMP9	SOAK9		SOAK9
RAMP10	SOAK10		SOAK10
RAMP11	SOAK11		SOAK11
RAMP12	SOAK12		SOAK12

NOTE: Return to page 2 of this form and continue configuration at parameter **NOCINPTS**.

CTLALGID Control Algorithm Identifier	Null RatioCtl	No affect Ratio Control
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HPM Ratio Control CTL Algorithm

Unit:	Tagname:	Engineer:	Date:
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PV Configuration Display

PV Configuration Display				
PVEUHI		<i>range = >=PVEULO</i>		
PV High Range In Engr Units	_ _ _ _ _ _ _	<i>default = 100.0</i>		
PVEULO		<i>range = <=PVEUHI</i>		
PV Low Range In Engr Units	_ _ _ _ _ _ _	<i>default = 0.0</i>		
PVFORMAT	DO	D2	D0=9999.	D2=99.99
PV Decimal Point Format	D1	D3	D1=999.9	D3 = 9.999

PVSRCOPT PV Source Option	Only Auto All	PV Source selection is not available Selection from PVSOURCE is available
PVSOURCE PV Source	Auto Man Sub	Field-wiring /memory -fetch supplies PV PV is supplied by Operator or program A value is substituted by a CL program
OVERVAL Overview Display Value	_ _ _ _	<i>range =</i> 0 to 100 <i>default =</i> 25

Mode Configuration Display

RCASOPT Remote Cascade Option	None Spc	Local cascade is the only valid cascade AM strategy stores to SP within SP limits
SHEDTIME Remote Cascade Shed Time	_ _ _ _	<i>range =</i> 0 to 1000 (seconds) <i>default =</i> 0 (mode shed disabled)
SHEDMODE Shed Mode	Man Auto Bcas	Supervisory point will shed to Manual, Supervisory point will shed to Automatic, Supervisory point will shed to BackupCas
NMODE (RegCtl) RegCtl Point's Normal Mode	None Man Auto Cas Bcas	No configured "normal" operating mode Manual is the "normal" operating mode Automatic is "normal" operating mode Cascade is "normal" operating mode Backup Cas is "normal" operating mode

N/A if NMODE =None

NMODATTR Normal Mode Attribute	None Operator Program	Parameter NMODE sets "normal" mode Operator sets "normal" operating mode Program sets "normal" operating mode
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MODEPERM Mode Permissive	Permit NotPerm	Operator can change this point's mode Optr. cannot change this point's mode
\$MODESEL (R630 and later) Mode Selection List	Man Prog Auto Cas Bcas	Man, Auto, Cas and Bcas mode, and Prog mode attribute are allowed for this data point.
\$BADOCMD (R680 and later) Bad Output Connection Mode	SHEDMAN 15_SEC 60_SEC NO_SHED	Point will shed to Manual immediately Point will shed to Manual after 15 seconds Point will shed to Manual after 60 seconds Do not shed mode
EXTSWOPT External Mode Switching	None Ems	No external mode switching is allowed Ext. source can change point's mode

NOCINPTS		range = 0 to 3
#Of Control Input Connections		default = 2
CISRC (Control-Input-Connection Source)		
<u>Input#</u>	<u>CISRC (point.parameter)</u>	<u>Destination Parameter (CIDSTN)</u>
1		PVAUTO
2		X2
3		SP

HPM Ratio Control CTL Algorithm

Unit:	Tagname:	Engineer:	Date:
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Output Configuration Display

NOCOPTS ☐ *range = 0 to 4*
#Of Control Output Connections *default = 1*

CODSTN (Control-Output-Connection Destination)
Output# (N/A if NOCOPTS = 0) CODSTN (point.parameter)

1	_____
2	_____
3	_____
4	_____

(N/A if NOCOPTS > 0)

CVEUHI Output Engineering Unit High Range	_ _ _ _	<i>range</i> = >=CVEULO <i>default</i> = 100.0
CVEULO Output Engineering Unit Low Range	_ _ _ _	<i>range</i> = >=CVEUHI <i>default</i> = 0.0
OPHILM Output High Limit In Percent	_ _ _ _	<i>range</i> = OPLOLM – 106.9 & NaN <i>default</i> = 105.0
OPLOLM Output Low Limit In Percent	_ _ _ _	<i>range</i> = -6.9 – OPHIL,& NaN <i>default</i> = -5.0
SAFEOP Safe Output Limit In Percent	_ _ _ _	<i>range</i> = -6.9 to 106.9, & NaN <i>default</i> = NaN
OPMCHLM Output Minimum Change In %	_ _ _ _	<i>range</i> = >=0.0, and NaN <i>default</i> = 0.0
OPROCLM OP Rate Of Change Limit In %	_ _ _ _	<i>range</i> = >=0.0, and NaN <i>default</i> = 0.0
\$OPTOL Output Tolerance in %	_ _ _ _	<i>range</i> = 0.0 to 106.9 <i>default</i> = 0.0
OUTIND (R620 or later) Reverse Output Indicator	Direct Reverse	4 - 20mA = 0 – 100% 20 – 4mA = 0 – 100% Note: With R670 and later, when this parameter is set to Reverse, the values of the displayed output and the related output parameters are changed. Refer to the definition of the OUTIND parameter in the PM Family Parameter Dictionary, XPM09.

HPM Ratio Control CTL Algorithm

Unit:	Tagname:	Engineer:	Date:
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Alarming Display – N/A if PNTFORM = Component

As you enter alarm trip points and alarm priorities in the next section please refer to the following alarm Priority definitions and the note.

NoAction	Alarm is not reported to the system and not annunciated (R670 & later)
Journal	Alarm is historized, not reported to US, not annunciated
Low	Alarm is historized, reported to US, and annunciated
High	Alarm is historized, reported to US, and annunciated
Emergency	Alarm is historized, reported to US, and annunciated
Printer	Not historized, reported to the printer, not annunciated
JnlPrint	Alarm historized, reported to the printer, not annunciated

Note: If you do not enter a trip point value for a given pair of trip-points and alarm-priorities, do not Circle an alarm priority option for that pair (that is, no trip point, no alarm priority).

\$AUXUNIT

Auxiliary Unit

|_|_|_|

range = Any configured LCN Unit ID
default = Null(--)

\$ALDLOPT (R620 or later)

Alarm Delay Option

None

OffDlyLo
OffDlyHi
OffDlyEm
OnDlyLo
OnDlyHi
OnDlyEm

Off Delay Lo
Off Delay Hi
Off Delay Emergency
On Delay Lo
On Delay Hi
On Delay Emergency

\$DLBADAL (R681 and later)

Delay for BADPV and BADCTL

None

Offdelay

range = Offdelay, None
default = None

DLTIME (R620 or later)

Delay Time

|_|_|_|_|

range = 0 to 9999 seconds
default = 0 seconds

\$REALARM (R660 or later)

Re-alarming Interval

|_|_|_|_|

range = 0 to 9999 minutes
default = 0 minutes

BADOCOPT

BADOC Alarm Option

Off

On

Bad Output Connection Alarm is not initially permitted
Bad Output Connection Alarm is initially permitted

BADOCPR

BADOC Alarm Priority

NoAction
(R670 and later)

Journal Low High Emergency Printer JnlPrint

OPALDB

Output Alarm Deadband

|_|_|_|_|

range = 0 – 25.0 but $\leq$ (OPHITP-OPLOTP)/2
default = 5.0 Output Units

OPHITP

Output High Alarm Trip Point

|_|_|_|_|

range = OPLOTP TO OPHILM and NaN
default = NaN

OPHIPR

Output High Alarm Priority

NoAction
(R670 and later)

Journal Low High Emergency Printer JnlPrint

OPLOTP

Output Low Alarm Trip Point

|_|_|_|_|

range = OPLOLM TO OPHITP and NaN
default = NaN

OPLOPR

Output Low Alarm Priority

NoAction
(R670 and later)

Journal Low High Emergency Printer JnlPrint

PVALDB

PV Alarm Deadband In %

Half **One** Two Three

Four Five

EU

PVALDBEU

PV Alarm Deadband in EU

|_|_|_|_|

range $\geq$ 0.0
default = 1.0

}

HPM Ratio Control CTL Algorithm

Unit:	Tagname:	Engineer:	Date:
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PVHITP PV High Alarm Trip Point	<u> </u>	<i>range = PVLOTP - PVHHTP, & NaN</i> <i>default = NaN</i>					
PVHIPR PV High Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
\$HISELCT (R681 and later) PV High Selective Cutout	Cutout Allow	<i>range = Cutout, Allow</i> <i>default = Cutout</i>					
PVLOTP PV Low Alarm Trip Point	<u> </u>	<i>range = PVLLTP - PVHITP, & NaN</i> <i>default = NaN</i>					
PVLOPR PV Low Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
\$LOSELCT (R681 and later) PV Low Selective Cutout	Cutout Allow	<i>range = Cutout, Allow</i> <i>default = Cutout</i>					
PVHHTP PV Hi Hi Alarm Trip Point	<u> </u>	<i>range = PVHITP to PVEUHI, & NaN</i> <i>default = NaN</i>					
PVHHPR PV High High Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
\$HHSELCT (R681 and later) PV High High Selective Cutout	Cutout Allow	<i>range = Cutout, Allow</i> <i>default = Cutout</i>					
PVLLTP PV Low Low Alarm Trip Point	<u> </u>	<i>range = PVEULO – PVLOTP, & NaN</i> <i>default = NaN</i>					
PVLLPR PV Low Low Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
\$LLSELCT (R681 and later) PV Low Low Selective Cutout	Cutout Allow	<i>range = Cutout, Allow</i> <i>default = Cutout</i>					
PVSGCHTP PV Significant Change Alarm Trip Point	<u> </u>	<i>range = >=0.0, and NaN</i> <i>default = NaN</i>					
PVROCPTP PV Pos Rate Of Chg Trip Point	<u> </u>	<i>range = >=0.0, and NaN</i> <i>default = NaN</i>					
PVROCPPR PV Pos Rate Of Chg Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
\$RPSELCT (R681 and later) PV Positive Rate of Change Selective Cutout	Cutout Allow	<i>range = Cutout, Allow</i> <i>default = Cutout</i>					
PVROCNTP PV Neg Rate Of Chg Trip Point	<u> </u>	<i>range = >=0.0, and NaN</i> <i>default = NaN</i>					

[illegible]

[illegible]

HPM Summer PV Algorithm

Unit:	Tagname:	Engineer:	Date:
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PV Algorithm Display

PVALGID PV Algorithm Identifier	NULL Summer	No affect Summer
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PV Configuration Display

PVEUHI PV High Range In Engr Units	_ _ _ _	<i>range</i> = PVEULO to PVEXEUHI <i>default</i> = 100.0
PVEULO PV Low Range In Engr Units	_ _ _ _	<i>range</i> = PVEXEULO to PVEUHI <i>default</i> = 0.0
PVFORMAT PV Decimal Point Format	DO D1	D2 D3 D0 = 9999. D1 = 999.9 D2 = 99.99 D3 = 9.999
PVEXEUHI PV Extended Hi Range in EUs	_ _ _ _ _ _	<i>range</i> = >=PVEUHI <i>default</i> = 102.9
PVEXEULO PV Extended Lo Range in EUs	_ _ _ _ _ _	<i>range</i> = <=PVEULO <i>default</i> = -2.9
PV CLAMP PV Clamping Option	NoClamp Clamp	No clamping of the PV value Clamp PV value at range extension limit

PV Algorithm Display (Continued)

N/A if PNTFORM = Componnt.

PVSRCOPT PV Source Option	Only Auto All	PV Source selection is not available Selection from PVSOURCE is available
PVSOURCE PV Source	Auto Man Sub	Field wiring/ memory fetch supplies PV PV is supplied by Operator or program A value is substituted by a CL program

TF PV Filter Lag Time in Minutes	_ _ _ _	<i>range</i> = 0.0 to 60.0 (minutes) <i>default</i> = 0.0 (PV will not be delayed)
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N/A if PNTFORM = Componnt

PVTV PV Target Value in EUs	_ _ _ _ _ _	<i>range</i> = PVEXEULO-PVEXEUHI, & NaN <i>default</i> = NaN
OVERVAL Overview Display Value	_ _ _	<i>range</i> = 0 to 100 <i>default</i> = 25

Unit:	Tagname:	Engineer:	Date:
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PVEQN	EqA	P1 input is scaled and biased
PV Equation Type	EqB	Up to 6 inputs are scaled and summed
C	_ _ _ _ _ _ _	<i>range = Not Applicable</i>
Scaling Constant		<i>default = 1.0</i>
D	_ _ _ _ _ _ _	<i>range = Not Applicable</i>
Overall Bias		<i>default = 0.0</i>
N (EQB only)		<i>range = 2 to 6</i>
# Inputs To Summer Algo		<i>default = 2</i>
C1 (If N ≥ 2)	_ _ _ _ _ _ _	<i>range = Not Applicable</i>
Scaling Constant for Input P1		<i>default = 1.0</i>
C2 (If N ≥ 2)	_ _ _ _ _ _ _	<i>range = Not Applicable</i>
Scaling Constant for Input P2		<i>default = 1.0</i>
C3 (If N ≥ 3)	_ _ _ _ _ _ _	<i>range = Not Applicable</i>
Scaling Constant for Input P3		<i>default = 1.0</i>
C4 (If N ≥ 4)	_ _ _ _ _ _ _	<i>range = Not Applicable</i>
Scaling Constant for Input P4		<i>default = 1.0</i>
C5 (If N ≥ 5)	_ _ _ _ _ _ _	<i>range = Not Applicable</i>
Scaling Constant for Input P5		<i>default = 1.0</i>
C6 (If N ≥ 6)	_ _ _ _ _ _ _	<i>range = Not Applicable</i>
Scaling Constant for Input P6		<i>default = 1.0</i>
PISRC (PV-Input-Connection Source)	PISRC (point.parameter)	Destination Parameter
<u>Input#</u>		<u>(PIDSTN)</u>
1	_ _ _ _ _ _ _	P1
2	_ _ _ _ _ _ _	P2
3	_ _ _ _ _ _ _	P3
4	_ _ _ _ _ _ _	P4
5	_ _ _ _ _ _ _	P5
6	_ _ _ _ _ _ _	P6

HPM Summer PV Algorithm

Unit:	Tagname:	Engineer:	Date:
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Alarming Display – N/A if PNTFORM = Comppnt

As you enter alarm trip points and alarm priorities in the next section please refer to the following alarm Priority definitions and the note.

NoAction	Alarm is not reported to the system and not annunciated (R670 & later)
Journal	Alarm is historized, not reported to US, not annunciated
Low	Alarm is historized, reported to US, and annunciated
High	Alarm is historized, reported to US, and annunciated
Emergency	Alarm is historized, reported to US, and annunciated
Printer	Not historized, reported to the printer, not annunciated
JnlPrint	Alarm historized, reported to the printer, not annunciated

Note: If you do not enter a trip point value for a given pair of trip-points and alarm-priorities, do not Circle an alarm priority option for that pair (that is, no trip point, no alarm priority).

Alarming Display - N/A if PNTFORM = Comppnt

\$AUXUNIT Auxiliary Unit	_ _ _	<i>range</i> = Any configured LCN Unit ID <i>default</i> = Null (- -)
\$ALDLOPT (R620 and later) Alarm Delay Option	None OffDlyLo OffDlyHi OffDlyEm OnDlyLo OnDlyHi OnDlyEm	Off Delay Lo Off Delay Hi Off Delay Emergency On Delay Lo On Delay Hi On Delay Emergency
\$DLBADAL (R681 and later) Delay for BADPV	None Offdelay	<i>range</i> = Offdelay, None <i>default</i> = None
DLTIME (R620 and later) Delay Time	_ _ _ _	<i>range</i> = 0 to 9999 seconds <i>default</i> = 0 seconds
\$REALARM (R660 and later) Re-arming Interval	_ _ _ _	<i>range</i> = 0 to 9999 minutes <i>default</i> = 0 minutes
PVALDB PV Alarm Deadband In %	Half One Two Three Four Five EU	
PVALDBEU PV Alarm Deadband In EU	_ _ _ _	<i>range</i> > = 0.0 <i>default</i> = 1.0
PVHITP PV High Alarm Trip Point	_ _ _ _ _ _ _	<i>range</i> = PVLOTP - PVHHTP, & NaN <i>default</i> = NaN
PVHIPR PV High Alarm Priority	NoAction Journal Low High Emergency JnlPrint Printer (R670 and later)	
\$HISELCT (R681 and later) PV High Selective Cutout	Cutout Allow	<i>range</i> = Cutout, Allow <i>default</i> = Cutout
PVLOTP PV Low Alarm Trip Point	_ _ _ _ _ _ _	<i>range</i> = PVLLTP - PVHITP, & NaN <i>default</i> = NaN
PVLOPR PV Low Alarm Priority	NoAction Journal Low High Emergency JnlPrint Printer (R670 and later)	

Unit:	Tagname:	Engineer:	Date:
\$LOSELCT (R681 and later) PV Low Selective Cutout	Cutout Allow	<i>range</i> = Cutout, Allow <i>default</i> = Cutout	
PVHHTP PV Hi Hi Alarm Trip Point	_____	<i>range</i> = PVHITP to PVEUHI, & NaN <i>default</i> = NaN	
PVHHPR PV High High Alarm Priority	NoAction Journal Low (R670 and later)	High	Emergency JnlPrint Printer
\$HHSELCT (R681 and later) PV High High Selective Cutout	Cutout Allow	<i>range</i> = Cutout, Allow <i>default</i> = Cutout	
PVLLTP PV Low Low Alarm Trip Point	_____	<i>range</i> = PVEULO - PVLOTP, & NaN <i>default</i> = NaN	
PVLLPR PV Low Low Alarm Priority	NoAction Journal Low (R670 and later)	High	Emergency JnlPrint Printer
\$LLSELCT (R681 and later) PV Low Low Selective Cutout	Cutout Allow	<i>range</i> = Cutout, Allow <i>default</i> = Cutout	
PVSGCHTP PV Significant Change Alarm Trip Point	_____	<i>range</i> = >=0.0, and NaN <i>default</i> = NaN	
PVROCTP PV Pos Rate Of Change Trip Point	_____	<i>range</i> = >=0.0, and NaN <i>default</i> = NaN	
PVROCPPR PV Pos Rate Of Chg Alarm Priority	NoAction Journal Low (R670 and later)	High	Emergency JnlPrint Printer
\$RPSELCT (R681 and later) PV Positive Rate of Change Selective Cutout	Cutout Allow	<i>range</i> = Cutout, Allow <i>default</i> = Cutout	
PVROCNTP PV Neg Rate Of Change Trip Point	_____	<i>range</i> = >=0.0, and NaN <i>default</i> = NaN	
PVROCNPR PV Neg Rate Of Chg Alarm Priority	NoAction Journal Low (R670 and later)	High	Emergency JnlPrint Printer
\$RNSELCT (R681 and later) PV Negative Rate of Change Selective Cutout	Cutout Allow	<i>range</i> = Cutout, Allow <i>default</i> = Cutout	
BADPVPR Bad-PV Alarm Priority	NoAction Journal Low (R670 and later)	High	Emergency JnlPrint Printer
ALENBST Point Alarm Enable Status	Enable Disable Inhibit	<u>Displayed</u> Yes No No	<u>Logged</u> Yes Yes No <u>Reported To EIP</u> Yes Yes No

OVERVAL	. . .	<i>range = 0 to 100</i>
Overview Display Value		<i>default = 25</i>

HPM Switch CTL Algorithm

Unit:	Tagname:	Engineer:	Date:
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Mode Configuration Display

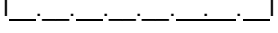
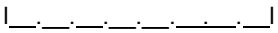
BADCTLOP Bad PV Mode Shed Option	No_Shed Shed Hold Shed Low Shed High Shed Safe	<i>range</i> = See APM Parameter Reference Dictionary <i>default</i> = No_Shed
RCASOPT Remote Cascade Option	None Ddc	Local cascade is the only valid cascade AM strategy stores to SP within SP limits
SHEDTIME Remote Cascade Shed Time		<i>range</i> = 0 to 1000 (seconds) <i>default</i> = 0 (mode shed disabled)
SHEDMODE (If Shedtime ≥ 1) Shed Mode	Man Bcas	Supervisory point will shed to Manual, Backup Cascade mode.
NMODE RegCtl Point's Normal Mode	None Man Cas Bcas	Supervisory point will shed to BackupCas No configured "normal" operating mode Manual is the "normal" operating mode Cascade is "normal" operating mode Backup Cas is "normal" operating mode

Mode Configuration Display (Continued) N/A if NMODE = None.

NMODATTR Normal Mode Attribute	None Operator Program	Parameter NMODE sets "normal" mode Operator sets "normal" operating mode Program sets "normal" operating mode
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MODEPERM Mode Permissive	Permit NotPerm	Operator can change this point's mode Opr. cannot change this point's mode
\$MODESEL (R630 and later) Mode Selection List	Man Prog Cas Bcas	Man, Cas and Bcas mode, and Prog mode attribute are allowed for this data point.
\$BADOCMD (R680 and later) Bad Output Connection Mode	SHEDMAN 15_SEC 60_SEC NO_SHED	Point will shed to Manual immediately Point will shed to Manual after 15 seconds Point will shed to Manual after 60 seconds Do not shed mode
EXTSWOPT External Mode Switching	None Ems	No external mode switching is allowed Ext. source can change point's mode

Reg Control Switch Algo Display

CTLEQN Control Equation Type	EqA EqB	Operator controls switch position Pgm. Or logic controls sw. position
TRACKING Selected Input Tracking	Off On	<i>range</i> = Off-On <i>default</i> = Off (Tracking disabled)
M # Of Initializable Control Algorithm Inputs		<i>range</i> = 2 to 4 <i>default</i> = 2
XEUHI X Input High Range in EUs		<i>range</i> = ≥ XEULO <i>default</i> = 100.0
XEULO X Input Low Range in EUs		<i>range</i> = ≤ XEUHI <i>default</i> = 0.0

N/A if CTLEQN = EqB, or if M < 2

SELXINP Selected X Input	SelectX1	SelectX2	SelectX3	SelectX4
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NOCINPTS		<u> </u>	<i>range</i> = 0 to 4 <i>default</i> = 0
#Of Control Input Connections			
CISRC (Control-Input-Connection Source)		CISRC (point.parameter)	<u>Destination Para-</u> <u>-meter (CIDSTN)</u>
<u>Input#</u>			
1			
2			
3			
4			

NOCOPTS		<i>range</i> = 0 to 4
# Of Control Output Connections		<i>default</i> = 1
CODSTN (Control-Output-Connection Destination)		
<u>Output #</u>	<u>CODSTN (point.parameter)</u>	
1		
2		
3		
4		
OPHILM		
Output High Limit In Percent		<i>range</i> = OPLOLM - 106.9 & NaN <i>default</i> = 105.0
OPLOLM		
Output Low Limit In Percent		<i>range</i> = -6.9 - OPHILM, & NaN <i>default</i> = -5.0
SAFEOP		
Safe Output Limit In Percent		<i>range</i> = -6.9 to 106.9 & NaN <i>default</i> = NaN
OPMCHLM		
Output Minimum Change In %		<i>range</i> = >=0.0, and NaN <i>default</i> = 0.0
OPROCLM		
OP Rate Of Change Limit In %		<i>range</i> = >=0.0, and NaN <i>default</i> = NaN
\$OPTOL		
Output Tolerance in %		<i>range</i> = 0.0 to 106.9 <i>default</i> = 0.0
OUTIND (R620 and later)	Direct	4 - 20mA = 0 – 100%
Reverse Output Indicator	Reverse	20 - 4mA = 0 – 100%
		Note: With R670 and later, when this parameter is set to Reverse, the values of the displayed output and the related output parameters are changed. Refer to the definition of the OUTIND parameter in the PM Family Parameter Dictionary, XPM09.

HPM Switch CTL Algorithm

Unit:	Tagname:	Engineer:	Date:
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Alarming Display - N/A if PNTFORM = Componnt

As you enter alarm trip points and alarm priorities in the next section please refer to the following alarm priority definitions and the note.

NoAction	Alarm is not reported to the system and not annunciated (R670 & later)
Journal	Alarm is historized, not reported to US, not annunciated
Low	Alarm is historized, reported to US, and annunciated
High	Alarm is historized, reported to US, and annunciated
Emergency	Alarm is historized, reported to US, and annunciated Not
Printer	historized, reported to the printer, not annunciated Alarm
JnlPrint	historized, reported to the printer, not annunciated

Note: If you do not enter a trip point value for a given pair of trip-points and alarm-priorities, do not circle an alarm priority option for that pair (that is, no trip point, no alarm priority).

Alarming Display - N/A if PNTFORM = Componnt

\$AUXUNIT Auxiliary Unit	_ _ _	<i>range = Any configured LCN Unit ID default = Null (- -)</i>			
\$DLBADAL (R681 and later) Delay for BADCTL	None Offdelay	<i>range = Offdelay, None default = None</i>			
DLTIME (R620 and later) Delay Time	_ _ _ _	<i>range = 0 to 9999 seconds default = 0 seconds</i>			
\$REALARM (R660 and later) Re-alarming Interval	_ _ _ _	<i>range = 0 to 9999 minutes default = 0 minutes</i>			
BADOCOPT BADOC Alarm Option	Off On	Bad Output Connection Alarm is not initially permitted Bad Output Connection Alarm is Initially permitted			
BADOCPR BADOC Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency JnlPrint Printer
OPALDB Output Alarm Deadband	_ _ _ _	<i>range = 0 - 25.0 but ≤ (OPHITP-OPLOTP)/2 default = 5.0 Output Units</i>			
OPHITP Output High Alarm Trip Point	_ _ _ _	<i>range = OPLOTP to OPHILM and NaN default = NaN</i>			
OPHIPR Output High Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency JnlPrint Printer
OPLOTP Output Low Alarm Trip Point	_ _ _ _	<i>range = OPLOLM to OPHITP and NaN default = NaN</i>			
OPLOPR Output Low Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency JnlPrint Printer
BADCTLPR Bad-control Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency JnlPrint Printer
ALENBST Point Alarm Enable Status	Enable Disable Inhibit	<u>Displayed</u> Yes No No	<u>Logged</u> Yes Yes No	<u>Reported To EIP</u> Yes Yes No	

PRIMMOD	N/A if PNTFORM = Component.
Primary Module ID _____	range = Not Applicable default = Null
USERID	Sixteen Character String
User ID Reservation _____	default = dashes

HPM Totalizer PV Algorithm

Unit:	Tagname:	Engineer:	Date:
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PV Algorithm Display

PVALGID PV Algorithm Identifier	NULL Totalizr	No affect Totalizer
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PV Configuration Display

PVEUHI PV High Range In Engr Units	_ _ _ _ _ _ _	<i>range</i> = PVEULO to PVEXEUHI <i>default</i> = 100.0
PVEULO PV Low Range In Engr Units	_ _ _ _ _ _ _	<i>range</i> = PVEXEULO to PVEUHI <i>default</i> = 0.0
PVFORMAT PV Decimal Point Format	D0 D2 D1 D3	D0 = 9999. D2 = 99.99 D1 = 999.9 D3 = 9.999
PVEXEUHI PV Extended Hi Range In EUs	_ _ _ _ _ _ _ _ _ _ _ _ _	<i>range</i> = >=PVEUHI <i>default</i> = 102.9
PVEXEULO PV Extended Lo Range In EUs	_ _ _ _ _ _ _ _ _ _ _ _ _	<i>range</i> = <=PVEULO <i>default</i> = -2.9
PVCLAMP PV Clamping Option	NoClamp Clamp	No clamping of the PV value Clamp PV value at range extension limit

N/A if PNTFORM = Componnt.

PVSRCOPT PV Source Option	OnlyAuto All	PV source selection is not available Selection from PVSOURCE is available
PVSOURCE PV Source	Auto Man Sub	Field-wiring/memory-fetch supplies PV PV is supplied by Operator or program A value is substituted by a CL program

TF PV Filter Lag Time In Minutes	_ _ _ _ _ _ _	<i>range</i> = 0.0 to 60.0 (minutes) <i>default</i> = 0.0 (PV will not be delayed)
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Alarming Display – N/A if PNTFORM = Component

As you enter alarm trip points and alarm priorities in the next section please refer to the following alarm Priority definitions and the note.

NoAction	Alarm is not reported to the system and not annunciated (R670 & later)
Journal	Alarm is historized, not reported to US, not annunciated
Low	Alarm is historized, reported to US, and annunciated
High	Alarm is historized, reported to US, and annunciated
Emergency	Alarm is historized, reported to US, and annunciated
Printer	historized, reported to the printer, not annunciated
JnlPrint	historized, reported to the printer, not annunciated

Note: If you do not enter a trip point value for a given pair of trip-points and alarm-priorities, do not circle an alarm priority option for that pair (that is, no trip point, no alarm priority).

HPM Totalizer PV Algorithm

Unit:	Tagname:	Engineer:	Date:
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Alarming Display - N/A if PNTFORM = Component

\$AUXUNIT Auxiliary Unit		<i>range = Any configured LCN Unit ID default = Null (- -)</i>
\$ALDLOPT (R620 and later) Alarm Delay Option	None OffDlyLo OffDlyHi OffDlyEm OnDlyLo OnDlyHi OnDlyEm	Off Delay Lo Off Delay Hi Off Delay Emergency On Delay Lo On Delay Hi On Delay Emergency
\$DLBADAL (R681 and later) Delay for BADPV	None Offdelay	<i>range = Offdelay, None default = None</i>
DLTIME (R620 and later) Delay Time		<i>range = 0 to 9999 seconds default = 0 seconds</i>
\$REALARM (R660 and later) Re-alarming Interval		<i>range = 0 to 9999 minutes default = 0 minutes</i>
PVALDB PV Alarm Deadband In %	Half One Two Three Four Five EU	
PVALDBEU PV Alarm Deadband In EU		<i>range > = 0.0 default = 1.0</i>
PVHITP PV High Alarm Trip Point		<i>range = PVLOTP - PVHHTP, & NaN default = NaN</i>
PVHIPR PV High Alarm Priority	NoAction Journal Low High Emergency Printer JnlPrint (R670 and later)	
\$HISELCT (R681 and later) PV High Selective Cutout	Cutout Allow	<i>range = Cutout, Allow default = Cutout</i>
PVLOTP PV Low Alarm Trip Point		<i>range = PVLLTP - PVHITP, & NaN default = NaN</i>
PVLOPR PV Low Alarm Priority	NoAction Journal Low High Emergency Printer JnlPrint (R670 and later)	
\$LOSELCT (R681 and later) PV Low Selective Cutout	Cutout Allow	<i>range = Cutout, Allow default = Cutout</i>
PVHHTP PV Hi Hi Alarm Trip Point		<i>range = PVHITP to PVEUHI, & NaN default = NaN</i>
PVHHPR PV High High Alarm Priority	NoAction Journal Low High Emergency Printer JnlPrint (R670 and later)	
\$HHSELCT (R681 and later) PV High High Selective Cutout	Cutout Allow	<i>range = Cutout, Allow default = Cutout</i>
PVLLTP PV Low Low Alarm Trip Point		<i>range = PVEULO - PVLOTP, & NaN default = NaN</i>

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Unit:	Tagname:	Engineer:	Date:
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PVLLPR PV Low Low Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
\$LLSELCT (R681 and later) PV Low Low Selective Cutout	Cutout Allow				<i>range</i> = Cutout, Allow <i>default</i> = Cutout		
PVSGCHTP PV Significant Change Alarm Trip Point	_____				<i>range</i> = >=0.0, and NaN <i>default</i> = NaN		
PVROCPTP PV Pos Rate Of Change Trip Point	_____				<i>range</i> = >=0.0, and NaN <i>default</i> = NaN		
PVROCPPR PV Pos Rate Of Chg Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
\$RPSELCT (R681 and later) PV Positive Rate of Change Selective Cutout	Cutout Allow				<i>range</i> = Cutout, Allow <i>default</i> = Cutout		
PVROCNTP PV Neg Rate Of Change Trip Point	_____				<i>range</i> = >=0.0, and NaN <i>default</i> = NaN		
PVROCNPR PV Neg Rate Of Chg Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
\$RNSELCT (R681 and later) PV Negative Rate of Change Selective Cutout	Cutout Allow				<i>range</i> = Cutout, Allow <i>default</i> = Cutout		
BADPVPR Bad-PV Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
ALENBST Point Alarm Enable Status	Enable Disable Inhibit			<u>Displayed</u> Yes No No	<u>Logged</u> Yes Yes No	<u>Reported To EIP</u> Yes Yes No	

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[illegible]

HPM Variable Dead Time with Lead Lag PV Algorithm

Unit:	Tagname:	Engineer:	Date:
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PV Algorithm Display

PVALGID PV Algorithm Identifier	NULL VdtLdLag	No affect Variable Dead Time with Lead Lag
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PV Configuration Display

PVEUHI PV High Range In Engr Units	_ _ _ _	<i>range</i> = PVEULO to PVEXEUHI <i>default</i> = 100.0	
PVEULO PV Low Range In Engr Units	_ _ _ _	<i>range</i> = PVEXEULO to PVEUHI <i>default</i> = 0.0	
PVFORMAT PV Decimal Point Format	D0 D2 D1 D3	D0 = 9999. D2 = 99.99 D1 = 999.9 D3 = 9.999	
PVEXEUHI PV Extended Hi Range In EUs	_ _ _ _ _ _ _	<i>range</i> = >=PVEUHI <i>default</i> = 102.9	
PVEXEULO PV Extended Lo Range In EUs	_ _ _ _ _ _ _	<i>range</i> = <=PVEULO <i>default</i> = -2.9	
PVCLAMP PV Clamping Option	NoClamp Clamp	No clamping of the PV value Clamp PV value at range extension limit	

PV Configuration Display (Continued)

N/A if PNTFORM = Componnt.

PVSRCOPT PV Source Option		OnlyAuto All	PV source selection is not available Selection from PVSOURCE is available
PVSOURCE PV Source		Auto Man Sub	Field-wiring/memory-fetch supplies PV PV is supplied by Operator or program A value is substituted by a CL program

N/A if PNTFORM = Componnt.

TF PV Filter Lag Time In Minutes	_ _ _ _	<i>range</i> = 0.0 to 60.0 (minutes) <i>default</i> = 0.0 (PV will not be delayed)
PVTV PV Target Value In EUs	_ _ _ _ _ _ _	<i>range</i> = PVEXEULO-PVEXEUHI, & NaN <i>default</i> = NaN
OVERVAL Overview Display Value	_ _ _ _	<i>range</i> = 0 to 100 <i>default</i> = 25

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Reg PV Vdt Lead/Lag Algo Display

[illegible]

HPM Variable Dead Time with Lead Lag PV Algorithm

Unit:	Tagname:	Engineer:	Date:
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Alarming Display – N/A if PNTFORM = Componnt

As you enter alarm trip points and alarm priorities in the next section please refer to the following alarm Priority definitions and the note.

NoAction	Alarm is not reported to the system and not annunciated (R670 & later)
Journal	Alarm is historized, not reported to US, not annunciated
Low	Alarm is historized, reported to US, and annunciated
High	Alarm is historized, reported to US, and annunciated
Emergency	Alarm is historized, reported to US, and annunciated
Printer	Not historized, reported to the printer, not annunciated
JnlPrint	Alarm historized, reported to the printer, not annunciated

Note: If you do not enter a trip point value for a given pair of trip-points and alarm-priorities, do not circle an alarm priority option for that pair (that is, no trip point, no alarm priority).

\$AUXUNIT Auxiliary Unit	_ _	<i>range = Any configured LCN Unit ID default = Null (--)</i>
\$ALDLOPT (R620 and later) Alarm Delay Option	None OffDlyLo OffDlyHi OffDlyEm OnDlyLo OnDlyHi OnDlyEm	Off Delay Lo Off Delay Hi Off Delay Emergency On Delay Lo On Delay Hi On Delay Emergency
\$DLBADAL (R681 and later) Delay for BADPV	None Offdelay	<i>range = Offdelay, None default = None</i>
DLTIME (R620 and later) Delay Time	_ _ _ _	<i>range = 0 to 9999 seconds default = 0 seconds</i>
\$REALARM (R660 and later) Re-alarming Interval	_ _ _ _	<i>range = 0 to 9999 minutes default = 0 minutes</i>
PVALDB PV Alarm Deadband In %	Half One Two Three	Four Five EU
PVALDBEU PV Alarm Deadband in EU	_ _ _ _	<i>range >= 0.0 default = 1.0</i>
PVHITP PV High Alarm Trip Point	_ _ _ _ _ _ _	<i>range = PVLOTP - PVHHTP, & NaN default = NaN</i>
PVHIPR PV High Alarm Priority	NoAction (R670 and later)	Journal Low High Emergency Printer JnlPrint
\$HISELCT (R681 and later) PV High Selective Cutout	Cutout Allow	<i>range = Cutout, Allow default = Cutout</i>
PVLOTP PV Low Alarm Trip Point	_ _ _ _ _ _ _	<i>range = PVLLTP - PVHITP, & NaN default = NaN</i>
PVLOPR PV Low Alarm Priority	NoAction (R670 and later)	Journal Low High Emergency Printer JnlPrint

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Unit:	Tagname:	Engineer:	Date:
\$LOSELCT (R681 and later) PV Low Selective Cutout	Cutout Allow		<i>range</i> = Cutout, Allow <i>default</i> = Cutout
PVHHTP PV Hi Hi Alarm Trip Point	_____		<i>range</i> = PVHITP to PVEUHI, & NaN <i>default</i> = NaN
PVHHPR PV High High Alarm Priority	NoAction (R670 and later)	Journal Low High Emergency Printer JnlPrint	
\$HHSELCT (R681 and later) PV High High Selective Cutout	Cutout Allow		<i>range</i> = Cutout, Allow <i>default</i> = Cutout
PVLLTP PV Low Low Alarm Trip Point	_____		<i>range</i> = PVEULO – PVLOTP, & NaN <i>default</i> = NaN
PVLLPR PV Low Low Alarm Priority	NoAction (R670 and later)	Journal Low High Emergency Printer JnlPrint	
\$LLSELCT (R681 and later) PV Low Low Selective Cutout	Cutout Allow		<i>range</i> = Cutout, Allow <i>default</i> = Cutout
PVSGCHTP PV Significant Change Alarm Trip Point	_____		<i>range</i> = >=0.0, and NaN <i>default</i> = NaN
PVROCTP PV Pos Rate Of Chg Trip Point	_____		<i>range</i> = >=0.0, and NaN <i>default</i> = NaN
PVROCPPR PV Pos Rate Of Chg Alarm Priority	NoAction (R670 and later)	Journal Low High Emergency Printer JnlPrint	
\$RPSELCT (R681 and later) PV Positive Rate of Change Selective Cutout	Cutout Allow		<i>range</i> = Cutout, Allow <i>default</i> = Cutout
PVROCNTP PV Neg Rate Of Chg Trip Point	_____		<i>range</i> = >=0.0, and NaN <i>default</i> = NaN
PVROCNPR PV Neg Rate Of Chg Alarm Priority	NoAction (R670 and later)	Journal Low High Emergency Printer JnlPrint	
\$RNSELCT (R681 and later) PV Negative Rate of Change Selective Cutout	Cutout Allow		<i>range</i> = Cutout, Allow <i>default</i> = Cutout
BADPVPR Bad-PV Alarm Priority	NoAction (R670 and later)	Journal Low High Emergency Printer JnlPrint	
ALENBST Point Alarm Enable Status	Enable Disable Inhibit	<u>Displayed</u> Yes No No	<u>Logged</u> Yes Yes No <u>Reported To EIP</u> Yes Yes No

STRLEN	8	<i>range = Not applicable</i>	
Length of Displayed Strings	16	<i>default = 8</i>	
	32		
	64		
SEQSLTSZ		<i>range = >=0</i>	
Sequence Slot Size	_._._._	<i>default = 0</i>	
CNTLOCK	Operator	Engineer	Lowest access lock that permits certain stores
Control Lock	Supervis	Program	

SEQPR	NoAction	Alarm is not reported to the system and not annunciated
Sequence Alarm Priority	(R670 and later)	Alarm is historized, not reported to US, not annunciated
Journal		Alarm is historized, reported to US, and annunciated
Low		Alarm is historized, reported to US, and annunciated
High		Alarm is historized, reported to US, and annunciated
Emergency		Not historized, reported to the printer, not annunciated
Printer		Alarm historized, reported to the printer, not annunciated
JnlPrint		

HPM Flag Point

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Unit:	Tagname:	Engineer:	Date:
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NAME Tag Name 		<i>range</i> = Not applicable <i>default</i> = Blank
NODETYPE Node Type HPM		An HPM point is being configured If a PM, APM, LM or SM point is being configured, go to the appropriate forms for those devices.
PNTFORM Point Form	Full Componnt	Point is fully displayed and alarmed Pt. Is partially displayed but not alarmed
PTDESC Point Descriptor 		<i>range</i> = Not applicable <i>default</i> = Blank
EUDESC Engineering Units Descriptor		<i>range</i> = Not applicable <i>default</i> = Blank
KEYWORD Keyword Descriptor		<i>range</i> = Not applicable <i>default</i> = Blank
ASSOCDSP Associated Display		<i>range</i> = Not applicable <i>default</i> = Blank
\$CDETAIL (R620 and later) Point Custom Display		<i>range</i> = Not applicable <i>default</i> = Blank
UNIT Process Unit Point Is Assigned To	..	<i>range</i> = Two Alphanumeric Characters <i>default</i> = Not applicable
NTWKNUM NIM's UCN That This Point Is In	..	<i>range</i> = 1 to 20 <i>default</i> = Not applicable
NODENUM HPMM's Address On The UCN	..	<i>range</i> = 1 to 64 <i>default</i> = Not applicable
SLOTNUM Slot Number Within HPMM		<i>range</i> = 1-2047 <i>default</i> = Not applicable

N/A if PNTFORM = Componnt

PRIMMOD Primary Module ID		<i>range</i> = Not Applicable <i>default</i> = Null
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Unit:	Tagname:	Engineer:	Date:
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STATETXT(1) State 1 Descriptor	_._._._._._.__	<i>range</i> = Not applicable <i>default</i> = ON
STATETXT(0) State 0 Descriptor	_._._._._._.__	<i>range</i> = Not applicable <i>default</i> = OFF

PV		<i>range</i> = Statetxt(0) – Statetxt(1)
Current PV State	_._._._._._.__	<i>default</i> = OFF

Operating Configuration Display				
BOXCLR (1) State 1 Box Color	RED CYAN	GREEN <i>YELLOW</i>	WHITE BLUE	BLACK MAGENTA
BOXCLR (0) State 0 Box Color	RED CYAN	<i>GREEN</i> YELLOW	WHITE BLUE	BLACK MAGENTA

\$AUXUNIT Auxiliary Unit	_____	<i>range</i> = Any configured LCN Unit ID <i>default</i> = Null (--)
\$REALARM (R660 and later) Re-arming Interval	_____	<i>range</i> = 0 to 9999 minutes <i>default</i> = 0 minutes
OFFNRMPR Offnormal Alarm Priority	NoAction (R670 and later) Journal Low High Emergency Printer JnlPrint	Alarm is not reported to the system and not annunciated Alarm is historized, not reported to US, not annunciated Alarm is historized, reported to US, and annunciated Alarm is historized, reported to US, and annunciated Alarm is historized, reported to US, and annunciated Not historized, reported to the printer, not annunciated Alarm historized, reported to the printer, not annunciated
\$OFSELECT (R681 and later) Off Normal Selective Cutout	Cutout Allow	<i>range</i> = Cutout, Allow <i>default</i> = Cutout
ALENBST Alarm Enable State	Enable Disable Inhibit	<i>default</i> = Enable
EIPPCODE EIP Point ID	_____	<i>range</i> = Not Applicable <i>default</i> = Blank

TIMEBASE	SECONDS	<i>range = Seconds/Minutes</i>
Timer Base Period	MINUTES	<i>default = Seconds</i>
SP		<i>range = 0 - 32000</i>
Set Time	_ _ _ _	<i>default = 0</i>

HPM Numeric Point

Unit:	Tagname:	Engineer:	Date:
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1 of 2 Numeric Points - Point Assignment Display

NAME Tag Name		<i>range</i> = Not applicable <i>default</i> = Blank
NODE TYP Node Type	HPM	An HPM point is being configured If a PM, APM, LM or SM point is being configured, go to the appropriate forms for those devices.
PNTFORM Point Form	Full Componnt	Point is fully displayed and alarmed
PTDESC Point Descriptor		<i>range</i> = Not applicable <i>default</i> = Blank
EUDESC Engineering Units Descriptor		<i>range</i> = Not applicable <i>default</i> = Blank
KEYWORD Keyword Descriptor		<i>range</i> = Not applicable <i>default</i> = Blank
ASSOCDSP Associated Display		<i>range</i> = Not applicable <i>default</i> = Blank
\$CDETAIL (R620 and later) Point Custom Display		<i>range</i> = Not applicable <i>default</i> = Blank
UNIT Process Unit Point Is Assigned To	..	<i>range</i> = Two Alphanumeric Characters <i>default</i> = Not applicable
NTWKNUM NIM's UCN That This Point Is In	..	<i>range</i> = 1 to 20 <i>default</i> = Not applicable
NODENUM HPMM's Address On The UCN	..	<i>range</i> = 1 to 64 <i>default</i> = Not applicable
SLOTNUM Slot Number Within HPMM		<i>range</i> = 1-2047 <i>default</i> = Not applicable

PRIMMOD Primary Module ID		<i>range</i> = Not Applicable <i>default</i> = Null
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Operating Configuration Display

PV Numeric PV Value		<i>range</i> = Not Applicable <i>default</i> = 0.0
PVFORMAT PV Decimal Format	D0 D2	D1 D3 <i>range</i> = D0 - D3 <i>default</i> = D1

HPM Numeric Point

Unit:	Tagname:	Engineer:	Date:
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Alarming Display - N/A if PNTFORM = Component

\$AUXUNIT (R660 and later) Auxiliary Unit	_ _	<i>range = A – Z, 0-9 and NULL (Any configured NCF Unit ID) default = Null (--)</i>					
\$REALARM (R660 and later) Re-alarming Interval	_ _ _	<i>range = 0 to 9999 minutes default = 0 minutes</i>					
PVALDBEU (R660 and later) PV Alarm Deadband In Real	_ _	<i>range > = 0.0 default = NaN</i>					
PVHITP (R660 and later) PV High Alarm Trip Point	_ _ _ _ _ _ _	<i>range = > PVLOTP, NaN default = NaN</i>					
PVHIPR (R660 and later) PV High Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
\$HISELCT (R681 and later) PV High Selective Cutout	Cutout Allow	<i>range = Cutout, Allow default = Cutout</i>					
PVLOTP (R660 and later) PV Low Trip Point	_ _ _ _ _ _ _	<i>range = < PVHITP, NaN default = NaN</i>					
PVLOPR (R660 and later) PV Low Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
\$LOSELCT (R681 and later) PV Low Selective Cutout	Cutout Allow	<i>range = Cutout, Allow default = Cutout</i>					
ALENBST (R660 and later) Alarm Enable State	Enable Disable Inhibit	Displayed Yes No No		Logged Yes Yes No			

Alarming Display – N/A if PNTFORM = Component

As you enter alarm trip points and alarm priorities in the next section please refer to the following alarm Priority definitions and the note.

NoAction	Alarm is not reported to the system and not annunciated (R670 & later)
Journal	Alarm is historized, not reported to US, not annunciated
Low	Alarm is historized, reported to US, and annunciated
High	Alarm is historized, reported to US, and annunciated
Emergency	Alarm is historized, reported to US, and annunciated
Printer	Not historized, reported to the printer, not annunciated
JnlPrint	Alarm historized, reported to the printer, not annunciated

Note: If you do not enter a trip point value for a given pair of trip-points and alarm-priorities, do not circle an alarm priority option for that pair (that is, no trip point, no alarm priority).

USERID	_._._._._._._._._._._._._.__	<i>range</i> = Sixteen Character String <i>default</i> = Dashes
User ID Reservation		
EXTDATA	<i>None</i>	No data Mapped from SI device
External Data Option	IO_FL	Flags are mapped from SI device
	IO_NN	Numerics are mapped from SI device
	IO_STR	Strings are mapped from SI device

HPM Array Point

Unit:	Tagname:	Engineer:	Date:
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Configuration Display – N/A if EXTDATA = None

IOPNUM	_ _	<i>range = 1 to 40</i>
SI IOP Number		<i>default = blank</i>
FTANUM	1	<i>range = 1 or 2</i>
SI FTA Number	2	<i>default = 1</i>
DEVADDR	_ _ _ _	<i>range = 0 to 99999</i>
Serial Link Device Address		<i>default = 0</i>
SCANPRI	Low	<i>range = Low or High</i>
SI Data FTA Scan Priority	High	<i>default = Low</i>

FTA-Driver Program Auxiliary Data

AUXDATA1	_ _ _ _
Auxiliary Data 1	
AUXDATA2	_ _ _ _
Auxiliary Data 2	
AUXDATA3	_ _ _ _
Auxiliary Data 3	
AUXDATA4	_ _ _ _
Auxiliary Data 4	

range = Not applicable
default = NaN

FTA-Driver Program Allen-Bradley Data

AB_DATA1	_ _ _ _
A – B PLC type	
AB_DATA2	_ _ _ _
A – B PLC file #	
AB_DATA3	_ _ _ _
A – B Data type	
AB_DATA3	_ _ _ _
A – B Scan Period	

range = Not applicable
default = NaN

HPM Array Point

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Configuration Display (Continued)

FLSTIX					
Flag Array Start Index	_ _ _ _ _ _ _		<i>range =</i>	0 to 99999 when EXTDATA = IO_FL	
			<i>default =</i>	0 to 16384 all other EXTDATA options	
				0	
NFLAG					
Number of Flags	_ _ _ _ _ _ _		<i>range =</i>	0 to 512 when EXTDATA = IO_FL	
			<i>default =</i>	0 to 1023 all other EXTDATA options	
				0	
NNSTIX					
Numeric Array Start Index	_ _ _ _ _ _ _		<i>range =</i>	0 to 99999 when EXTDATA = IO_NN	
			<i>default =</i>	0 to 16384 all other EXTDATA options	
				0	
NNUMERIC					
Number of Numerics	_ _ _ _		<i>range =</i>	0 to 64 dep. on device, if EXTDATA = IO_NN	
			<i>default =</i>	0 to 240 all other EXTDATA options	
				0	
STRSTIX					
String Array Start Index	_ _ _ _ _ _ _		<i>range =</i>	0 to 99999 when EXTDATA = IO_STR	
			<i>default =</i>	0 to NSTRING for all other EXTDATA options (16384 max.)	
				0	
STRLEN					
String Length in Characters	8 16 32 64		<i>range =</i>	Not applicable	
			<i>default =</i>	8	
NSTRING					
Number of Strings	_ _ _ _		<i>range =</i>	0 to 240 when EXTDATA <> IO_STR	
			<i>default =</i>	0 to 64/STRLEN when EXTDATA = IO_STR	
				0	
TIMESTIX					
Time Array Start Index	_ _ _ _ _ _ _		<i>range =</i>	0 to 4096	
			<i>default =</i>	0	
NTIME					
Number of Times	_ _ _ _		<i>range =</i>	0 to 240	
			<i>default =</i>	0	
SPLOCK					
Data Array Setpoint Lock	Operator Supervis Engineer Program		<i>range =</i>	Not applicable	
			<i>default =</i>	Operator	

HPM Array Point

Unit:	Tagname:	Engineer:	Date:
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Configuration Display (Continued)

Descriptors for Flag, Numeric String, and Time Arrays

FLDESC Flag Descriptor	<i>range = Sixty-four Character String</i> <i>default = Blank</i>
_____ _____	
NDESC Numeric Descriptor	<i>range = Sixty-four Character String</i> <i>default = Spaces</i>
_____ _____	
STRDESC String Descriptor	<i>range = Sixty-four Character String</i> <i>default = Spaces</i>
_____ _____	
TIMEDESC Time Descriptor	<i>range = Sixty-four Character String</i> <i>default = Spaces</i>
_____ _____	

NOSTATES	2	2 digital switch states can be configured
Number of Digital States	3	3 digital switch states can be configured
NODINPTS	0	0 digin connections can be configured
# Of Digital Input Connections	1	1 digin connections can be configured
	2	2 digin connections can be configured

Unit:	Tagname:	Engineer:	Date:
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NODOPTS	0	0 digout connections can be configured
# Of Digital Output Connections	1	1 digout connections can be configured
	2	2 digout connections can be configured
	3	3 digout connections can be configured

NOLINPTS	__	<i>range</i> = 0 to 12 <i>default</i> = 0
Number of Logic Inputs		
LIABADOPT	Off On <i>Hold</i>	<i>range</i> = “Off” state is substituted for bad input <i>default</i> = “On” state is substituted for bad input last good value is substituted for bad input
Logic Bad Input Handling Option		

Refer to the following explanations when selecting the Primary Gate Algorithm :

- | | | | | | |
|--------|---|--------------------------------|--------|---|--------------------------------|
| GT | - | Greater Than PINN | GT2 | - | Greater Than PISRC |
| GE | - | Greater Than or Equal to PINN | GE2 | - | Greater Than or Equal to PISRC |
| LT | - | Less Than PINN | LT2 | - | Less Than PISRC |
| LE | - | Less Than or Equal to PINN | LE2 | - | Less Than or Equal to PISRC |
| EQ | - | Equal to PINN | EQ2 | - | Equal to PISRC |
| NE | - | Not Equal to PINN | NE2 | - | Not Equal to PISRC |
| IN_SET | - | Compare to In Set table values | INVERT | - | Invert the signal |

LISRC (n) *range* = Tagname Parameter
 Logic Input #n Connection *default* = Null.null

|-----|
 Point.Parameter

PIALGID (n)
 Primary Input Gate # Algorithm **NULL** INVERT IN_SET

PINN (n)
 Constant Numeric

PISRC (n)
 Second Input State

GT GE LT LE EQ NE GT2 GE2 LT2 LE2 EQ2 NE2

default = 0.0 range = <=> NaN

range = L1 through L12 correspond to the logic Inputs LISRC (1) – (12)
 default = Null

HPM Device Control Point

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N/A for PIALGID = NULL, INVERT, or IN_SET

PIDEADB (n)
Deadband

range = ≥ 0
default = 0.0

Refer to the following definitions when selecting the Secondary Input Gate Algorithm and destination :

Algorithms		Destinations	
DLY	- Delay	SI0, I0, I1, I2	- Interlocks
ONDLY	- On delay	P0, P1, P2	- Permissive Interlocks
OFFDLY	- Off delay	OPCMD	- Output Command
PULSE	- Pulse	SOCMD0, 1, 2	- Output Status Command 0, 1, 2
MAXPULSE	- Max pulse	SG1, SG2	- Secondary Gates 1, 2
MINPULSE	- Min pulse	PG1, 2, 3, 4	- Primary Gates 1 - 4

Input Connections Display (Continued)

SIALGID (n) Secondary Input
Gate # Algorithm =

NULL DLY ONDLY OFFDLY PULSE MAXPULSE MINPULSE

SIDLYTIM (n)
Delay or Pulse Width

range = 0 to 8000 seconds
default = 0

SIDSTN (n)	NONE	SI0	I0	I1	OPCMD	SOCMD0	SOCMD1	SOCMD2
Destination	I2	P0	P1	P2	SG1 PG3	SG2 PG4	PG1	PG2

In_Set Algorithm Configuration Display Applies if PIALGID(n) = In_Set

NNINSET (1)		NNINSET (6)	
NNINSET (2)		NNINSET (7)	
NNINSET (3)		NNINSET (8)	
NNINSET (4)		NNINSET (9)	
NNINSET (5)		NNINSET (10)	
Numerics 1-10			range = 0 to 32, 767 default = 0

HPM Device Control Point

Unit:	Tagname:	Engineer:	Date:
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Primary Gate Configuration Display

NOPGATE

Number of Primary Gates

range = 0 to 4

default = 0

Use a separate section of this form for each Primary Gate (n). Circle the Algorithm type (for example, AND, OR, etc.) and enter/select the parameters that apply for the algorithm. Up to four copies of this form are needed to configure all four possible Primary Gates and Destinations.

Refer to the following definitions when selecting the Primary Gate Algorithm and destination:

	<u>Algorithms</u>	<u>Destinations</u>
PAND	- AND, pulsed output	SI0, I0, I1, I2 - Interlocks
POR	- OR, pulsed output	P0, P1, P2 - Permissive Interlocks
PNAND	- NAND, pulsed output	OPCMD - Output Command
PNOR	- NOR, pulsed output	SOCMD0,1,2 - Output Status Command 0, 1, 2
PXOR	- XOR, pulsed output	SG1, SG2 - Secondary Gates 1, 2

PGALGID (n) Primary Gate # ____	Algorithm	AND	OR	NAND	NOR	XOR	PAND	POR	PNAND	PNOR	PXOR	
PGPLSWTH (n) Primary Gate Pulse Width												range = 0 to 8000 seconds default = 0
PGDSTN (n) Primary Gate #n Destination		SI0	I0	I1	OPCMD	SOCMD0	SOCMD1	SOCMD2				
		I2	P0	P1	P2	SG1	SG2					

HPM Device Control Point

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Secondary Gate Configuration Display

NOSGATE

Number of Secondary Gates

range = 0 to 2
default = 0

Use a separate section of this form for each Secondary Gate (n). Circle the Algorithm type (for example, AND, OR, etc.) and enter/select the parameters that apply for the algorithm. Up to two copies of this form are needed to configure the two possible Secondary Gates and Destinations.

Refer to the following definitions when selecting the Secondary Gate Algorithm and destination:

	<u>Algorithms</u>		<u>Destinations</u>
PAND	- AND, pulsed output	SI0, I0, I1, I2	- Interlocks
POR	- OR, pulsed output	P0, P1, P2	- Permissive Interlocks
PNAND	- NAND, pulsed output	OPCMD	- Output Command
PNOR	- NOR, pulsed output	SOCMD0,1,2	- Output Status Command 0, 1, 2
PXOR	- XOR, pulsed output		

SGALGID (n) Secondary Gate # ____	Algorithm	AND	OR	NAND	NOR	XOR	PAND	POR	PNAND	PNOR	PXOR
SGPLSWTH (n) Secondary Gate Pulse Width											
							range = 0 to 8000 seconds default = 0				
SGDSTN (n) Secondary Gate #n Destination	SI0	I0	I1	OPCMD	SOCMD0	SOCMD1	SOCMD2				
	I2	P0	P1	P2							

HPM Device Control Point

Unit:	Tagname:	Engineer:	Date:
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Output Connections Display

NOLOPTS

Number of Logic Outputs

|_|_|

range = 0 to 2

default = 0

Refer to the following parameter explanations when filling out the output-connections table below :

LOSRC

LODSTN Output-Connection Source

L1...L12
 FL1...FL12
 NN1...NN8
 D1, D2
 P0...P2
 SI0, IO...I2
 PISO1...PISO12
 SISO1...SISO12
 PGSO1...PGSO4
 SGSO1...SGSO2
 PINN1...PINN8

SECVAR

LOENBL

LODSTN Output Enable

L1...L12
 FL1...FL12
 D1 or D2
 SI0, IO...I2
 P0...P2
 PISO1...PISO12
 SISO1...SISO12
 PGSO1...PGSO4
 SGSO1...SGSO2

Device Control Output Destination Table

LODSTN (Logic Output Connection
 Destination) Destination Point.Parameter
 defaults = null.null

LOSRC

defaults = FL1

01|_|_|_|_|

02|_|_|_|_|

LOENBL

defaults = FL2

|_|_|_|_|

|_|_|_|_|

Flags and Numerics – Initial Values

Numerics 1-8 are used as local variables. Entry is a real number. Default = NaN

NN1 |_|_|_|_|_|_|_|

NN2 |_|_|_|_|_|_|_|

NN3 |_|_|_|_|_|_|_|

NN4 |_|_|_|_|_|_|_|

NN5 |_|_|_|_|_|_|_|

NN6 |_|_|_|_|_|_|_|

NN7 |_|_|_|_|_|_|_|

NN8 |_|_|_|_|_|_|_|

Flags 7-12 are used as local variables

FL7 Off On

FL8 Off On

FL9 Off On

FL10 Off On

FL11 Off On

FL12 Off On

Device Control Point	Interlock	Component
LOGICSRC Logic Source _____		Logic Point driving the interlock to this Device Control Point

HPM Device Control Point

Unit:	Tagname:	Engineer:	Date:
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PV / Input Configuration Display

DISRC (Digital-Input-Connection Source)		
Input #	DISRC (point.parameter)	Destination
1		D1
2		D2

Only if NODINPTS = 1

D1_1 D1_One PV State	PVState0 PVState1	Use & display contents of STATETXT0 Use & display contents of STATETXT1
--------------------------------	-----------------------------	--

D2D1_00 D2_D1 Zero_Zero PV State D2 = Off, D1 = Off	PVState1 PVState0 PVState2 MovPV BadPV	Use & display contents of STATETXT1 Use & display contents of STATETXT0 Use & display contents of STATETXT2* Use & display contents of MOVPTXT Use & display contents of BADPTXT
D2D1_01 D2D1 Zero_One PV State D2 = Off, D1 = On	PVState1 PVState0 PVState2 MovPV BadPV	Use & display contents of STATETXT1 Use & display contents of STATETXT0 Use & display contents of STATETXT2* Use & display contents of MOVPTXT Use & display contents of BADPTXT
D2D1_10 D2D1 One_Zero PV State D2 = On, D1 = Off	PVState1 PVState0 PVState2 MovPV BadPV	Use & display contents of STATETXT1 Use & display contents of STATETXT0 Use & display contents of STATETXT2* Use & display contents of MOVPTXT Use & display contents of BADPTXT
D2D1_11 D2D1 One_One PV State D2 = On, D1 = On	PVState1 PVState0 PVState2 MovPV BadPV	Use & display contents of STATETXT1 Use & display contents of STATETXT0 Use & display contents of STATETXT2* Use & display contents of MOVPTXT Use & display contents of BADPTXT

*Only applies if NOSTATES = 3

N/A if NODINPTS = 0

PVSRCOPT PV Source Option	Only Auto	PV Source selection is not available and Field wiring or memory fetch supplies PV.
	All	Selection from PVSOURCE is available & Operator/CL-program may supply PV.
PVSOURCE PV Source	Auto	Field wiring /memory -fetch supplies PV
	Man	PV is supplied by Operator or program
	Sub	A value is substituted by a CL program
	Track	PV tracks the output

HPM Device Control Point

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Output Configuration Display – N/A if NODOPTS = 0

		<u>NOSTATES</u>	<u>NODOPTS</u>
ST1_OP1	Off	2 - 3	1 - 3
State 1 - Output 1	On		
ST1_OP2	Off	2 - 3	2 - 3
State 1 - Output 2	On		
ST1_OP3	Off	2 - 3	3
State 1 - Output 3	On		
ST0_OP1	Off	1 - 3	1 - 3
State 0 - Output 1	On		
ST0_OP2	Off	1 - 3	2 - 3
State 0 - Output 2	On		
ST0_OP3	Off	1 - 3	3
State 0 - Output 3	On		
ST2_OP1	Off	3	1 - 3
State 2 - Output 1	On		
ST2_OP2	Off	3	2 - 3
State 2 - Output 2	On		
ST2_OP3	Off	3	3
State 2 - Output 3	On		
DODSTN (Digital Output Connection Destination)			
<u>Output #</u>	<u>DODSTN (point.parameter)</u>		
1			
2			
3			

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PULSEWTH		<i>range</i> = 0 0 to 60.0
Pulse Width For DevCtl Point Outputs	_._._._	<i>default</i> = 1.0

STCHGOPT State Change Option	None State 0	<i>range</i> = Not Applicable <i>default</i> = None
PAUSTIM State 0 Pause Time		<i>range</i> = 0 – 1000 seconds <i>default</i> = 0

SEALOPT Seal-in Circuit Option	None Sealin	<i>range</i> = Not Applicable <i>default</i> = None
--	-----------------------	--

LMREV Local Manual Reverse	Direct Reverse	<i>range</i> = Not Applicable <i>default</i> = Direct
LMSRC Local Manual Source		<i>range</i> = See Parameter Ref. Guide <i>default</i> = Null.null

|-----|
point.parameter

[illegible]

HPM Device Control Point

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Mode Configuration Display

N/A if NODOPTS = 0

NMODATTR Normal Mode Attribute	None Operator Program	Parameter NMODE sets "normal" mode Operator sets "normal" operating mode Program sets "normal" operating mode
MODEPERM Mode Permissive	Permit NotPerm	Operator can change this point's mode Optr. cannot change this point's mode
\$MODESEL (R630 and later) Mode Selection List	Man Prog	Man mode and Prog mode attribute are allowed for this data point.
OROPT Interlock Bypass Option	Off On	Optr. Cannot bypass this point's interlocks Operator can bypass this point's interlocks If BYPASS = ON
BYPASS Interlock Bypass State	Off On	Outputs not allowed to ignore interlocks Stores to OP & SO are allowed to ignore the interlocks

Alarming Display

N/A if PNTFORM = Componnt.

\$AUXUNIT Auxiliary Unit	_ _ _	<i>range</i> = Any configured LCN Unit ID <i>default</i> = Null (--)
PVNORMAL PV Value's Normal State Text	_ _ _ _ _ _ _ _ _	<i>range</i> = STATETXT (0) – STATETXT(2) <i>default</i> = Text equal to "None"
FBTIME Feedback Time	_ _ _	<i>range</i> = 0 to 1000 <i>default</i> = 0

N/A if FBTIME = 0

CMDFALTM Command Failure Timeout	_ _ _	<i>range</i> = 0 to 1000 seconds <i>default</i> = 0												
ALENBST Point Alarm Enable Status	Enable Disable Inhibit	<table> <tr> <td><u>Displayed</u></td><td><u>Logged</u></td><td><u>Reported to EIP</u></td></tr> <tr> <td>Yes</td><td>Yes</td><td>Yes</td></tr> <tr> <td>No</td><td>Yes</td><td>Yes</td></tr> <tr> <td>No</td><td>No</td><td>No</td></tr> </table>	<u>Displayed</u>	<u>Logged</u>	<u>Reported to EIP</u>	Yes	Yes	Yes	No	Yes	Yes	No	No	No
<u>Displayed</u>	<u>Logged</u>	<u>Reported to EIP</u>												
Yes	Yes	Yes												
No	Yes	Yes												
No	No	No												

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N/A if PNTFORM = Componnt or if NODINPTS = 0

SIOALOPT Safety Override Alarm	None Auto_Rtn Cnfm_Rqd	No Override Alarming Return to normal when Override is Cleared Confirm to Clear after Interlock is Cleared SIOALPPR must not be set to NoAction or Journal.						
OVRDALOP Override Alarm Option	None Auto_Rtn Cnfm_Rqd	No Override Alarming Return to normal when Override is Cleared Confirm to Clear after Interlock is Cleared OVRDALPR must not be set to NoAction or Journal.						
BADPVPR Bad PV Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	

Alarming Display (Continued) N/A if FBTIME = 0.

CMDDISPR Command Disagree Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
--	--	---------	-----	------	-----------	---------	----------	--

N/A if SIOALOPT = None

SIOALPR Override SIO Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
---	--	---------	-----	------	-----------	---------	----------	--

N/A if OVRDALOP = None.

OVRDALPR Override Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
--	--	---------	-----	------	-----------	---------	----------	--

EVTOPT Event Reporting Option	None Eip	No Reporting Digital PV changes invoke Event-Initiated Processing						
EIPPCODE Event Init Proc Destination								range = Not applicable default = Null

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CTLALGID Control Algorithm Identifier	<i>Null</i> PID Pos Pr	No affect PID with Position Proportional
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HPM PID With Position Proportional CTL Algorithm

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PV Configuration Display

PVEUHI PV High Range In Engr Units		<i>range</i> = PVEULO to PVEXEUHI <i>default</i> = 100.0	
PVEULO PV Low Range In Engr Units		<i>range</i> = PVEXEULO to PVEUHI <i>default</i> = 0.0	
PVFORMAT PV Decimal Point Format	D0 D2 D1 D3	D0=9999 D2=99.99 D1=999.9 D3 = 9.999	

PV Configuration Display (Continued)

PVSRCOPT PV Source Option	Only Auto All	PV Source selection is not available Selection from PVSOURCE is available
PVSOURCE PV Source	Auto Man Sub	Field-wiring /memory -fetch supplies PV PV is supplied by Operator or program A value is substituted by a CL program

OVERAL Overview Display Value		<i>range</i> = 0 to 100 <i>default</i> = 25
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Mode Configuration Display

BADCTLOP Bad PV Mode Shed Option	No_Shed Shed Hold Shed Low Shed High Shed Safe	<i>range</i> = See Parameter Reference Dictionary <i>default</i> = No_Shed
RCASOPT Remote Cascade Option	None Spc Rsp	Local cascade is the only valid cascade AM strategy stores to SP within SP limits.
SHEDTIME Remote Cascade Shed Time		<i>range</i> = 0 to 1000 (seconds) <i>default</i> = 0 (mode shed disabled)
SHEDMODE (If Shedtime ≥ 1) Shed Mode	Man Auto	Supervisory point will shed to Manual or Automatic mode.
NMODE RegCtl Point's Normal Mode	None Man Auto Cas Bcas	No configured "normal" operating mode Manual is the "normal" operating mode Automatic is "normal" operating mode Cascade is "normal" operating mode Backup Cascade is "normal" operating mode
		N/A if NMODE =None
NMODATTR Normal Mode Attribute	None Operator Program	Parameter NMODE sets "normal" mode Operator sets "normal" operating mode Program sets "normal" operating mode

HPM PID with Position Porportional CTL Algorithm

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MODEPERM Mode Permissive	Permit NotPerm	Operator can change this point's mode Optr. cannot change this point's mode
\$MODESEL (R630 and later) Mode Selection List	Man Prog Auto Cas Bcas \$HEDMAN	Man, Auto, Cas and Bcas mode, and Prog mode attribute are allowed for this data point.
\$BADOCMD (R680 and later) Bad Output Connection Mode	15_SEC 60_SEC NO_SHED	Point will shed to Manual immediately Point will shed to Manual after 15 seconds Point will shed to Manual after 60 seconds Do not shed mode
EXTSWOPT External Mode Switching	None Ems	No external mode switching is allowed Ext. source can change point's mode

Setpoint Display

SPHILM SP High Limit		<i>range</i> = SPLOLM – SPEUHI, and NaN <i>default</i> = NaN
SPLOLM SP Low Limit		<i>range</i> = SPEULO – SPHILM, and NaN <i>default</i> = NaN
\$SPTOL Set Point Tolerance		<i>range</i> = ≥ 0.0 <i>default</i> = 0.0
SP Set Point In Engineering Units Of The PV		<i>range</i> = SPLOLM to SPHILM <i>default</i> = 0.0
SPOPT Set Point Option Asp is invalid if PNTFORM=Componnt.	None Tv Asp	No specialized options are available Target value processing Advisory deviation alarming
RB OPT Ratio Bias Option	NoRatBi FixRatBi AutoRat AutoBi	No ratio/bias is used to calculate the SP Fixed ratio (R) and fixed bias(B) are used R is back calculated during initialization B is back calculated during initialization
RTHILM Ratio High Limit		<i>range</i> = RTLOLM – 100.0, & NaN <i>default</i> = 100.0
RTLOLM Ratio Low Limit		<i>range</i> = 0.0001 - RTHILM, & NaN <i>default</i> = 0.0001
BSHILM Bias High Limit		<i>range</i> = $\geq$ BSLOLM, & NaN <i>default</i> = 50.0
BSLOLM Bias Low Limit		<i>range</i> = $\leq$ BSHILM, & NaN <i>default</i> = -50.0
RATIO Ratio		<i>range</i> = RTLOLM to RTHILM <i>default</i> = 1.0
BIAS Bias Parameter		<i>range</i> = BSLOLM to BSHILM <i>default</i> = 0.0

Unit:	Tagname:	Engineer:	Date:
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PIDFORM PID Controller Form	Interact Ideal	(Proportional + Integral) x (Derivative) (Proportional) +(Integral) + (Derivative)
CTLEQN Control Equation Type	EqA EqB EqC EqD	P, I & D act on Error P & I act on Error, D acts on PV I acts on Error, P & D act on PV Integral-only control
PVTRACK PV Tracking Option	No Track Track	SP will never be set equal to PV Man mode or init. causes SP to track PV
CTLACTN Control Action	Reverse Direct	As PV increases, Output decreases As PV increases, Output increases

N/A if CTLEQN = EqD.

GAPLOPT	Ext	NonLin	Gap	K = KFF*KEXT (gain from external source) K = KFF*((NLFM)+(NLGAIN*[PVP-SPP]/[100])) K = KLIN*KGAP if PV>(SP-GAPLO) or if PV<(SP+GAPHI), else K = KLIN. K = the value entered by the user
Gain (K) Option				
K		Lin		
Overall Gain In % Per %				range = 0.0 – 240.0 default = Not applicable
KLIN				range = 0.0 – 240.0 default = 1.0
Linear Gain Factor				
KGAP				range = 0.0 to 1.0 default = 1.0
Gap Gain Factor				
GAPHI				range = >=0.0 default = 0.0
Gap High Limit				
GAPLO				range = >=0.0 default = 0.0
Gap Low Limit				
NLGAIN				range = 0.0 – 240.0 default = 0.0
Non Linear Gain				
NLFM		0		K can go to zero
Non Linearity Form		1		K cannot go to < 1
KEXT				range = 0.0 to 240.0 default = 1.0
External Gain Modifier				

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[illegible]

HPM PID With Position Proportional CTL Algorithm

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OPHISRC	_____	
OPHIFL Input Connection	point.parameter	range = See Parameter Ref.Guide default = Null.null
OPLOSRC	_____	
OPLOFL Input Connection	point.parameter	range = See Parameter Ref.Guide default = Null.null
SAFOPCMD	Idle Raise Lower	range = Not Applicable default = Idle
Safe OP Command		
LMSRC	_____	
Local Manual Source	point.parameter	range = See Parameter Ref.Guide default = Null.null

Reg Control PidPosPr Algo Display (Continued)

NOCINPTS	_	range = 1 to 2 default = 1
#Of Control Input Connections		
CISRC (Control-Input-Connection Source)		<u>Destination Para</u>
<u>Input#</u>	<u>CISRC(n)</u> (point.parameter)	<u>-meter (CIDSTN)</u>
1	_____	PVAUTO
2	_____	SP

Alarming Display – N/A if PNTFORM = Componnt.

\$AUXUNIT	_ _	range = Any configured LCN Unit ID default = Null (--)
Auxiliary Unit		
\$ALDLOPT (R620 and later)	None	Off Delay Lo
Alarm Delay Option	OffDlyLo	Off Delay Hi
	OffDlyHi	Off Delay Emergency
	OffDlyEm	On Delay Lo
	OnDlyLo	On Delay Hi
	OnDlyHi	On Delay Emergency
	OnDlyEm	
\$DLBADAL (R681 and later)	None	range = Offdelay, None
Delay for BADPV	Offdelay	default = None
DLYTIME (R620 and later)	_ . _ . _ .	range = 0 to 9999 seconds default = 0 seconds
Delay Time		
\$REALARM (R660 and later)	_ . _ . _ .	range = 0 to 9999 minutes default = 0 minutes
Re-alarming Interval		
BADOCPT	Off	Bad Output Connection Alarm is not initially permitted
BADOC Alarm Option	On	Bad Output Connection Alarm is initially permitted
BADOCPR	NoAction	Journal Low High Emergency Printer JnlPrint
BADOC Alarm Priority	(R670 and later)	

HPM PID With Position Proportional CTL Algorithm

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Alarming Display N/A if PNTFORM = Compnnt (Continued)

PVALDB PV Alarm Deadband In %	Half	One	Two	Three	Four	Five	EU	
PVALDBEU PV Alarm Deadband in EU	_ _ _ _ _ _ _				range >= 0.0 default = 1.0			
PVHITP PV High Alarm Trip Point	_ _ _ _ _ _ _				range = PVLOTP - PVHHTP, & NaN default = NaN			
PVHIPR PV High Alarm Priority	NoAction (R670 and later)		Journal	Low	High	Emergency	Printer	JnlPrint
\$HISELCT (R681 and later) PV High Selective Cutout	Cutout Allow		range = Cutout, Allow default = Cutout					
PVLOTP PV Low Alarm Trip Point	_ _ _ _ _ _ _				range = PVLLTP - PVHITP, & NaN default = NaN			
PVLOPR PV Low Alarm Priority	NoAction (R670 and later)		Journal	Low	High	Emergency	Printer	JnlPrint
\$LOSELCT (R681 and later) PV Low Selective Cutout	Cutout Allow		range = Cutout, Allow default = Cutout					
PVHHTP PV Hi Hi Alarm Trip Point	_ _ _ _ _ _ _				range = PVHITP to PVEUHI, & NaN default = NaN			
PVHHPR PV High High Alarm Priority	NoAction (R670 and later)		Journal	Low	High	Emergency	Printer	JnlPrint
\$HHSELCT (R681 and later) PV High High Selective Cutout	Cutout Allow		range = Cutout, Allow default = Cutout					
PVLLTP PV Low Low Alarm Trip Point	_ _ _ _ _ _ _				range = PVEULO - PVLOTP, & NaN default = NaN			
PVLLPR PV Low Low Alarm Priority	NoAction (R670 and later)		Journal	Low	High	Emergency	Printer	JnlPrint
\$LLSELCT (R681 and later) PV Low Low Selective Cutout	Cutout Allow		range = Cutout, Allow default = Cutout					
PVSGCHTP PV Significant Change Alarm Trip Point	_ _ _ _ _ _ _				range = >=0.0, and NaN default = NaN			

Unit:	Tagname:	Engineer:	Date:
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Alarming Display (Continued)

PVROCTPT PV Pos Rate Of Change Trip Point	_____							<i>range = >=0.0, and NaN default = NaN</i>
PVROCPPR PV Pos Rate of Chg Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
\$RPSELCT (R681 and later) PV Positive Rate of Change Selective Cutout	Cutout Allow							<i>range = Cutout, Allow default = Cutout</i>
PVROCNTP PV Neg Rate Of Change Trip Point	_____							<i>range = >=0.0, and NaN default = NaN</i>
PVROCNPR PV Neg Rate Of Chg Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
\$RNSELCT (R681 and later) PV Negative Rate of Change Selective Cutout	Cutout Allow							<i>range = Cutout, Allow default = Cutout</i>
BADPVPR Bad-PV Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
DEVHITP Deviation High Alarm Trip Point	_____							<i>range = >=0.0, and NaN default = NaN</i>
DEVHIPR Deviation High Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
\$DHSELCT (R681 and later) Deviation High Selective Cutout	Cutout Allow							<i>range = Cutout, Allow default = Cutout</i>
DEVLOTP Deviation Low Alarm Trip Point	_____							<i>range = >=0.0, and NaN default = NaN</i>
DEVLOPR Deviation Low Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
\$DLSELCT (R681 and later) Deviation Low Selective Cutout	Cutout Allow							<i>range = Cutout, Allow default = Cutout</i>
ADVDEVTP Advisory Deviation Alarm Trip Point (When SPOPT = ASP)	_____							<i>range = >=0.0, and NaN default = NaN</i>
ADVDEVPR Advisory Deviation Alarm Priority (When SPOPT = ASP)	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
ALENBST Point Alarm Enable Status	Enable Disable Inhibit			<u>Displayed</u> Yes No No		<u>Logged</u> Yes Yes No	<u>Reported to EIP</u> Yes Yes No	

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Point Assignment Display

Point Assignment Display		
NAME Tag Name		<i>range</i> = Not applicable <i>default</i> = Blank
NODETYPE Node Type	PM APM HPM	A PM Point Is Being Configured An APM Point Is Being Configured An HPM Point Is Being Configured
PNTFORM Point Form	FuLL Componnt	Point is fully displayed and alarmed Point is partially displayed but not alarmed

PTDESC Point Descriptor	_____	<i>range</i> = Not applicable <i>default</i> = Blank
EUDESC Engineering Units Descriptor	_____	<i>range</i> = Not applicable <i>default</i> = Blank
KEYWORD Keyword Descriptor	_____	<i>range</i> = Not applicable <i>default</i> = Blank
ASSOCDSP Associated Display	_____	<i>range</i> = Not applicable <i>default</i> = Blank
\$CDETAIL (R620 and later) Point Custom Display	_____	<i>range</i> = Not applicable <i>default</i> = Blank
UNIT Process Unit Point Is Assigned To	____	<i>range</i> = Two Alphanumeric Characters <i>default</i> = Not applicable
NTWKNUM NIM's UCN That This Point Is In	____	<i>range</i> = 1 to 20 <i>default</i> = Not applicable
NODENUM HPMM's Address On The UCN	____	<i>range</i> = 1 to 64 <i>default</i> = Not applicable
MODNUM I/O Module Number	____	<i>range</i> = 0 <i>default</i> = 0 (Control Processor Point)
SLOTNUM Slot Number	____	<i>range</i> = 1-250 (but <= NCTLSLOT) <i>default</i> = Not applicable

		N/A if PNTFORM = Componnt.
PRIMMOD		<i>range</i> = Not Applicable
Primary Module ID	_._._._._._._._._._._._._._._._	<i>default</i> = Null
USERID		Sixteen Character String
User ID Reservation	_._._._._._._._._._._._._._._._	<i>default</i> = dashes

HPM MulDiv CTL Algorithm

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Control Algorithm Display

CTLALGID Control Algorithm Identifier	NULL MulDiv	No affect Multiply Divide
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PV Configuration Display

OVERAL Overview Display Value	. . .	range = 0 to 100 default = 25
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Mode Configuration Display

BADCTLOP Bad PV Mode Shed Option	No_Shed Shed Hold Shed Low Shed High Shed Safe	range = See HPM Parameter Reference Dictionary default = No_Shed
RCASOPT Remote Cascade Option	None DDC	Local cascade is the only valid cascade AM strategy stores to SP within SP limits
SHEDTIME Remote Cascade Shed Time		range = 0 to 1000 (seconds) default = 0 (mode shed disabled)
SHEDMODE (If Shedtime ≥ 1) Shed Mode	Man BCas	Supervisory point will shed to Manual or Automatic mode.
NMODE RegCtl Point's Normal Mode	None Man Cas Bcas	No configured "normal" operating mode Manual is the "normal" operating mode Cascade is "normal" operating mode Backup Cascade is "normal" operating mode
N/A if NMODE =None		
NMODATTR Normal Mode Attribute	None Operator Program	Parameter NMODE sets "normal" mode Operator sets "normal" operating mode Program sets "normal" operating mode

MODEPERM Mode Permissive	Permit NotPerm	Operator can change this point's mode Optr. cannot change this point's mode
\$MODESEL (R630 and later) Mode Selection List	Man Cas Prog Bcas	Man, Cas and Bcas mode, and Prog mode attribute are allowed for this data point.
\$BADOCMD (R680 and later) Bad Output Connection Mode	\$SHEDMAN 15_SEC 60_SEC NO_SHED	Point will shed to Manual immediately Point will shed to Manual after 15 seconds Point will shed to Manual after 60 seconds Do not shed mode
EXTSWOPT External Mode Switching	None Ems	No external mode switching is allowed Ext. source can change point's mode

range = <=XEUHI
default = 0.0

HPM MulDiv CTL Algorithm

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Reg Control MulDiv Algo Display

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HPM MulDiv CTL Algorithm

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Alarming Display - N/A if PNTFORM = Compoent (Continued)

As you enter alarm trip points and alarm priorities in the next section please refer to the following alarm priority definitions and the note.

NoAction

Journal

Low

High

Emergency

Printer

JnlPrint

Alarm is not reported to the system and not annunciated (R670 & later)

Alarm is historized, not reported to US, not annunciated

Alarm is historized, reported to US, and annunciated

Alarm is historized, reported to US, and annunciated

Alarm is historized, reported to US, and annunciated Not

historized, reported to the printer, not annunciated Alarm

historized, reported to the printer, not annunciated

Note: If you do not enter a trip point value for a given pair of trip-points and alarm-priorities, do not circle an alarm priority option for that pair (that is, no trip point, no alarm priority).

\$AUXUNIT Auxiliary Unit	_ _	<i>range</i> = Any configured LCN Unit ID <i>default</i> = Null (- -)					
\$DLBADAL (R681 and later) Delay for BADCTL	None Offdelay	<i>range</i> = Offdelay, None <i>default</i> = None					
DLTIME (R620 and later) Delay Time	_ _ _ _	<i>range</i> = 0 to 9999 seconds <i>default</i> = 0 seconds					
\$REALARM (R660 and later) Re-alarming Interval	_ _ _ _	<i>range</i> = 0 to 9999 minutes <i>default</i> = 0 minutes					
BADOCOPT BADOC Alarm Option	Off On	Bad Output Connection Alarm is not initially permitted Bad Output Connection Alarm is Initially permitted					
BADOCPR BADOC Alarm Priority	NoAction Journal Low High Emergency Printer JnlPrint (R670 and later)						
OPALDB Output Alarm Deadband	_ _ _ _ _ _ _	<i>range</i> = 0 - 25.0 but $\leq$ (OPHITP-OPLOTP)/2 <i>default</i> = 5.0 Output Units					
OPHITP Output High Alarm Trip Point	_ _ _ _ _ _ _	<i>range</i> = OPLOTP TO OPHILM and NaN <i>default</i> = NaN					
OPHIPR Output High Alarm Priority	NoAction Journal Low High Emergency Printer JnlPrint (R670 and later)						
OPLOTP Output Low Alarm Trip Point	_ _ _ _ _ _ _	<i>range</i> = OPLOLM TO OPHITP and NaN <i>default</i> = NaN					
OPLOPR Output Low Alarm Priority	NoAction Journal Low High Emergency Printer JnlPrint (R670 and later)						
BADCTLPR Bad-control Alarm Priority	NoAction Journal Low High Emergency Printer JnlPrint (R670 and later)						
ALENBST Point Alarm Enable Status	Enable Disable Inhibit	<u>Displayed</u> Yes No No	<u>Logged</u> Yes Yes No	<u>Reported To EIP</u> Yes Yes No			

	N/A if PNTFORM = Component
PRIMMOD Primary Module ID	<i>range</i> = Not Applicable <i>default</i> = Null
USERID User ID Reservation	Sixteen Character String <i>default</i> = dashes

HPM Summer CTL Algorithm

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Control Algorithm Display

CTLALGID Control Algorithm Identifier	NULL Summer	No affect Summer
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PV Configuration Display

OVERAL Overview Display Value	. . .	range = 0 to 100 default = 25
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Mode Configuration Display

BADCTLOP Bad PV Mode Shed Option	No_Shed Shed Hold Shed Low Shed High Shed Safe	range = See Parameter Reference Dictionary default = No_Shed
RCASOPT Remote Cascade Option	None DDC	Local cascade is the only valid cascade AM strategy stores to SP within SP limits
SHEDTIME Remote Cascade Shed Time		range = 0 to 1000 (seconds) default = 0 (mode shed disabled)
SHEDMODE (If Shedtime ≥ 1) Shed Mode	Man BCas	Supervisory point will shed to Manual, or Automatic mode.
NMODE RegCtl Point's Normal Mode	None Man Cas Bcas	No configured "normal" operating mode Manual is the "normal" operating mode Cascade is "normal" operating mode Backup Cascade is "normal" operating mode
		N/A if NMODE =None
NMODATTR Normal Mode Attribute	None Operator Program	Parameter NMODE sets "normal" mode Operator sets "normal" operating mode Program sets "normal" operating mode

MODEPERM Mode Permissive	Permit NotPerm	Operator can change this point's mode Optr. cannot change this point's mode
\$MODESEL (R630 and later) Mode Selection List	Man Prog Cas Bcas	Man, Cas and Bcas mode, and Prog mode attribute are allowed for this data point.
\$BADOCMD (R680 and later) Bad Output Connection Mode	SHEDMAN 15_SEC 60_SEC NO_SHED	Point will shed to Manual immediately Point will shed to Manual after 15 seconds Point will shed to Manual after 60 seconds Do not shed mode
EXTSWOPT External Mode Switching	None Ems	No external mode switching is allowed Ext. source can change point's mode

HPM Summer CTL Algorithm

Unit:	Tagname:	Engineer:	Date:
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Mode Configuration Display

B Output Bias		<i>range</i> = Any Real number <i>default</i> = 0.0
K Overall Gain		<i>range</i> = Not applicable <i>default</i> = 1.0.
RATE1 Ramp Rate For Output Bias		<i>range</i> = ≥ 0.0 , NaN <i>default</i> = 0.0
XEUHI X Input High Range In EUs		<i>range</i> = $\geq$ XEULO <i>default</i> = 100.0
XEULO X Input Low Range In EUs		<i>range</i> = $\leq$ XEUHI <i>default</i> = 0.0

HPM Summer CTL Algorithm

Unit:	Tagname:	Engineer:	Date:
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Reg Control Summer Algo Display - N/A if PNTFORM = Componnt

M		<i>range = 1 - 4</i> <i>default = 2</i>
No. of Inputs for M		
K1		<i>range = Not applicable</i> <i>default = 1.0</i>
Scaling For Input 1		
K2		<i>range = Not applicable</i> <i>default = 1.0</i>
Scaling For Input 2		
K3		<i>range =</i> <i>default = 1.0</i>
Scaling For Input 3		
K4		<i>range =</i> <i>default = 1.0</i>
Scaling For Input 4		
NOCINPTS		<i>range = M-1 - M</i> <i>default = 1.0</i>
# Of Control Input Connections		
CISRC (Control-Input-Connection Source)		<u>Destination Parameter</u>
<u>Input #</u>	<u>CISRC (point.parameter)</u>	<u>(CIDSTN)</u>
1		X2
2		X3
3		X4
4		X1

HPM Summer CTL Algorithm

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Unit:	Tagname:	Engineer:	Date:
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Output Configuration Display

NOCOPTS *range = 0 to 4*
Of Control Output Connections *default = 1*

CODSTN (Control-Output-Connection Destination)

Output #

CODSTN (point.parameter)

1	
2	
3	
4	

OPHILM *range = OPLOLM – 106.9 & NaN*
Output High Limit In Percent *default = 105.0*

OPLOLM *range = -6.9 – OPHILM, & NaN*
Output Low Limit In Percent *default = -5.0*

SAFEOP *range = -6.9 to 106.9 & NaN*
Safe Output Limit In Percent *default = NaN*

OPMCHLM *range = >=0.0, and NaN*
Output Minimum Change In % *default = 0.0*

OPROCLM *range = >=0.0, and NaN*
OP Rate Of Change Limit In % *default = NaN*

\$OPTOL *range = 0.0 to 106.9*
Output Tolerance in % *default = 0.0*

OUTIND (R620 and later) *Direct*
Reverse Output Indicator *Reverse*

4 - 20mA = 0 – 100%
20 – 4mA = 0 – 100%
Note: With R670 and later, when this parameter is set to Reverse, the values of the displayed output and the related output parameters are changed. Refer to the definition of the OUTIND parameter in the PM Family Parameter Dictionary, XPM09.

HPM Summer CTL Algorithm

Unit:	Tagname:	Engineer:	Date:
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Alarming Display - N/A if PNTFORM = Componnt

As you enter alarm trip points and alarm priorities in the next section please refer to the following alarm priority definitions and the note.

NoAction	Alarm is not reported to the system and not annunciated (R670 & later)
Journal	Alarm is historized, not reported to US, not annunciated
Low	Alarm is historized, reported to US, and annunciated
High	Alarm is historized, reported to US, and annunciated
Emergency	Alarm is historized, reported to US, and annunciated Not
Printer	historized, reported to the printer, not annunciated Alarm
JnlPrint	historized, reported to the printer, not annunciated

Note: If you do not enter a trip point value for a given pair of trip-points and alarm-priorities, do not circle an alarm priority option for that pair (that is, no trip point, no alarm priority).

Alarming Display - N/A if PNTFORM = Componnt (Continued)

\$AUXUNIT Auxiliary Unit	<u> </u>	<i>range</i> = Any configured LCN Unit ID <i>default</i> = Null (- -)					
\$DLBADAL (R681 and later) Delay for BADCTL	None Offdelay	<i>range</i> = Offdelay, None <i>default</i> = None					
DLTIME (R620 and later) Delay Time	<u> </u>	<i>range</i> = 0 to 9999 seconds <i>default</i> = 0 seconds					
BADOCOPT BADOC Alarm Option	Off On	Bad Output Connection Alarm is not initially permitted Bad Output Connection Alarm is Initially permitted					
BADOCPR BADOC Alarm Priority	NoAction Journal Low (R670 and later)	High	Emergency	Printer	JnlPrint		

OPALDB Output Alarm Deadband	<u> </u>	<i>range</i> = 0 - 25.0 but $\leq$ (OPHITP-OPLOTP)/2 <i>default</i> = 5.0 Output Units						
OPHITP Output High Alarm Trip Point	<u> </u>	<i>range</i> = OPLOTP TO OPHILM and NaN <i>default</i> = NaN						
OPHIPR Output High Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	
OPLOTP Output Low Alarm Trip Point	<u> </u>	<i>range</i> = OPLOLM TO OPHITP and NaN <i>default</i> = NaN						
OPLOPR Output Low Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint	

BADCTLPR Bad-control Alarm Priority	NoAction Journal Low High Emergency Printer JnlPrint (R670 and later)						
ALENBST Point Alarm Enable Status	Enable Disable Inhibit	<u>Displayed</u> Yes No No	<u>Logged</u> Yes Yes No	<u>Reported To EIP</u> Yes Yes No			

NIM Library Configuration

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Unit:	Tagname:	Engineer:	Date:
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Network Number(NTWKNUM)

Library Number (LIBRYNUM)

range = 1 - 3

NIM Library Configuration Display

Index	Entry(LIBRYTXT)	Index	Entry(LIBRYTXT)	Index	Entry(LIBRYTXT)	Index	Entry(LIBRYTXT)
_01*		_02		_03		_04	
_05		_06		_07		_08	
_09		_10		_11		_12	
_13		_14		_15		_16	
_17		_18		_19		_20	
_21		_22		_23		_24	
_25		_26		_27		_28	
_29		_30		_31		_32	
_33		_34		_35		_36	
_37		_38		_39		_40	
_41		_42		_43		_44	
_45		_46		_47		_48	
_49		_50					

NIM Library Configuration Display

_51		_52		_53		_54	
_55		_56		_57		_58	
_59		_60		_61		_62	
_63		_64		_65		_66	
_67		_68		_69		_70	
_71		_72		_73		_74	
_75		_76		_77		_78	
_79		_80		_81		_82	
_83		_84		_85		_86	
_87		_88		_89		_90	
_91		_92		_93		_94	
_95		_96		_97		_98	
_99		_00					

Note that each library can contain upto 1000 entries; therefore, upto 5 copies of this form may be needed to configure each library.

*No entry allowed for Library 1, Index 001

NIM Library Configuration

Unit:	Tagname:	Engineer:	Date:
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Network Number(NTWKNUM)

|_|_|

Library Number (LIBRYNUM)

|_|

range = 1 - 3

NIM Library Configuration Display

Index	Entry(LIBRYTXT)	Index	Entry(LIBRYTXT)	Index	Entry(LIBRYTXT)	Index	Entry(LIBRYTXT)
_01*	_ _ _ _ _ _ _	_02	_ _ _ _ _ _ _	_03	_ _ _ _ _ _ _	_04	_ _ _ _ _ _ _
_05	_ _ _ _ _ _ _	_06	_ _ _ _ _ _ _	_07	_ _ _ _ _ _ _	_08	_ _ _ _ _ _ _
_09	_ _ _ _ _ _ _	_10	_ _ _ _ _ _ _	_11	_ _ _ _ _ _ _	_12	_ _ _ _ _ _ _
_13	_ _ _ _ _ _ _	_14	_ _ _ _ _ _ _	_15	_ _ _ _ _ _ _	_16	_ _ _ _ _ _ _
_17	_ _ _ _ _ _ _	_18	_ _ _ _ _ _ _	_19	_ _ _ _ _ _ _	_20	_ _ _ _ _ _ _
_21	_ _ _ _ _ _ _	_22	_ _ _ _ _ _ _	_23	_ _ _ _ _ _ _	_24	_ _ _ _ _ _ _
_25	_ _ _ _ _ _ _	_26	_ _ _ _ _ _ _	_27	_ _ _ _ _ _ _	_28	_ _ _ _ _ _ _
_29	_ _ _ _ _ _ _	_30	_ _ _ _ _ _ _	_31	_ _ _ _ _ _ _	_32	_ _ _ _ _ _ _
_33	_ _ _ _ _ _ _	_34	_ _ _ _ _ _ _	_35	_ _ _ _ _ _ _	_36	_ _ _ _ _ _ _
_37	_ _ _ _ _ _ _	_38	_ _ _ _ _ _ _	_39	_ _ _ _ _ _ _	_40	_ _ _ _ _ _ _
_41	_ _ _ _ _ _ _	_42	_ _ _ _ _ _ _	_43	_ _ _ _ _ _ _	_44	_ _ _ _ _ _ _
_45	_ _ _ _ _ _ _	_46	_ _ _ _ _ _ _	_47	_ _ _ _ _ _ _	_48	_ _ _ _ _ _ _
_49	_ _ _ _ _ _ _	_50	_ _ _ _ _ _ _				

NIM Library Configuration Display

_51	_ _ _ _ _ _ _	_52	_ _ _ _ _ _ _	_53	_ _ _ _ _ _ _	_54	_ _ _ _ _ _ _
_55	_ _ _ _ _ _ _	_56	_ _ _ _ _ _ _	_57	_ _ _ _ _ _ _	_58	_ _ _ _ _ _ _
_59	_ _ _ _ _ _ _	_60	_ _ _ _ _ _ _	_61	_ _ _ _ _ _ _	_62	_ _ _ _ _ _ _
_63	_ _ _ _ _ _ _	_64	_ _ _ _ _ _ _	_65	_ _ _ _ _ _ _	_66	_ _ _ _ _ _ _
_67	_ _ _ _ _ _ _	_68	_ _ _ _ _ _ _	_69	_ _ _ _ _ _ _	_70	_ _ _ _ _ _ _
_71	_ _ _ _ _ _ _	_72	_ _ _ _ _ _ _	_73	_ _ _ _ _ _ _	_74	_ _ _ _ _ _ _
_75	_ _ _ _ _ _ _	_76	_ _ _ _ _ _ _	_77	_ _ _ _ _ _ _	_78	_ _ _ _ _ _ _
_79	_ _ _ _ _ _ _	_80	_ _ _ _ _ _ _	_81	_ _ _ _ _ _ _	_82	_ _ _ _ _ _ _
_83	_ _ _ _ _ _ _	_84	_ _ _ _ _ _ _	_85	_ _ _ _ _ _ _	_86	_ _ _ _ _ _ _
_87	_ _ _ _ _ _ _	_88	_ _ _ _ _ _ _	_89	_ _ _ _ _ _ _	_90	_ _ _ _ _ _ _
_91	_ _ _ _ _ _ _	_92	_ _ _ _ _ _ _	_93	_ _ _ _ _ _ _	_94	_ _ _ _ _ _ _
_95	_ _ _ _ _ _ _	_96	_ _ _ _ _ _ _	_97	_ _ _ _ _ _ _	_98	_ _ _ _ _ _ _
_99	_ _ _ _ _ _ _	_00	_ _ _ _ _ _ _				

HPM Output Select CTL Algorithm

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Point Assignment Display

NAME Tag Name	_____	<i>range = Not applicable</i> <i>default = Blank</i>
NODETYP Node Type	PM APM HPM	Output Select algorithm is applicable only to the HPM.
PNTFORM Point Form	FuLL Componnt	Point is fully displayed and alarmed Point is partially displayed but not alarmed

PTDESC Point Descriptor	_____	<i>range = Not applicable</i> <i>default = Blank</i>
EUDESC Engineering Units Descriptor	_____	<i>range = Not applicable</i> <i>default = Blank</i>
KEYWORD Keyword Descriptor	_____	<i>range = Not applicable</i> <i>default = Blank</i>
ASSOCDSP Associated Display	_____	<i>range = Not applicable</i> <i>default = Blank</i>
\$CDETAIL (R620 and later) Point Custom Display	_____	<i>range = Not applicable</i> <i>default = Blank</i>
UNIT Process Unit Point Is Assigned To	____	<i>range = Two Alphanumeric Characters</i> <i>default = Not applicable</i>
NTWKNUM NIM's UCN That This Point Is In	____	<i>range = 1 to 20</i> <i>default = Not applicable</i>
NODENUM HPMM's Address On The UCN	____	<i>range = 1 to 64</i> <i>default = Not applicable</i>
MODNUM I/O Module Number	____	<i>range = 0</i> <i>default = 0 (Control Processor Point)</i>
SLOTNUM Slot Number	____	<i>range = 1-250 (but <= NCTLSLOT)</i> <i>default = Not applicable</i>

N/A if PNTFORM = Componnt.

PRIMMOD Primary Module ID	_____	<i>range = Not Applicable</i> <i>default = Null</i>
USERID User ID Reservation	_____	Sixteen Character String <i>default = dashes</i>

HPM Output Select CTL Algorithm

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Control Algorithm Display

CTLALGID Control Algorithm Identifier	NULL Null	
	OPSelect Output Select	

PV Configuration Display

OVERVAL Overview Display Value	_ _ _ _	<i>range</i> = 0 to 100 <i>default</i> = 25
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Mode Configuration Display

BADCTLOP Bad PV Mode Shed Option	No_Shed Shed Hold Shed Low Shed High Shed Safe	<i>range</i> = See PM Family Param. Ref. Dict. <i>default</i> = No_Shed
NMODE RegCtl Point's Normal Mode	None Man Cas	No configured "normal" operating mode Manual is the "normal" operating mode Cascade is "normal" operating mode

Mode Configuration Display (Continued) N/A if NMODE = None.

MODEPERM Mode Permissive	Permit NotPerm	Operator can change this point's mode Operator cannot change this point's mode
\$MODESEL (R630 and later) Mode Selection List	Man Prog Cas	Man and Cas mode, and Prog mode attribute are allowed for this data point.
\$BADOCMD (R680 and later) Bad Output Connection Mode	SHEDMAN 15_SEC 60_SEC NO_SHED	Point will shed to Manual immediately Point will shed to Manual after 15 seconds Point will shed to Manual after 60 seconds Do not shed mode
EXTSWOPT External Mode Switching Option	None Ems	No external mode switching is allowed External source can change point's mode

Reg Control OPSelect Algo Display

CTLEQN Algorithm Equation Type	EqA EqB	Operator controls select position Program or logic controls select position
RATE1 Internal Bias Ramping Rate	_ _ _ _	<i>range</i> = >=0.0, NaN <i>default</i> = 0.0
B Output Bias	_ _ _ _	<i>range</i> = N/A <i>default</i> = 0.0
XEUHI X Input High Range in EUs	_ _ _ _	<i>range</i> = >=XEULO <i>default</i> = 100.0
XEULO X Input Low Range in Eus	_ _ _ _	<i>range</i> = <=XEUHI <i>default</i> = 0.0

HPM Output Select CTL Algorithm

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NOCINPTS *range = 0 or 1*
#Of Control Input Connections *default = 0*

CISRC (Control-Input-Connection Source) Destination Para-
-meter (CIDSTN)
Input# **CISRC** (point.parameter)

1

Output Configuration Display

\$NOPCONN *range = 1 to 4.*
Number of Output Connections *default = 1*

\$SELCTOP
Selected Output

SELOP1
SELOP2
SELOP3
SELOP4

NOCOPTS *range = 0 to \$NOPCONN*
Of Control Output Connections *default = 1*

CODSTN (Control-Output-Connection Destination)
Output # **CODSTN** (point.parameter)

1

2

3

4

\$CDEF1 *range = CVEULO to CVEUHI, NaN*
Output 1 Default *default = NaN*

\$CDEF2 *range = CVEULO to CVEUHI, NaN*
Output 2 Default *default = NaN*

\$CDEF3 *range = CVEULO to CVEUHI, NaN*
Output 3 Default *default = NaN*

\$CDEF4 *range = CVEULO to CVEUHI, NaN*
Output 4 Default *default = NaN*

HPM Output Select CTL Algorithm

Unit:	Tagname:	Engineer:	Date:
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OPHILM			<i>range</i> = OPLOLM - 106.9 & NaN <i>default</i> = 105.0
Output High Limit In Percent	_ . _ . _ . _ . _		
OPLOLM			<i>range</i> = -6.9 - OPHILM, & NaN <i>default</i> = -5.0
Output Low Limit In Percent	_ . _ . _ . _ . _		
SAFEOP			<i>range</i> = -6.9 to 106.9 & NaN <i>default</i> = NaN
Safe Output Limit In Percent	_ . _ . _ . _ . _		
OPMCHLM			<i>range</i> = >=0.0, and NaN <i>default</i> = 0.0
Output Minimum Change In %	_ . _ . _ . _ . _		
OPROCLM			<i>range</i> = >=0.0, and NaN <i>default</i> = NaN
OP Rate Of Change Limit In %	_ . _ . _ . _ . _		
\$OPTOL			<i>range</i> = 0.0 to 106.9 <i>default</i> = 0.0
Output Tolerance in %	_ . _ . _ . _ . _		
OUTIND	Direct		4 - 20mA = 0 – 100%
Reverse Output Indicator	Reverse		20 - 4mA = 0 – 100%
			Note: With R670 and later, when this parameter is set to Reverse, the values of the displayed output and the related output parameters are changed. Refer to the definition of the OUTIND parameter in the PM Family Parameter Dictionary, XPM09.

Alarming Display - N/A if PNTFORM = Componnt

As you enter alarm trip points and alarm priorities in the next section please refer to the following alarm priority definitions and the note.

NoAction	Alarm is not reported to the system and not annunciated (R670 & later)
Journal	Alarm is historized, not reported to US, not annunciated
Low	Alarm is historized, reported to US, and annunciated
High	Alarm is historized, reported to US, and annunciated
Emergency	Alarm is historized, reported to US, and annunciated
Printer	Not historized, reported to the printer, not annunciated
JnlPrint	Alarm historized, reported to the printer, not annunciated

Note: If you do not enter a trip point value for a given pair of trip-points and alarm-priorities, do not circle an alarm priority option for that pair (that is, no trip point, no alarm priority).

Alarming Display - N/A if PNTFORM = Componnt

\$AUXUNIT			<i>range</i> = Any configured LCN Unit ID <i>default</i> = Null (- -)
Auxiliary Unit	_ . _		
\$REALARM (R660 and later)	_ . _ . _ . _		<i>range</i> = 0 to 9999 minutes <i>default</i> = 0 minutes
Re-alarming Interval	_ . _ . _ . _		
BADOCOPT	Off		Bad Output Connection Alarm is not initially permitted
BADOC Alarm Option	On		Bad Output Connection Alarm is initially permitted

Unit:	Tagname:	Engineer:	Date:
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OPALDB Output Alarm Deadband		<i>range</i> = 0 - 25.0 but $\leq (\text{OPHITP}-\text{OPLOTP})/2$ <i>default</i> = 5.0 Output Units
OPHITP Output High Alarm Trip Point		<i>range</i> = OPLOTP to OPHILM and NaN <i>default</i> = NaN
OPLOTP Output Low Alarm Trip Point		<i>range</i> = OPLOLM to OPHITP and NaN <i>default</i> = NaN

BADCTLPR Bad-control Alarm Priority	NoAction (R670 and later)	Journal	Low	High	Emergency	Printer	JnlPrint
ALENBST Point Alarm Enable Status			<u>Displayed</u>		<u>Logged</u>		<u>Reported To EIP</u>
	Enable		Yes		Yes		Yes
	Disable		No		Yes		Yes
	Inhibit		No		No		No

